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Placement

Synopsys University Program

WEEK 6 Introduction to Placement
WEEK 7 Placement

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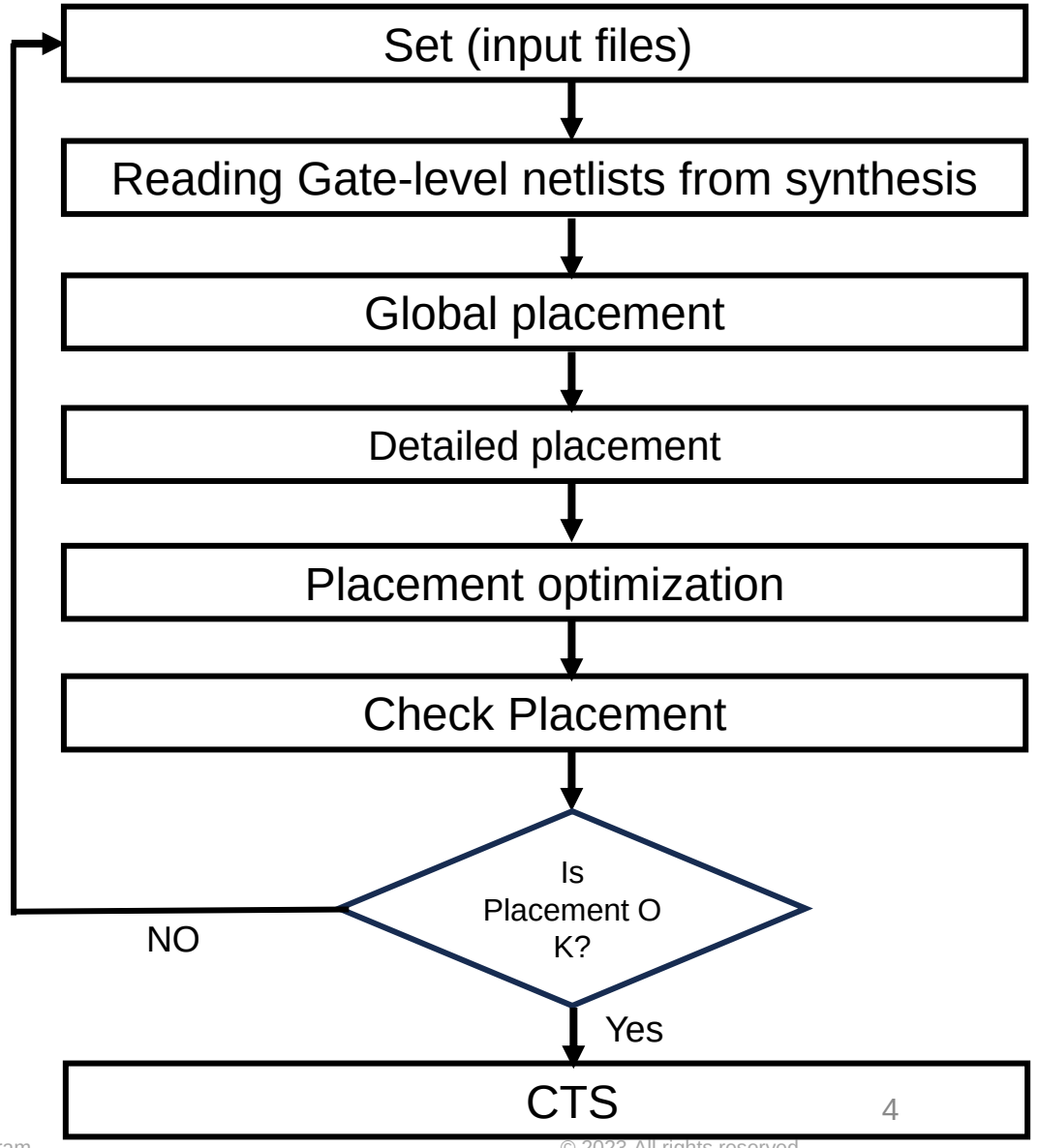
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WEEK 6 : Introduction to Placement

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Introduction to Placement

- Placement Overview
  - This module provides an overview of some of the key steps in R ead Design :
  - **Defining**
    - Tech file (.tf)
    - Netlist
    - Library files (.lib & .lef)
    - SDC
    - TLU+ files
    - Powerplan db
  - **Overview to Placement (directory, input files, script)**
  - **Placement**



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Introduction to Placement

- What is Placement
  - Placement is the process of placing STD cells in the design
  - ICC2 tool determines the location of each STD cell on the die

    (ICC2 places these based on the algorithms which it uses internally)
  - Placement does not just place the STD cells available in the synthesized netlist\

    (Placement also optimizes the design)
  - Placement determines the routability of design
  - Placement will be driven by different criteria like timing driven, congestion driven and power optimization

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Introduction to Placement

- Goals of Pre-Placement
  - Routability
  - Timing Performance
  - Power
- Goals of Placement
  - Timing, Power and Area optimization
  - Minimum congestion
  - Minimal cell & pin density
  - Minimal congestion hot-spots
  - Minimal timing DRVs

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Introduction to Placement

- Things to check before Placement
  - Checks for placement & routing blockages
  - Checks Don’t use cell list and whether it is properly applied in the tool (optional)
  - Checks Don't touch cells & nets
  - Better to have limit the local density (local congestion can create issue in routing or ECO step)
  - Check all optimization options and placement setting in the tool
  - There should not be any high WNS (Worst Negative Slack) timing violations (Make sure that clock is ideal network)
  - Fix all the hard Macros and pre-placed cells
  - Check the pin access

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Introduction to Placement : Method of Placement

- Timing Driven Placement

  - Timing Driven Placement is Net Based and optimize large sets of path delays
  - Timing Driven Placement Refine placement based on congestion, timing & power

  Timing-driven Placement can perform three types of placement, based on the setting of the application option **Plan.place.timing_driven_mode**

  - **-value option**
  - macro: performs timing-driven placement only for Macros
  - std_cell: Performs timing-driven placement only for STD cells
  - both: Performs timing-driven placement only to macros & STD cells

  The following example applies timing-driven placement only to STD cells, Macros are placed normally

```
set_app_options -name plan.place.timing_driven_mode \
   -value std_cell

create_placement -floorplan -timing_driven
```


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Introduction to Placement : Method of Placement

- Congestion Driven Placement
  - Congestion Driven Placement uses a method of distance STD cell instances from each other such that more routing tracks are created between them

Congestion-driven placement can perform three types of congestion reduction, based on the setting of the application option **plan.place.congestion_driven_mode**

- **-value option**
- macro: Reduces the congestion over and between the Macros only
- std_cell: Reduces congestion between STD cells
- both: Reduces congestion both between Macros and between STD cells

The following example applies congestion reduction only between hard macros, and performs coarse placement with high effort

```
set_app_options -name plan.place.congestion_driven_mode \
   -value macro

create_placement -floorplan -congestion \
   -congestion_effort high
```


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Introduction to Placement : Method of Placement

- Combined Congestion-Driven and Timing-Driven Placement
  - To perform congestion-driven placement and timing-driven placement with the same command, use both the and options and specify th e and application options

    (The following example performs congestion-driven and timing-driven placement on both macros and standard)

```
set_app_options -name plan.place.congestion_driven_mode \
   -value both

set_app_options -name plan.place.timing_driven_mode \
   -value both

create_placement -floorplan -congestion -timing_driven
```


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Introduction to Placement : Pre-Placement

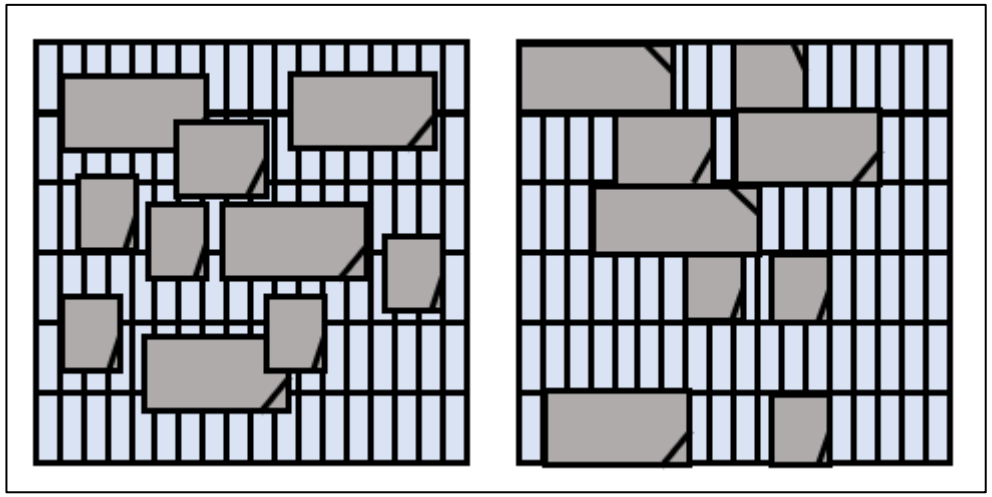
- Physical-Only Cells (Well Tap & End Cap)
  - These physical cells do not have signal connectivity and connect only to the power & ground rails (VDD VSS)
  - End Cap Cells
    - End-cap cells are preplaced physical-only cells required to meet certain design rules and placed at the ends of the site rows by satisfy ing well tie-off requirements for the core rows
    - End-cap cells are positioned in the design to guard against manufacturing damage to a regular cells gate that is located close to the b oundary
    - End-cap cells do not have any signal connectivity and they connect only to the power and ground rails
    - Each end of the core row, left and right, can have only one end cap cell specified
  - Well Tap (Tap Cell)
    - TAP cell is a special no logic cell with a well tie, substrate tie, or both
    - TAP cells reduce resistance between VDD/GND to wells of the Substrate
    - TAP Cells are do not have any signal connectivity and they usually placed on the Power Rails of the Standard Cells (VDD/GND)
    - TAP cells are used to prevent latch-up in CMOS

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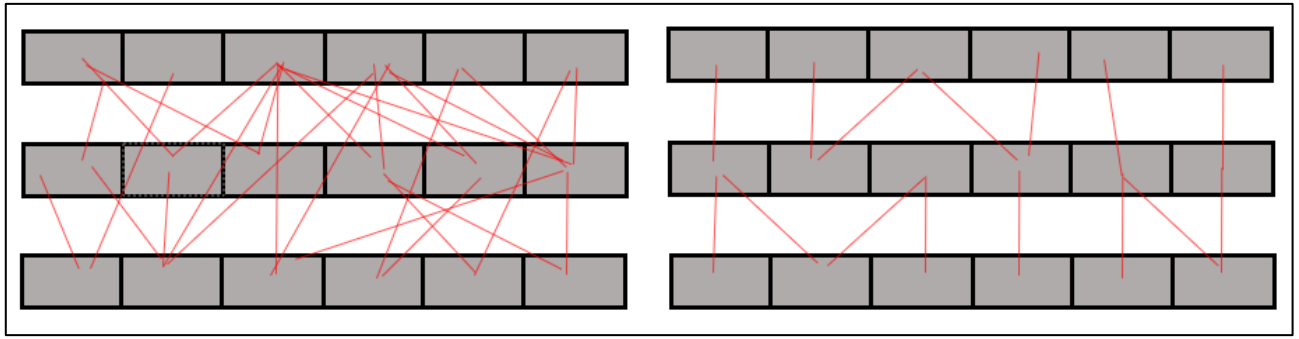
Introduction to Placement : Placement_Flow

- Global placement
  - Global placement is the first stage of STD cell placement
  - Global placement determines the approximate position of each cell, and the result allows partial overlap
- Detailed placement
  - Detailed placement is the final stage of placement
  - All the final quality improvements will occur at Detailed placement
  - Detailed placement further improves the legalized placement solution in an iterative

    manner by rearranging a small group of modules in a location region while keeping all other modules fixed



Global placement Detailed placement



BAD placement : High congestion

Good placement : Low congestion

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Introduction to Placement ：Placement_Flow (place_opt command)

- place_opt is a command that Place and optimize the current design
  - Place_opt consists of the (initial_place, nitial_drc, initial_opto, final_place, and final_opto) step
  - This option cannot be used with the -from or -to options

- **5 stages of place_opt command**

  - **initial_place**

    Places the current design
  - **initial_drc**

    Optimizes the placed design for timing, electrical DRC violations, area, power, and routability
  - **initial_opto**

    Performs incremental placement to optimize timing and routability
  - **final_place**

    Further optimizes the placed design
  - **final_opto**

    Legalizes the design placement at the end

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Introduction to Placement ：Placement_Flow (Global placement)

- Global placement (Core Placement + Timing Analysis + Congestion Analysis)
  - Global placement is the first stage of the placement where cells are placed inside the core area for the first time looking at the timing & congestion
  - Global Placement goals at generating a rough placement solution that may violate some placement constraints while ma intaining a global view of the whole Netlist (several overlap cells during global placement)

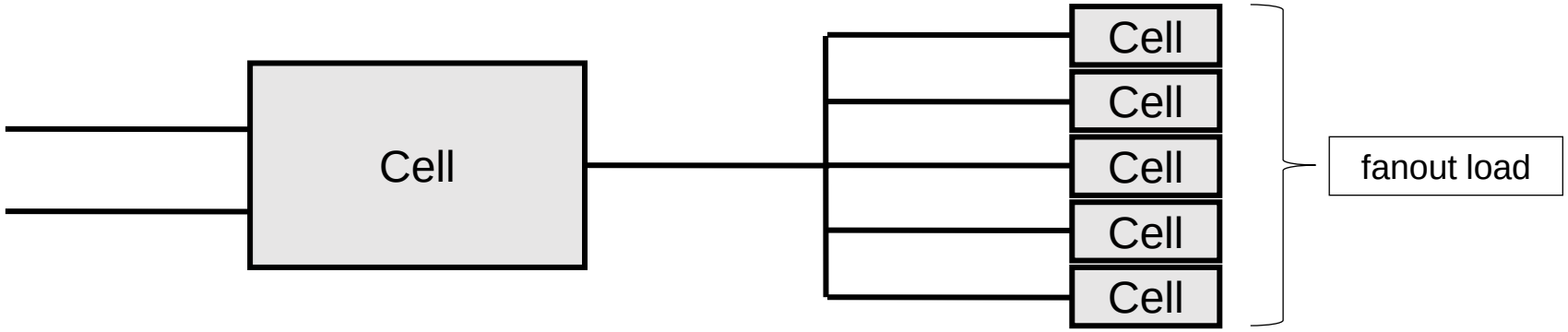
  - **What to do during Global Placement**
    - ❖ High Fanout synthesis
    - ❖ Logic Restructuring
    - ❖ Scan Reordering
    - ❖ Fix input transisition

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Introduction to Placement ：Placement Flow (Global placement)

- **High Fanout synthesis**
  - **What is Fanout**

    Fanout is Number of gates connected from an output of a driver cell (Fanout is basically a number of gate input which the output is saf ely connected)



  - **What is Fanout load**

    High fanout load means there are more number of loads which is being driven by a single output of a gate (Load more than desired ca uses to issue in the design)

  - **How to fix High Fanout**
    - Optimize the fanout on violating nets or upsizing driver instance
    - Splitting load by adding a similar instance at placement

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Introduction to Placement ：Placement Flow (Global placement)

- **Logic Restructuring**
- Placement engine restructure the netlist if there are timing benefit (restructuring means gate composition or decomposition)
- The optimization engine evaluates the impact of restructuring on congestion, power, area etc...
  - **Area restructuring**

    Area restructuring finds functionally equivalent cones of logic that improve area/leakage without degrading timing
  - **Timing-based restructuring**

    Restructuring solutions are evaluated in parallel with other transforms like sizing and buffering, to find the best overall timing solution

    ❖ **Example of Timing-based restructuring**

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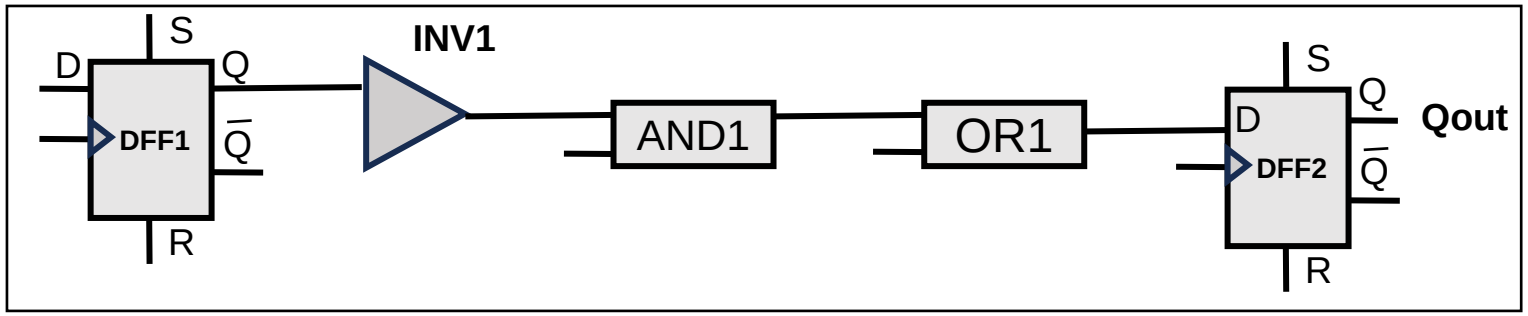
Introduction to Placement : Placement Flow (Global placement)

- **Scan Reordering**
  - **What is scan chain**

    The purpose of adding scan chain is to make the flip-flops (FF) in the design controllable and observable them

  - **Circuit which consists of 2 Dflip-flop (where there is small combinational circuit present in between the flops)**

  ❖ **Example of no scan chain inserted logic**

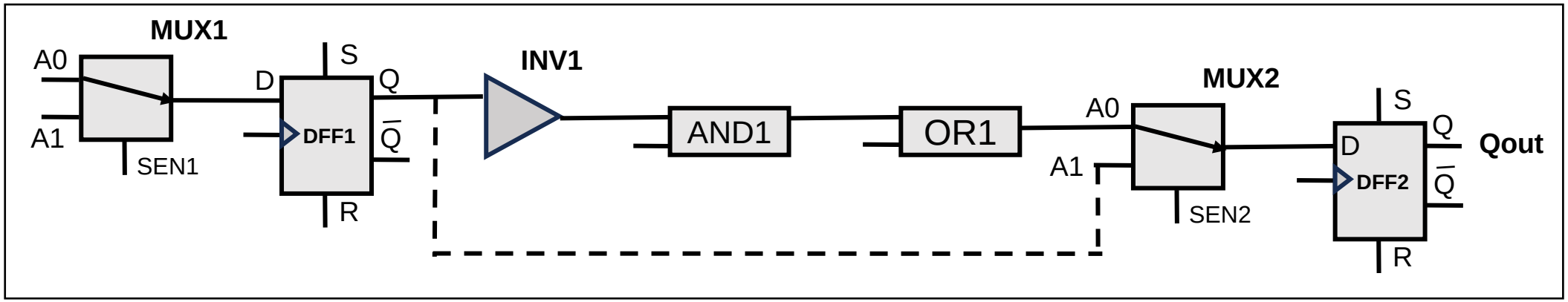


Typically when scan chain has not been inserted, the circuit fully operates for functional

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Introduction to Placement : Placement Flow (Global placement)

- **Scan Reordering**
  - **Circuit which consists of 2 Dflip-flop (where there is small combinational circuit present in between the flops)**
  - ❖ **Example of scan chain inserted logic**

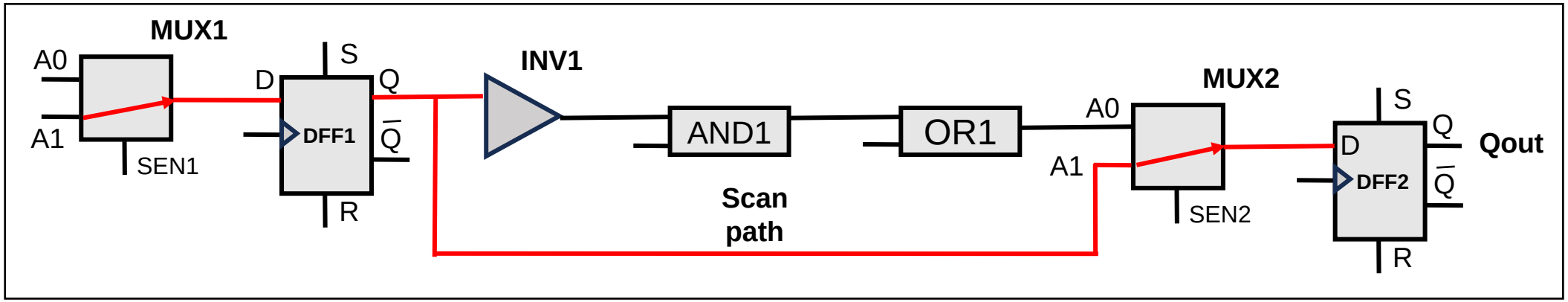


- After scan chain inserted then Muxs will be inferred and the select line of Mux will control the functional and scan propagation
- If SEN = 0, then A0 will be propagating at this time (functional propagation) and if SEN = 1 then A1 will be propagating, it will be sc an path
- For Example, we can see that SEN1 and SEN2 are in 0 state and hence A0 is getting propagated (which means functional propag ation happens here)
- A0 – DFF1 – INV1 – AND1 – OR1 - MUX2 – DFF2 is the path (how the logic propagation happens in functial mode)

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Introduction to Placement : Placement Flow (Global placement)

- **Scan Reordering**
  - **Circuit which consists of 2 Dflip-flop (where there is small combinational circuit present in between the flops)**
  - ❖ **Example of scan chain inserted logic**



  - If SEN1 and SEN2 is 1 then A1 will be propagating and the path will be a scan path
  - Path will be as A0 – DFF1 – MUX2 - DFF2

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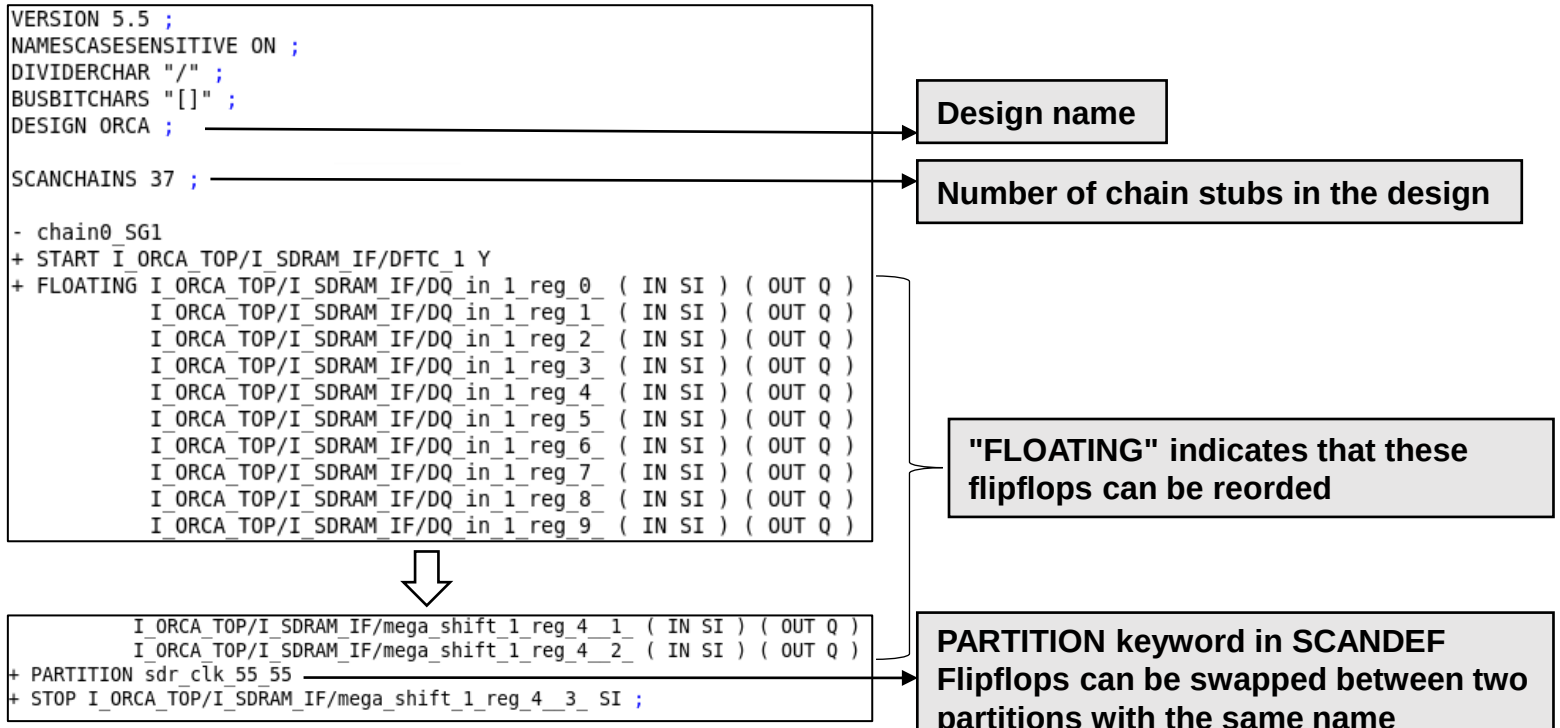
Introduction to Placement : Placement Flow

- **Scan Reordering**
  - **What is scan.def**

    Scan.def is a file, which containing scan chain information

    In scan.def, there are two can chains START and STOP keyword that denotes the starting and the ending point of the scan chain

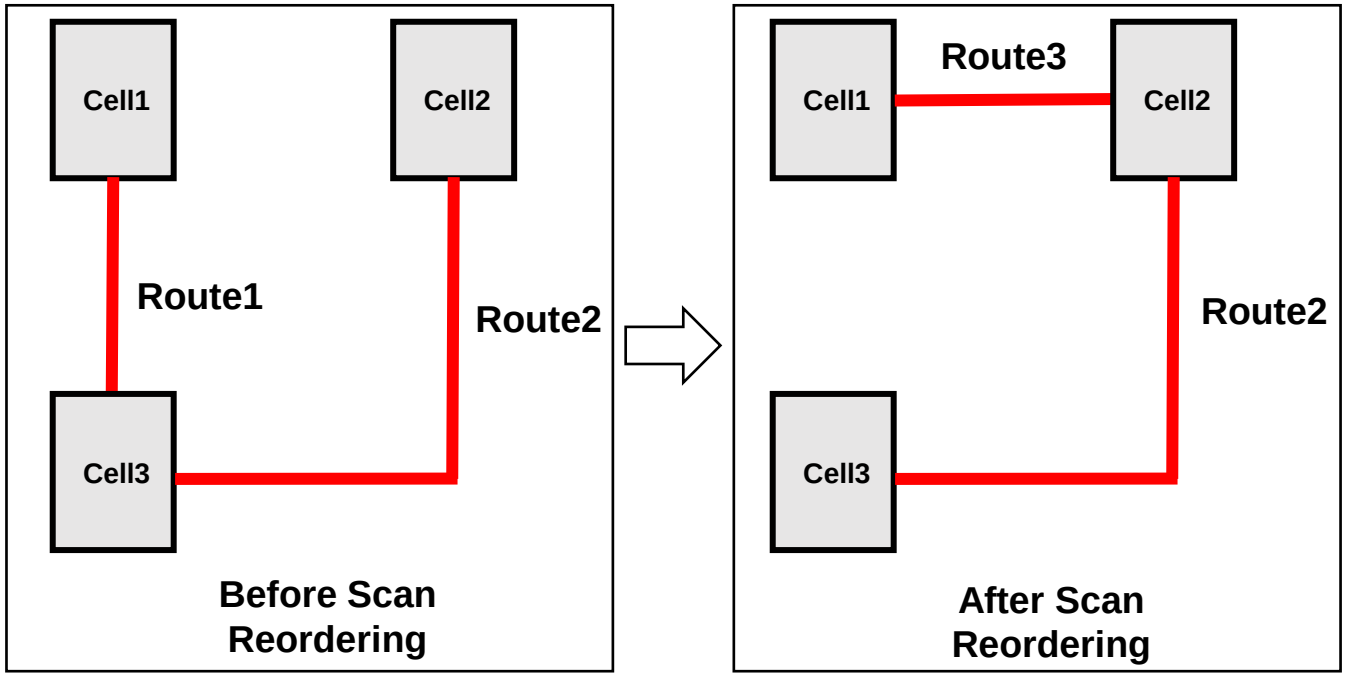
    ❖ **Example of scan.def**



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Introduction to Placement : Placement Flow (Global placement)

- **Scan Reordering**
  - Scan Reordering helps optimizing the routing resources and make design decongested
  - ❖ **Example of Before & After Scan Reordering**

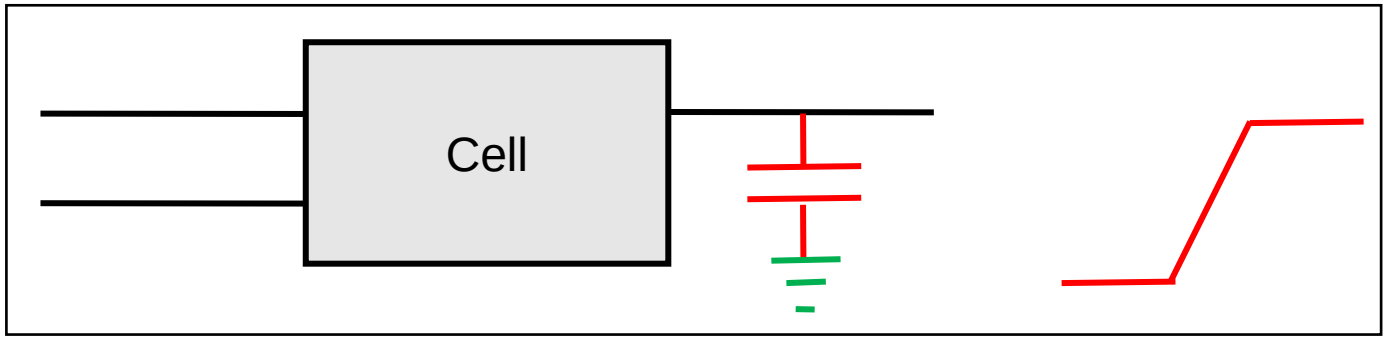


  - route 1 and route2 completes the scan chain routing which is comparatively longer route than after scan chain reordering where w e have small route length completing our requirement
  - During placement optimization, tool does these optimizations to save route tracks for the clock/signal routing at next stage
  - After Scan reordering, it helps optimizing the routing resources

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Introduction to Placement : Placement Flow (Global placement)

- **Fix input Transition**
  - Delay depends upon input transition and output capacitance
  - if high input transition is there, then we will see more delay in the design
  - SDC command “set_input_transition” sets the transition value, so if input pin transition go beyond the number then that creates violations
  - ❖ **Example of input Transition in VLSI**



  - **How to fix Transition**
    - Reduce wire length by placing cells near to each other
    - Add buffer in the transition violated path
    - Split buffer into inverters and place with some distance
    - Assign higher metal layers to all the nets to the tran violated nets
    - Upsize driver instance by increasing the drive strength (Cell sizing)

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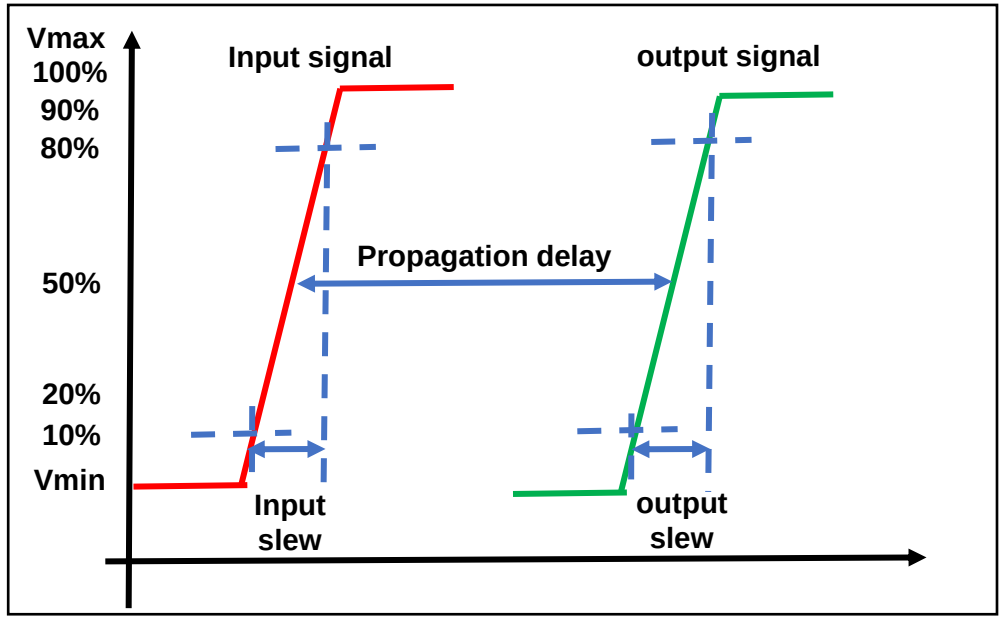
Introduction to Placement : Placement Flow (Global placement)

- **set max_transtion & max_capacitance**

  - **set_max_transition**
    - This command is used to change the max transition time restriction specified in a technology library
    - This command sets a maximum transition time for the nets attached to the identified ports or to all the nets in a design by setting the **max_transition** attribute on the named objects
  - **set_max_capacitance**
    - The transition time constraints specified above do not provide a direct way to control the actual capacitance of nets
    - This command sets the maximum capacitance constraint on input ports or designs
    - This command applies maximum capacitance limit to output pin or port of the design
    - This command can also be used to apply capacitance limit on any net

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Introduction to Placement : Placement Flow (Global placement)



- **Transition Delay**

  - Transition delay or slew id defined as the time taken by signal to rise from 10%(20%) to the 90%(80%) of its maximum value (rise time)
  - Fall time can be defined as the time taken by a signal to fall from 90%(80) to the 10%(20%) of its maximum value
  - Transition is the time it takes for the pin to change state

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Introduction to Placement : Placement Flow (Global placement)

- **Propagation Delay**

  - Propagation delay is the time required for a signal to propagate through a gate or net
  - If it is cell, call it as Gate cell delay of cell delay
  - If it is net, call it as Net delay
  - Propagation delay of a gate or cell is the time it takes for a signal at the input pin to affect the output signal at output pin
  - For any gate propagation delay is measured between 50% of input transition to the corresponding 50% of output transition
    - **4 possibilities of Propagation delay**

    1) Propagation delay between 50 % of Input rising to 50 % of output rising
    2) Propagation delay between 50 % of Input rising to 50 % of output falling
    3) Propagation delay between 50 % of Input falling to 50 % of output rising
    4) Propagation delay between 50 % of Input falling to 50 % of output falling

  - Each of delays has different values (Max & Min values are very important)
  - Max & Min propagation delay values are considered for timing analysis
  - For net propagation delay is the delay between the time a signal is first applied to the net and the time it reaches other devices connected t o that net
  - Propagation delay depends on the input transition time and the output load

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Introduction to Placement : Placement Flow (Refine Placement)

- **Refine Placement (placement legalization)**
  - Legalization makes the rough solution from global placement legal moving modules around locally
  - Check congestion, timing, and power the objects are moved within the region to achieve the goal of placement
  - Refine placement takes care of overlaps of cells/modules within the core area and correct it by moving the cell/module nearer
  - There are number of tool options which we can use for refining the placement

    - **-effort very_low | low | medium | high**

      Specifies the CPU effort level for coarse placement. If running very low effort it is advised to set the -congestion_effort to low

      (The default effort level is medium)
    - **-congestion_effort low | medium | high**

      Specifies the effort level of congestion removal

      The higher effort, the less attention is paid to other objects: timing, density and total wire-length (The default effort level is medium)
    - **-perturbation_level low | medium | high**

      Specifies the perturbation level for the placement to optimize congestion

      Specifying low produces the least amount of movement from the placement

      The higher the level specified, the more movement you might expect from the placement. So, specifying high produces the most movement from the placement

      (The default effort level is medium)
    - **-coordinates bbox**

      Specifies one rectangle that defines the region from which to run incremental congestion removal

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Introduction to Placement ：Placement Flow (Detailed Placement)

- **Detailed Placement**

  - Detailed placement further improves the legalized placement solution in iterative manner by rearranging a small group of moules in a local region while keeping all other modules fixed
  - Detailed placement is the final stage of placement
  - Detailed placement try & minimize wirelength
  - Detailed placement try to finish with uncongested design
  - All the final quality improvements will occur at this stage

  - **Taken care of during detailed placement**
    - **Timing**
    - **Congestion**
    - **Power**

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Introduction to Placement ：High Fanout Synthesis Port Punching

- **By default, port punching may add or remove ports at hierarchy boundaries for HF synthesis**

- **To disable port punching on specified cells**

```
set_freeze_ports -data|-clock|-all [get_cells "CELL_LIST"]
```

  - **Set_freeze_ports**

    Sets the freeze_ports attribute on cells to prevent synthesis from modifying their port signature during optimization

  - **To un-freeze the ports use**

```
set_freeze_ports -all [get_cells "scan_en_iopad"] false
```

  - **To query the current status**

```
icc2_shell> get_attribute [get_cells "scan_en_iopad"] freeze_data_ports
true
icc2_shell> get_attribute [get_cells "scan_en_iopad"] freeze_clock_ports
true
```

- Setting a dont_touch on a hierarchical cell will preserve the pins along the block boundary but will also prevent optimization from modifying the internal block logic

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Introduction to Placement : Distribution of Spare Cells

- **Add Spare Cells Not Already in the Netlist**
  - If your design doesn't contain spare cells already, or you want to add additional spare cells, you can add spare cells after place_opt as well

Original netlist

↓

Place_opt …
#Spare cells from design netlist are
#Placed and spread during place_opt

↓

```
set core_bdry "{425.543 425.846} {1014.459 1014.152}"

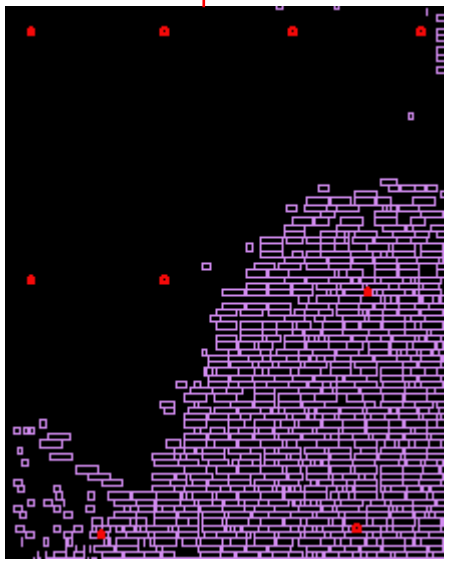
add_spare_cells -num_instances 30 \
        -boundary $core_bdry \
        -cell_name ORCA_spare_cell \
        -lib_cell {saed32rvt/NBUFFX2_RVT}

legalize_placement -incremental
set_placement_status legalize_only \
        [get_flat_cells -filter is_spare_cell]
```

↓

CTS & Routing

- Listed cells are spread evenly across the core area
- If you insert various cells, cells are placed close together as a set



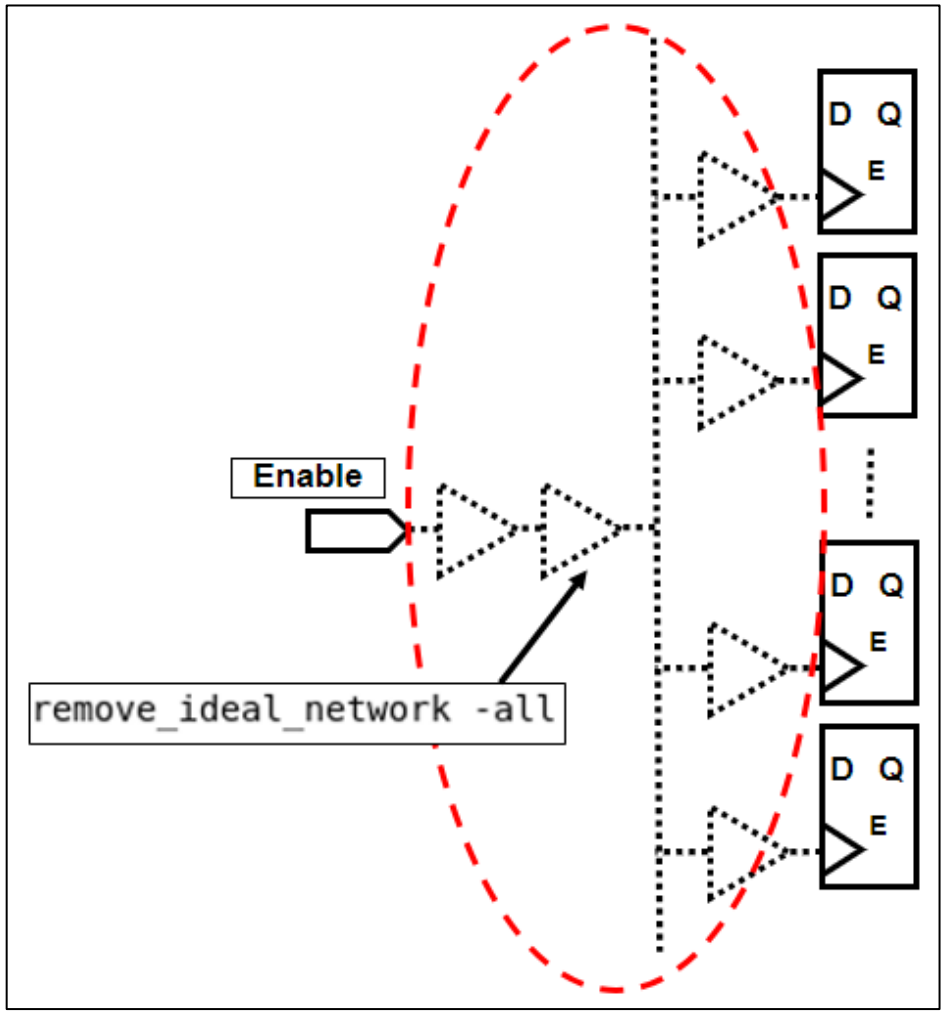
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Introduction to Placement ：Remove Unwanted ideal Networks

- **Ideal network constraints on high fanout nets (like set/reset, enable) prevent buffering during placeme nt**
  - Report_ideal_network

    Displays information about ports, pins, nets, and cells on ideal networks in the current design

- **Remove all ideal network constraints to allow HFN buffering during placement**



- An ideal network prevents Design Compiler from synthesizing buffer trees, to defer buffering to the physical design phase
- To remove pre-existing buffer trees, use **remove_buffer_trees** command

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Introduction to Placement : Library Cell Purpose

- **Every library cell can be restricted to certain optimization tasks using a library cell purpose**
  - **set_lib_cell_purpose**

    Specifies valid purposes for library cells

```
set_lib_cell_purpose -include cts|optimization|hold|power|all|none {lib_cell List}

set_lib_cell_purpose -exclude cts|optimization|hold {lib_cell List}
```

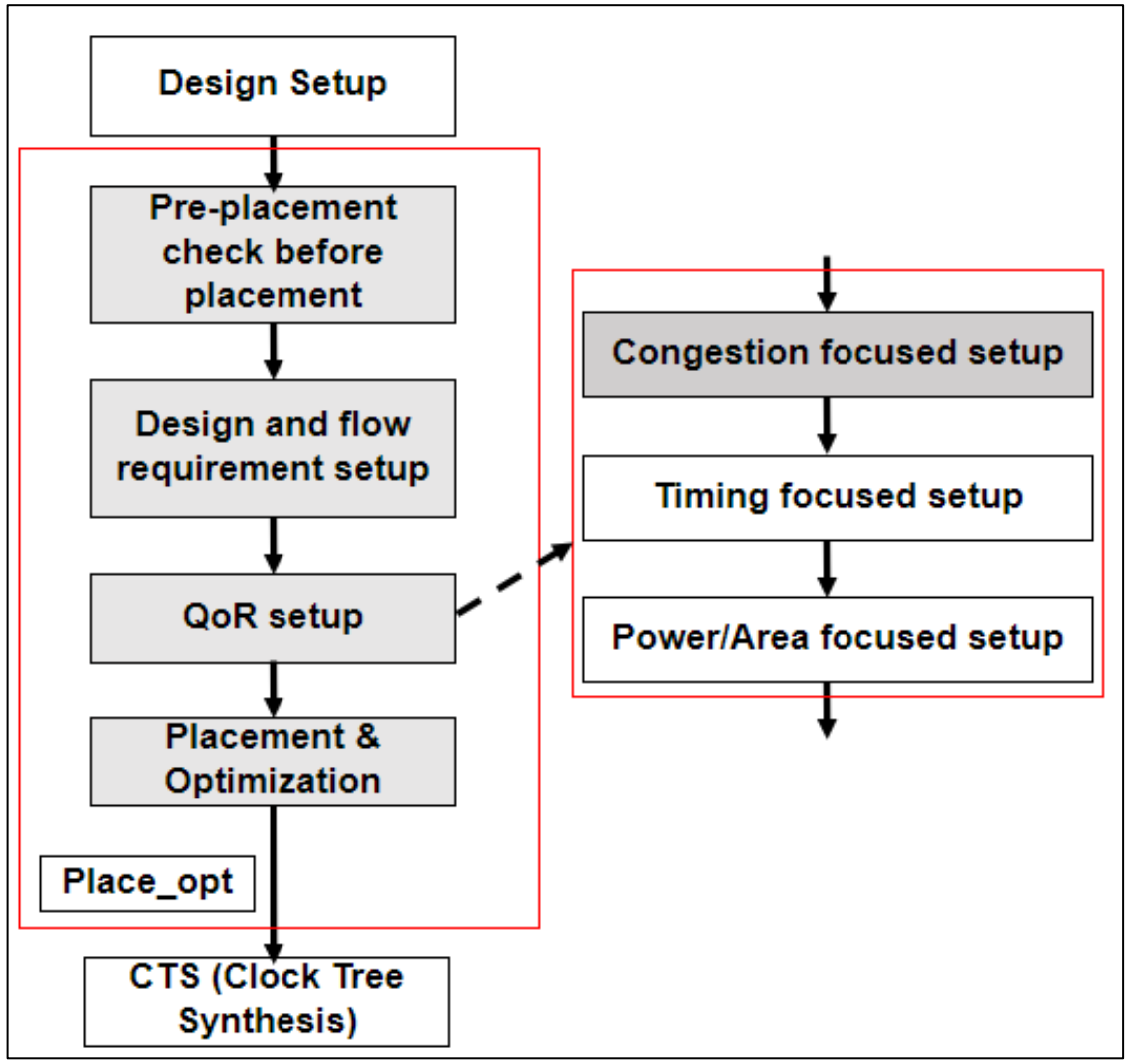
- **The following lib cell purposes are recommended:**
  - ❖ **Optimization –** for setup/hold, power and logical design rule optimization
  - ❖ **CTS –** for clock tree synthesis
    - ❖ **Hold purpose behaves like optimization**
    - ❖ **Power purpose is configurable but is not use by the tool**

- If the original cells in the liberty library have dont_use=true, then no purposes are set

- When the set_lib_cell_purpose command is used on the library cells, the valid_purposes attribute should be set to the corresponding values
- If you subsequently set the dont_use attribute to true on the same lib cell, the dont_use attribute setting overwrites the valid_purposes attribute

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Introduction to Placement : Congestion-Focused Setup Steps



- After running **place_opt** analyze congestion & cell/pin density
- Perform the following setup steps, as applicable, to improve routability
  - Increase placement and congestion effort levels
  - Enable global route-based HFS(High Fanout Synthesis) and DRC Fixing
  - Apply keepout margins and placement blockages
  - Enable pin access checking and optimization
  - Enable cell expansion using GR(Global Routing) congestion maps
  - Run congestion-driven restructuring (CDR)

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Introduction to Placement ：Increase Placement/Congestion Effort

- **Increasing the congestion effort to high (default is medium) generally improves congestion, at a cost of runtime and possible timing degradation**
  - Relevant during final_place, and during initial_place when using two-pass placement

> ❖ The place_opt command will run a two-pass flow to generate a better initial placement
>
> ❖ The two-pass flow performs wirelength-driven placement, followed by DRC fixing and another timing-driven placement

- **Increasing the final_place effort level to high may improve congestion**
  - May also improve timing

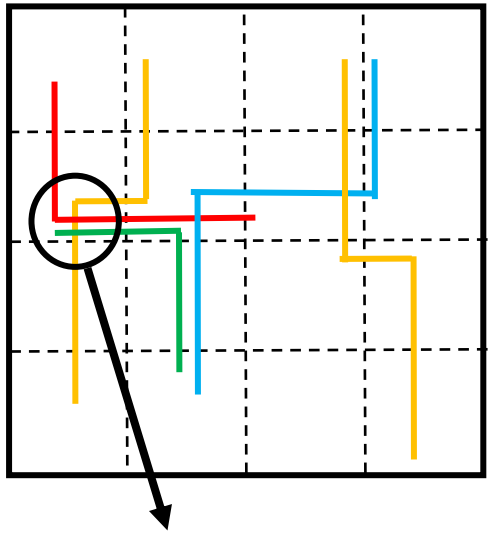
```
set_app_options -name place_opt.place.congestion_effort -value high

set_app_options -name place_opt.final_place.effort -value high
```


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Introduction to Placement : Congestion

- **The number of routing tracks available for routing in a given location is less than the number necessary, the area is considered congested**
- **Thus the increased circuit density introduces additional congestion**
- **Congestion in a layout means too many nets are routed in local regions**
- **Congestion causes detoured nets and un-routable nets in Detailed Routing**



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Introduction to Placement : Congestion Analysis

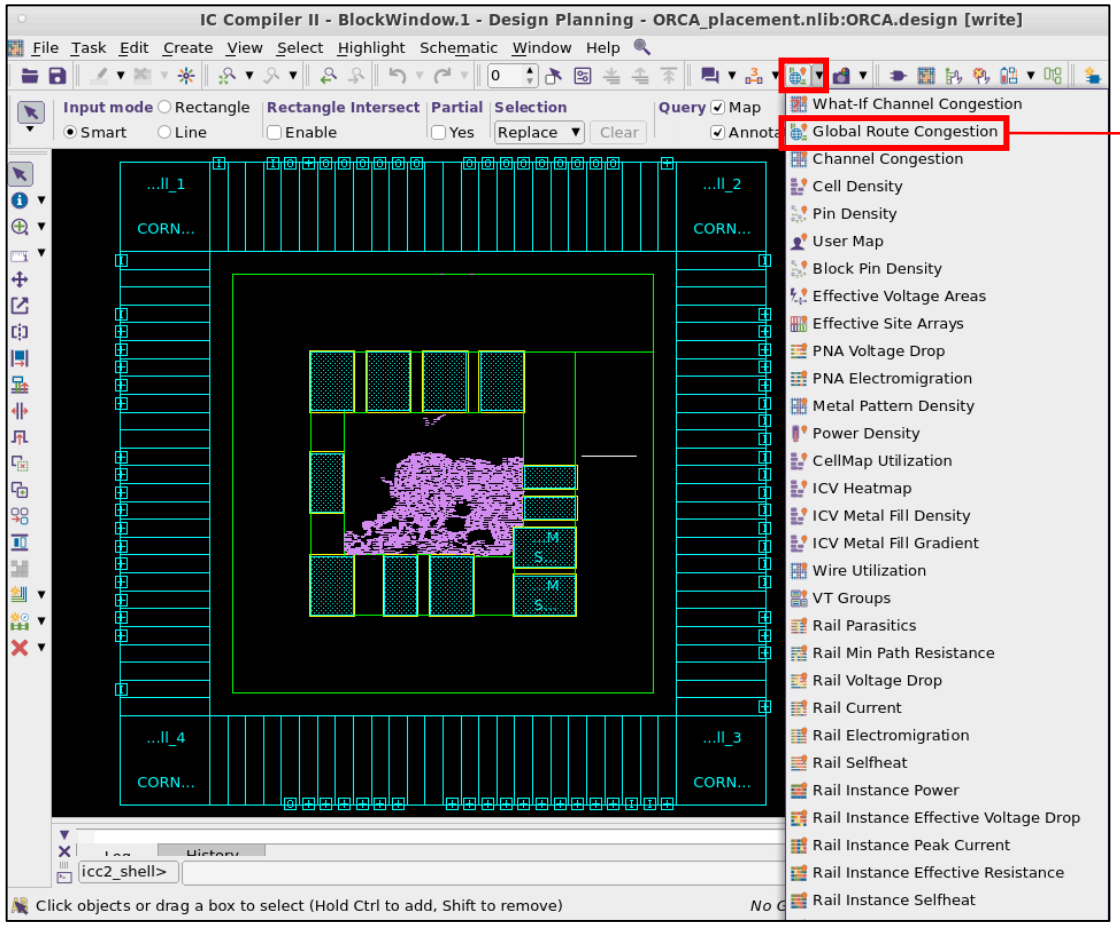
- **Routing Congestion Analysis**
  - Congestion in general referred to Routing Congestion
  - Routing congestion is the difference between supplied and available tracks
  - A track is nothing but a routing resource which files the entire Core
- **Placement Congestion Analysis**
  - Placement Congestion is due to overlap of STD Cells
  - Overlapping issue can be fixed by aligning cells to the Placement grid by legalization

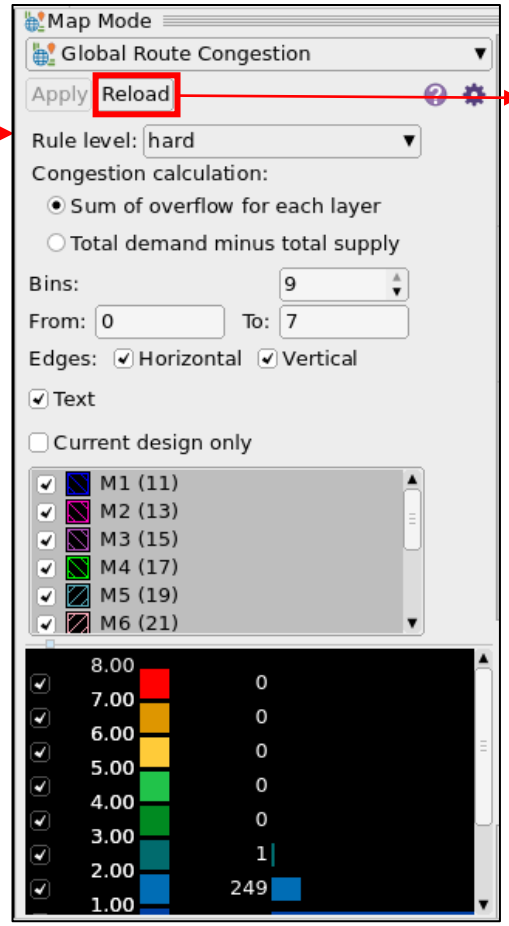
- They fall into two categories (Congestion estimation and removal during global routing stage, and Congestion estimation and removal during Placement stage)
- To estimate Congestion, tool does initial/global routing
- Congestion reports are generated after each Routing stages which shows difference between supplied and demanded Tracks or G-cells
- Overflow = Routing Demand – Routing supply
- Typically starts the initial Target Utilization with 65~70%

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Introduction to Placement : Global Route Congestion window

- **Tool supports the Global Route Congestion map to highlight areas of the design which contain a high demand for global routes for the specified layers (Congested areas might lead to problems during detail route)**





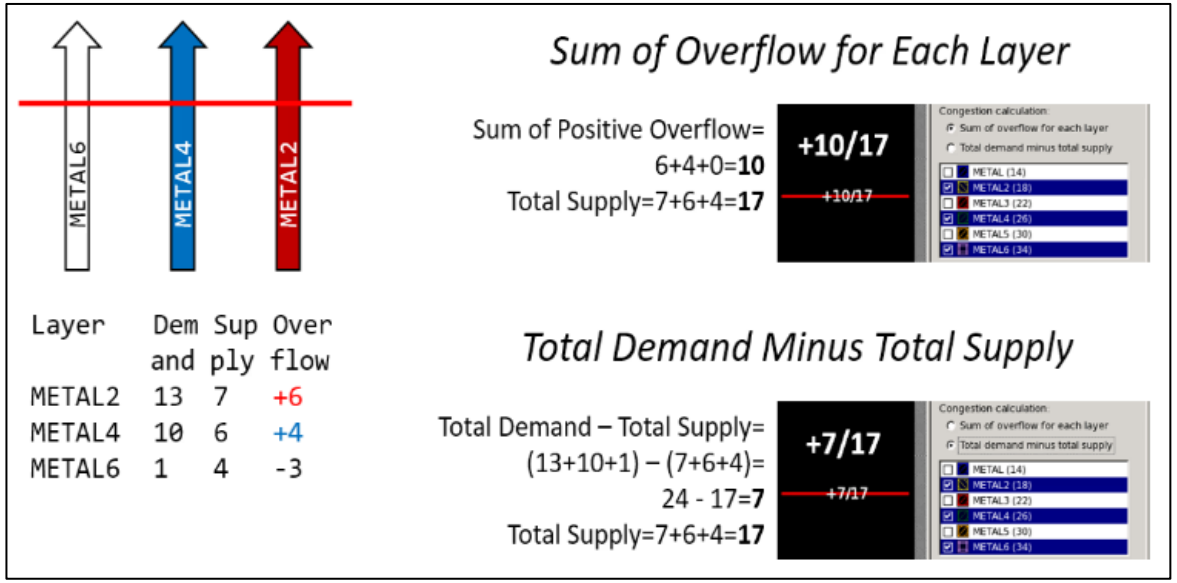
To generate the global route congestion debug map, click the reload button

- View location
  You can open the Global Route Congestion Map by choosing View > Map > Global Route Congestion

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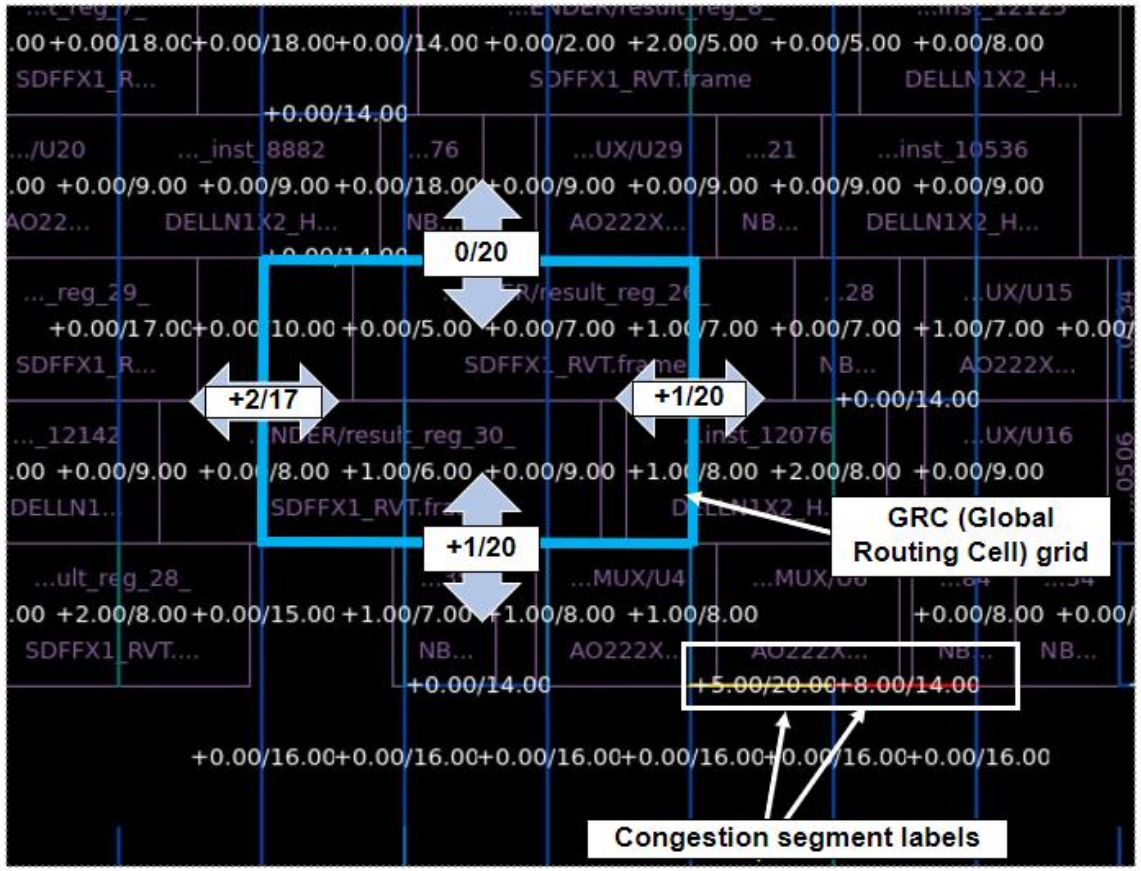
Introduction to Placement : Congestion map calculations

- The Global Route Congestion map supports two calculation modes: "Sum of overflow for each layer" and "Total demand minus total supply"
- The mode determines the congestion value that the tool displays when a global route cell has congestion; the sum-of-overflow mode is the def ault



Layer	Demand	Supply	Overflow
METAL2	13	7	+6
METAL4	10	6	+4
METAL6	1	4	-3

- Layer METAL2 is heavily congested with a demand of 13 routing tracks and a supply of only 7 tracks, for an overflow of 6
- Layer METAL4 is moderately congested with an overflow of 4, and layer METAL6 is not congested and contains underflow of 3
- Other layers are not used in the congestion calculation in this example



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Introduction to Placement : Global Route Congestion option

- **Rule level**

  Select the rule level to apply to the congestion calculation

- **Congestion calculation**

  Select Sum of overflow for each layer or Total demand minus total supply to select the method used to calculate congestion

- **Bins**

  Change the number of unique bins used to display the map by changing the Bins value; a larger number of bins partitions the map into a greater number of value ranges

- **From and To**

  Change the range of overflow values displayed in the map and the histogram by changing these values. These values represent the overflow in routing capacity for the route cell, where higher numbers represent more severe congestion

- **Edges**

  Select the edges, Horizontal and Vertical, of the global route cell for which to display congestion values

- **Text**

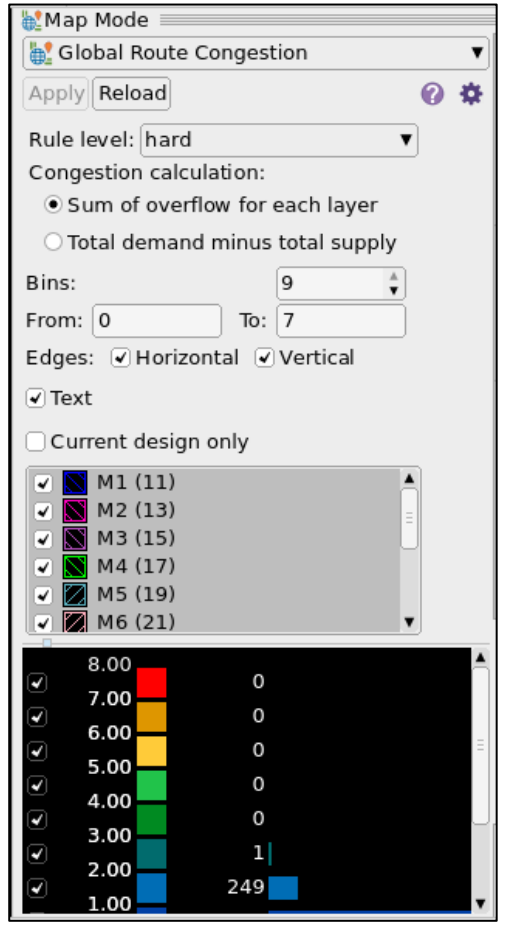
  Select this option to display in the layout a number that represents the calculated congestion values. If the values are not visible, you must zoom in to view them and make sure that range is selected in the histogram

- **Current design only**

  Only display map data for current design. When unchecked, the tool displays data for all expanded cells.

- **Layers**

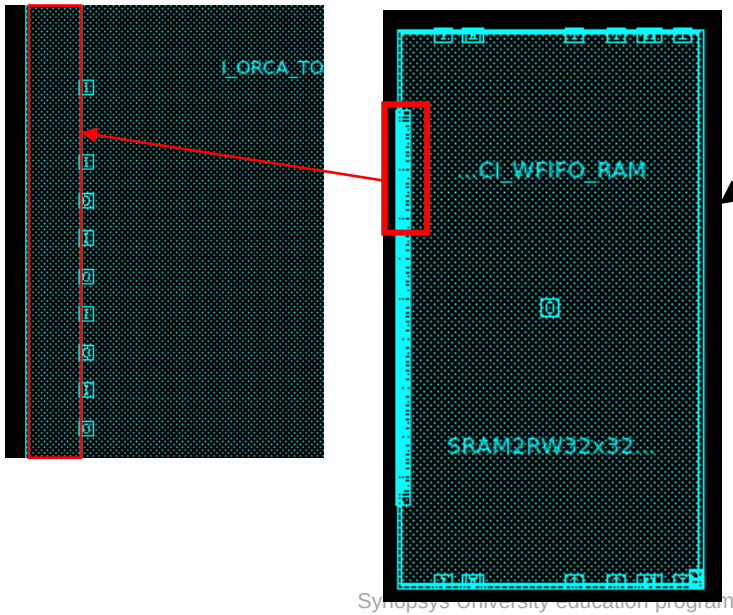
  Select the routing layers to consider when calculating congestion 0505 2788 341



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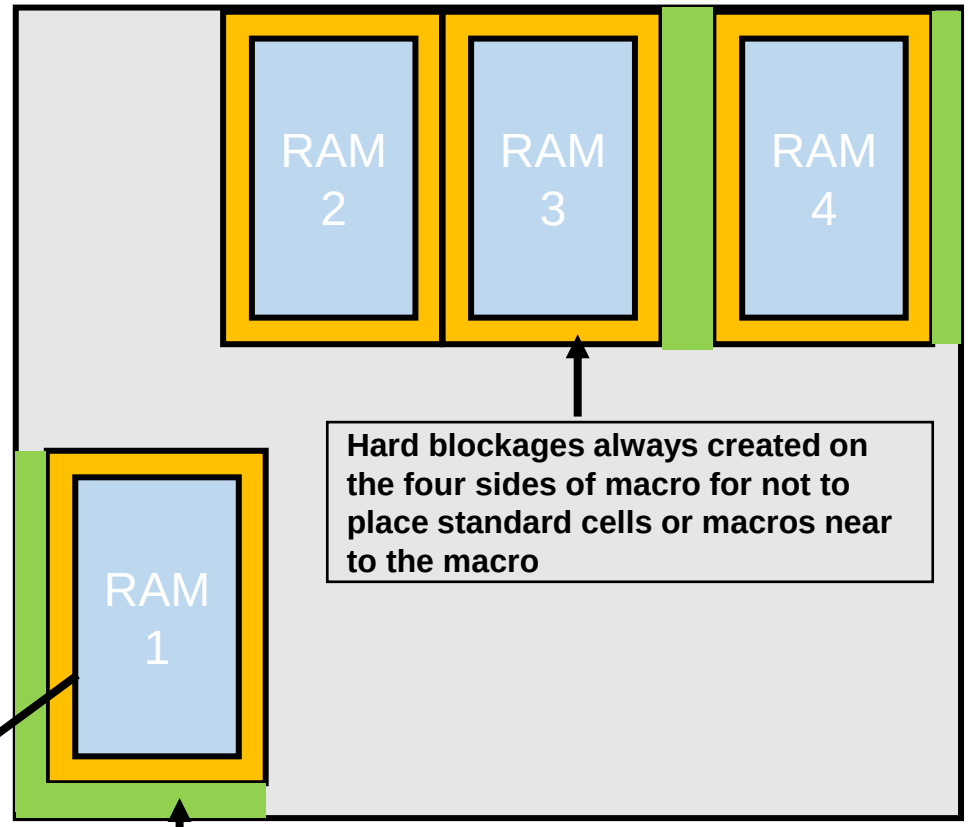
Introduction to Placement : How to fix Congestion issue

- **Method of Congestion Fixes**
  - Add placement blockages in channels and around Macro corners (soft, hard)
  - Review the floorplan (Macro placement)
  - Reduce local cell density
  - Reordering scan chain to reduce congestion
  - Congestion driven placement with high effort
  - Continue the iterations until good congestion results during placement
  - Density screen is applied to limit the density of STD cells in an area to reduce congestion due high pin density



If the design contains Macros that are not placed near another Macro (or) the edge of the core then Halo is created

(Standard cells can not be placed in this region which give more routing resources to the signal routes)



Hard blockages always created on the four sides of macro for not to place standard cells or macros near to the macro

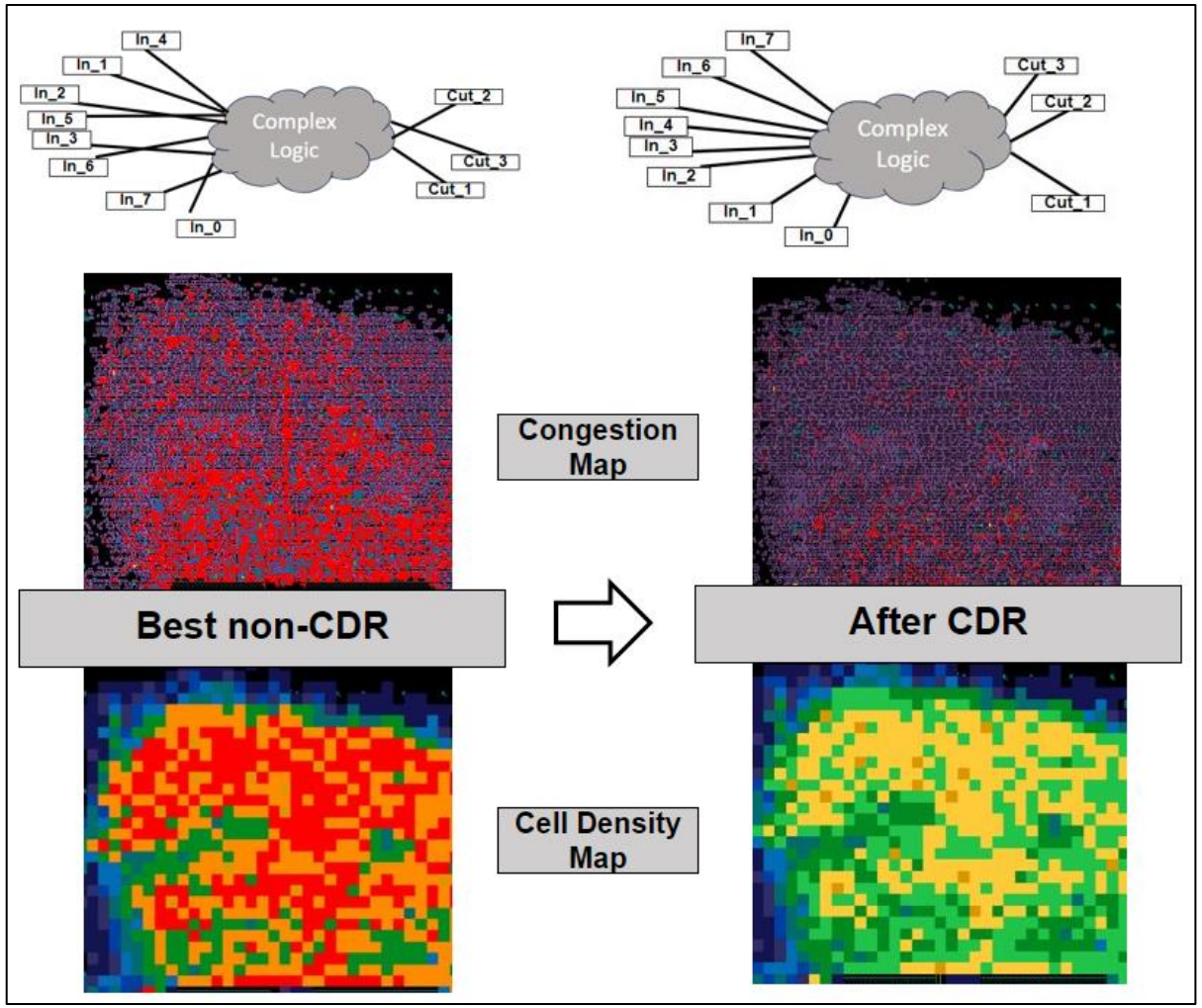
Soft Blockages created only for the channels between macros (or) Between macro & the core boundary to give enough space for routing

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Introduction to Placement ：CDR (Congestion-Driven Restructuring)

- Designs with complex AOI/OAI logic structures can cau se many net crossings, thereby creating substantial cor e congestion hotspots

- CDR identifies tangled nets that drive input pins of com mutative and associative logic trees ((N)AND/OR/XOR trees), reorders and places them more optimally
  - **Alleviates congestion**
  - **Reduces wire length**



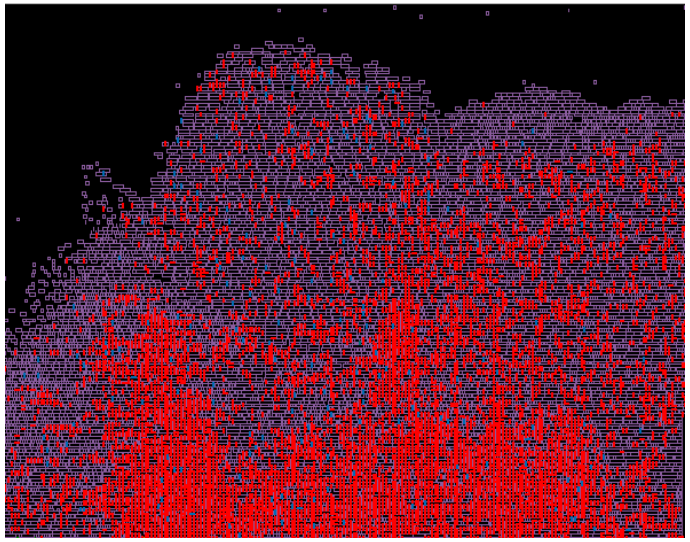
- **From the "non-CDR" congestion and cell density maps on the left, notice that in the area with high congestion, there is actually low cell density – the placer avoids placing cells in high congestion areas (This can lead to higher cell density, and therefore increased congestion in surrounding areas)**

- **On the right, once CDR untangles the net crossings and thereby eliminates the congestion, the placer is now able to place more cells in this same area, which can help to reduce congestion in surrounding cell density areas**

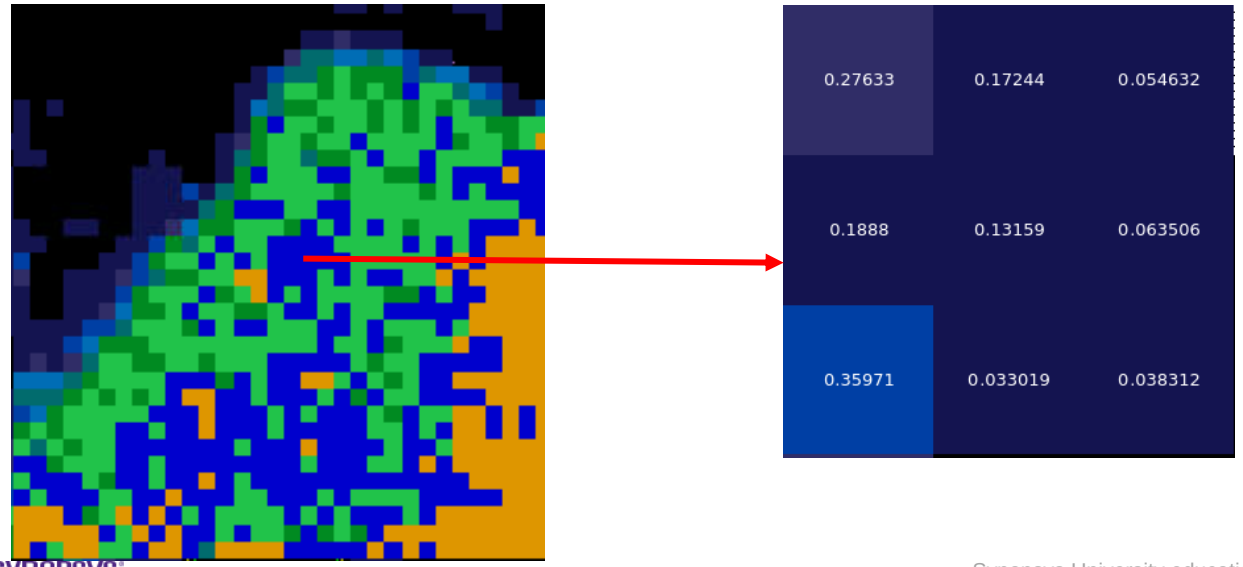
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Introduction to Placement ：High cell Density Causing Congestion

Congestion map



Cell density map



- **The placer may create high density areas**
  - When optimizing for timing (insert buffer of inverter), whi ch can cause congestion

- **The placer may create high density areas**
  - If cell density hotspots coincide with congestion hotspots (Lower cell density in that area)

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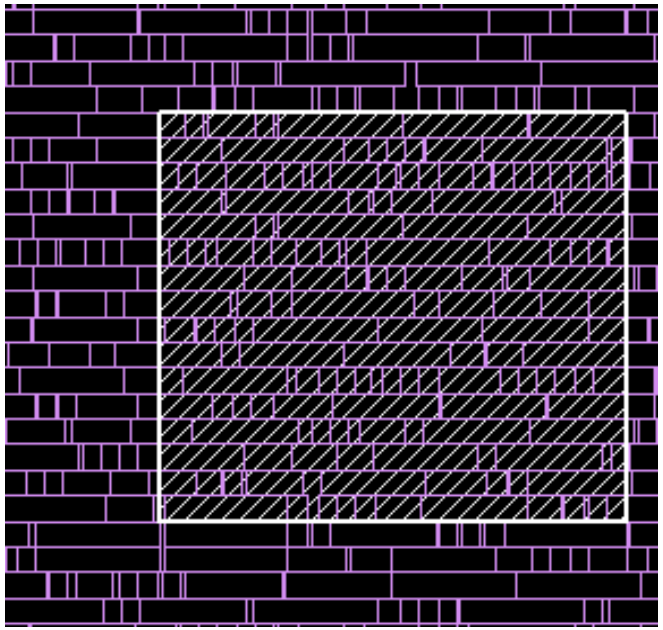
Introduction to Placement ：Limit Cell Density with Partial Blockages

- **A "category" type blockage can be used to create user-defined partial blockages**

```
# Define an area with a maximum utilization of 40%
create_placement_blockage -boundary {{llx lly} {urx ury}} \
                          -type partial -blocked_percentage 60

# Max utilization 30% allow buffers and inverters only
create_placement_blockage -boundary {{llx lly} {urx ury}} \
                          -type allow_buffer_only -blocked_percentage 70


# Max Utilization 80% partial blockage, no registers allowed
create_placement_blockage -boundary {{llx lly} {urx ury}} \
                          -type register -blocked_percentage 20
```



- **-type option**
  - **Partial**

    A partial blockage limits the cell density in the specified region

    (When you create a partial blockage, you must specify the blocked percentage value by using the -blocked_percentage option)
  - **Allow_buffer_only**

    An allow-buffer-only blockage allows only buffer and inverter cells to be placed in the specified region

    (When you create an allow-buffer-only blockage, you must specify the blocked percentage value by using the -blocked_percentage option)
  - **Register**

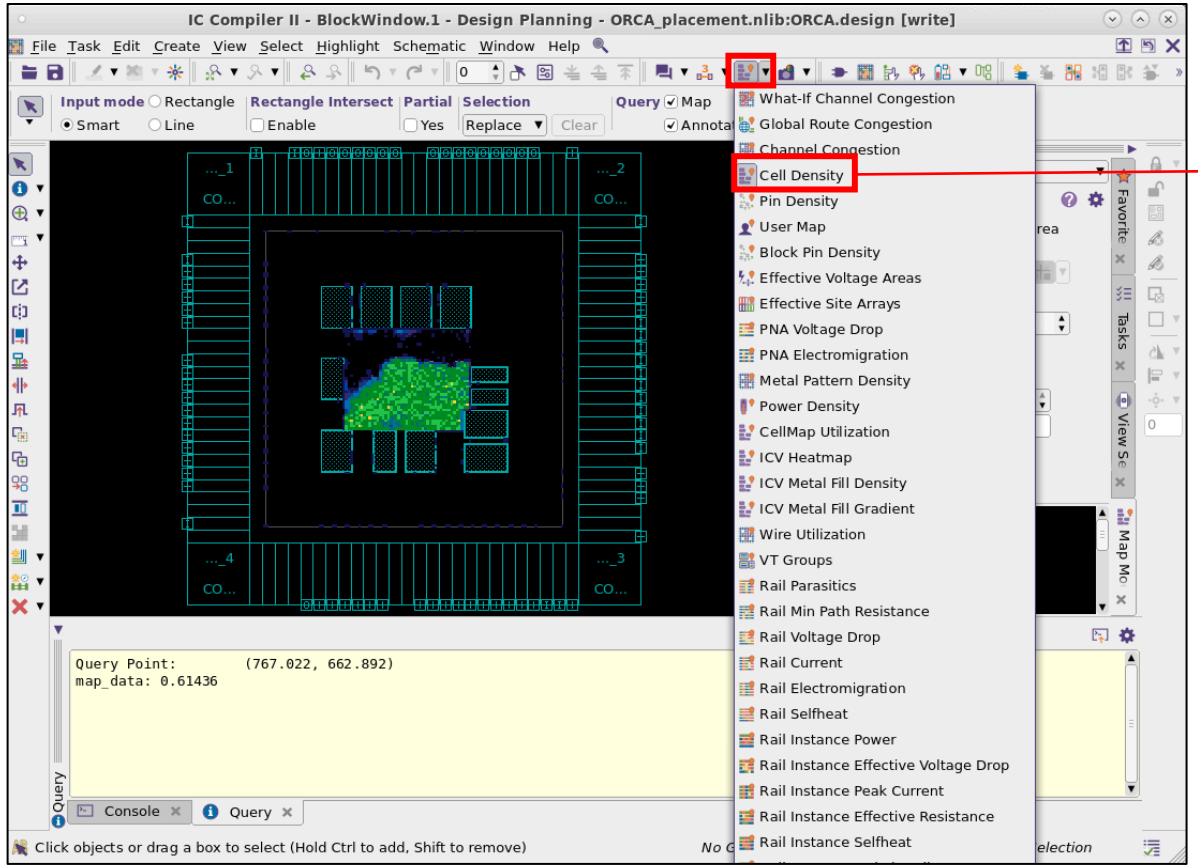
    A register blockage is a partial blockage that also prevents the placement of registers in the specified region, and It is honored by coarse placement, but not optimization, or clock tree Synthesis (When you create an allow-buffer-only blockage, you must specify the blocked percentage value by using the -blocked_percentage option)

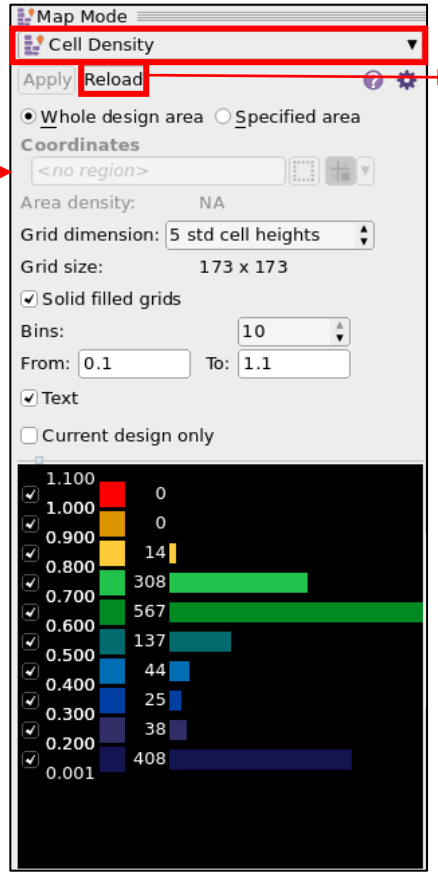
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Introduction to Placement : Cell Density Map window

- **Cell density map shows a heat map that represents the density value of the STD cells in the design**
- The tool only takes the whole cell area when cells are within the grid or the clipped area when cells are overlap the grid and displays the densi ty value and heat map color in the layout window

(The tool also displays a histogram of cell densities)





To generate Cell Density debug map, click the reload button

- View location
  You can open the Cell Density Map by choosing View > Map Mode and selecting Cell Density

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Introduction to Placement ：Cell Density Map option

- **threshold (From To)**

  Set the min and max threshold values to control the top and bottom range for the density bins

  (The default values are the 1.1 for max threshold and 0.1 for min threshold)

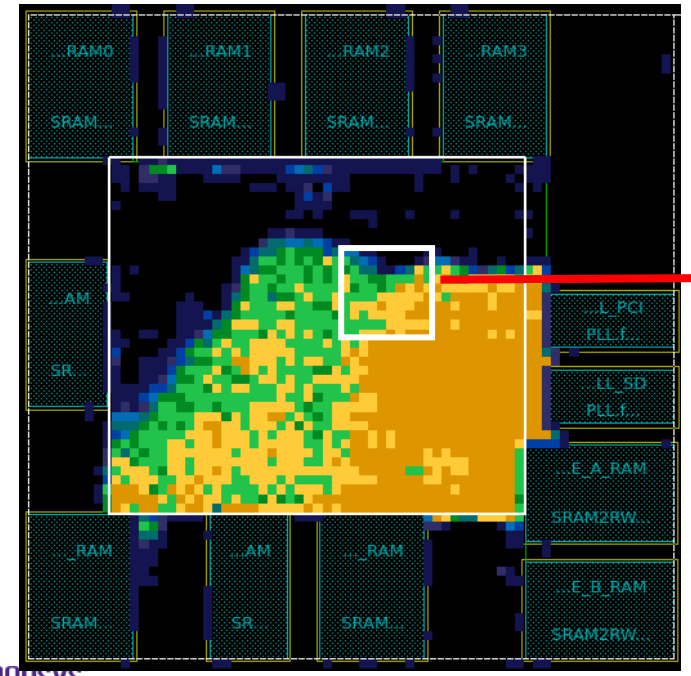
- **Bins**

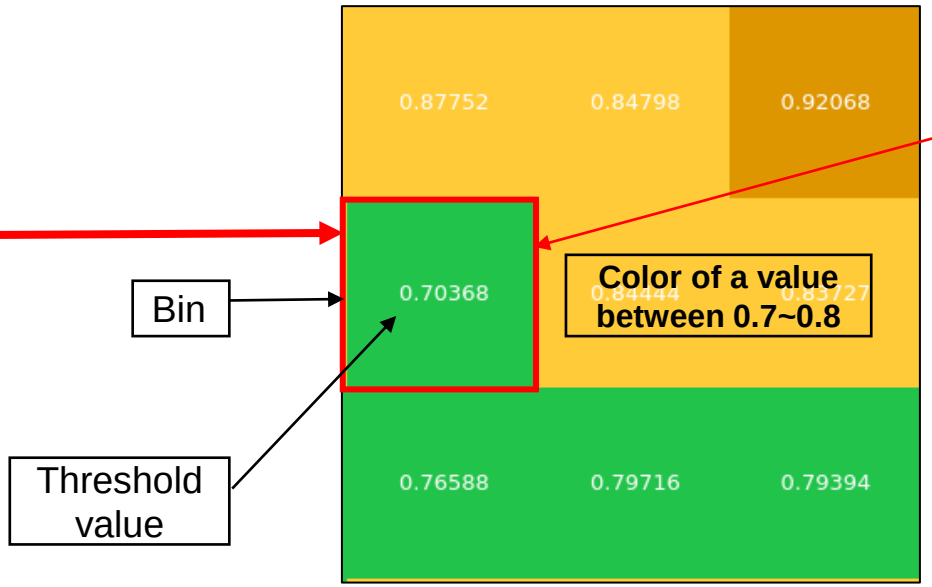
  Bins represent the number of intervals (The maximum number of bins is ten, the minimum is two,

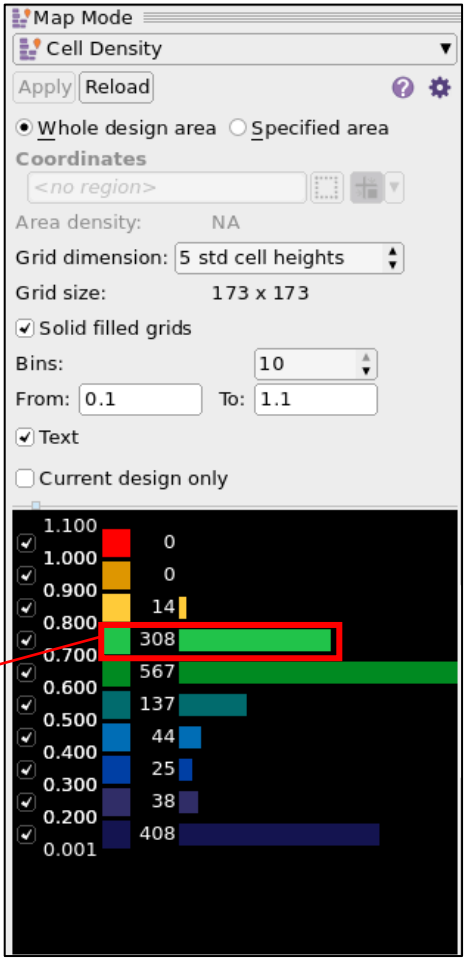
  and the default value and is ten bins)

- **Colors**

  Colors represent different bins in the layout window (By default, the cell density map uses the tool default color scheme)







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Introduction to Placement : Optimizing Pin Access

- **Enable pin density awareness during coarse placement**
  - **ICC2 will automatically calculate a maximum pin density based on the design and try to respect it during cell placement**

place.coarse.pin_density_aware

Enables pin density-aware coarse placement

Default : false

  - **Description**

When this value is set to true, the placer tries to control the maximum local pin density (possible values : true , false)

- **Enable pin optimization during legalization**
  - **Improves the routability of areas with high pin densities by redistributing the cells**

place.legalize.optimize_pin_access_using_cell_spacing

Enable that cell legalization optimizes pin access using cell spacing.

Default : false

  - **Description**

Use this app option to enable that cell legalization optimizes pin access using cell spacing

(This feature will redistribute the locally available space in areas with high pin densities, with the goal of improving routability)

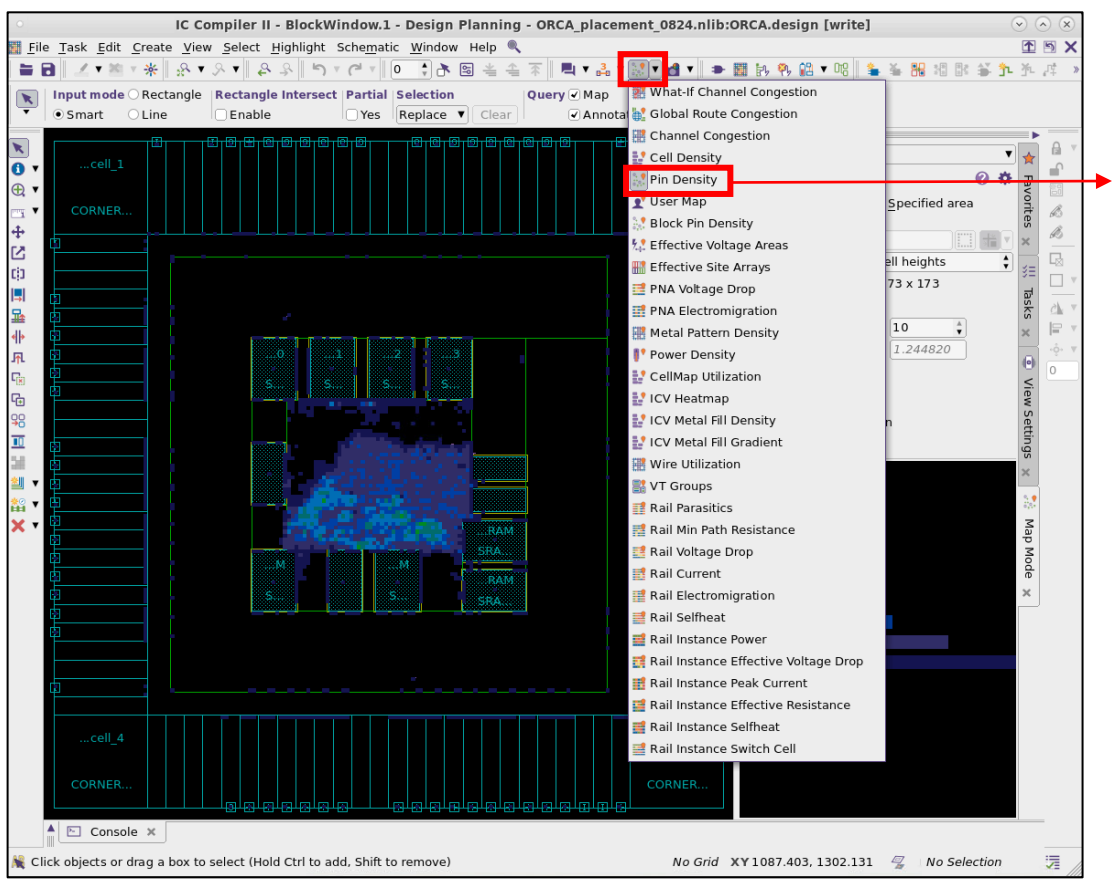
- Affects the "Classic Legalizer" only
- The Advance legalizer performs advanced pin access optimization by default

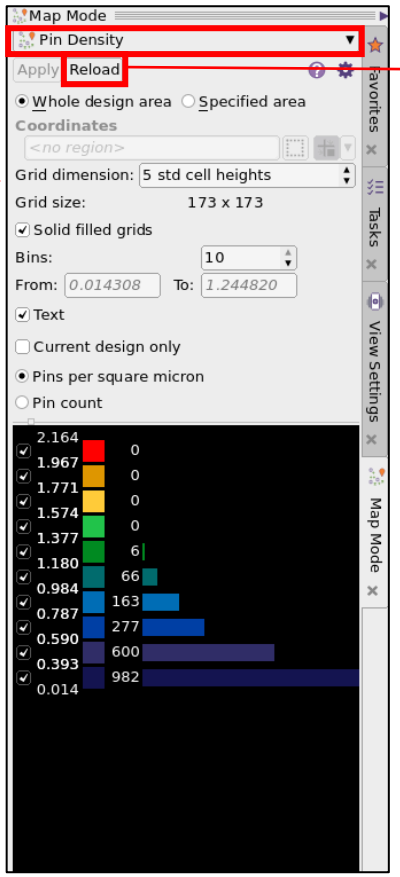
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Introduction to Placement : Pin Density Map window

- **The pin density map shows a heat map that represents the number of pins or pins per square micron within a grid in the design**
- The tool calculates the number of pins or pins per square micron that are within the grid or overlap the grid, and displays the pin count or pins per square micron and heat map color in the layout window

(The tool also displays a histogram of pin densities)





To generate Pin Density debug map, click the reload button

Pin Density

- pin density is considered as a total quantity of pins divided by total instance area of the block
- As cells geometry becoming smaller and smaller, Pin density per area is increased (Due to which more route resources will be used per area)

- View location
  You can open the Pin Density Map by choosing View > Map Mode and selecting Pin Density from the dropdown menu

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Introduction to Placement : Pin Density option

- **Thresholds**

  Set the min and max threshold values to control the top and bottom range for the density bins

  (The default values are the current max and min values for the specified area)

- **Bins**

  Bins represent the number of intervals. The maximum number of bins is ten, the minimum is two

  (The default value and is ten bins)

- **From and To**

  Change the range of overflow values displayed in the map and the histogram by changing these values. These values represent the overflow in routing capacity for the route cell, where higher numbers represent more severe congestion

- **Colors**

  Colors represent different bins in the layout window (By default, the pin density map uses the tool default color scheme)

- **Pins per square micron**

  The number shown in each grid is the number of pins per square micron within the grid or pins that overlap the grid

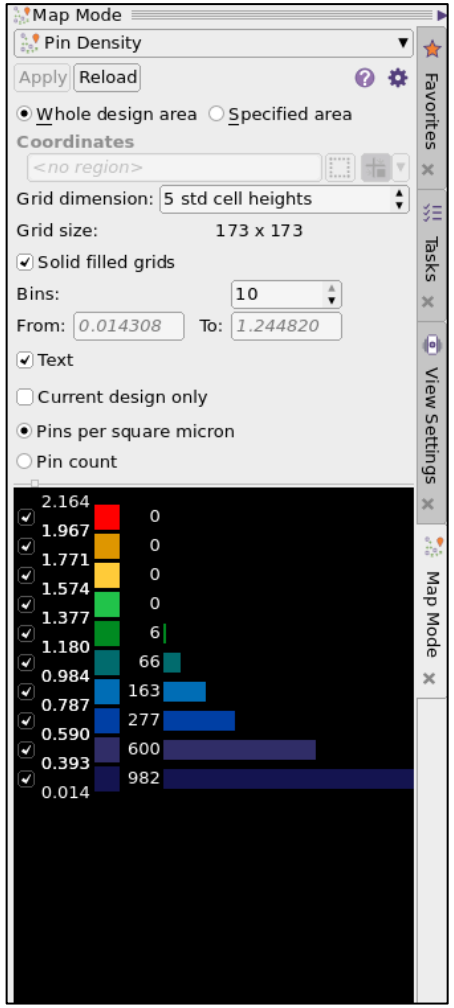
  (The number is a float value)

- **Pin count**

  The number shown in each grid is the number of pins that are within the grid or overlap the grid (The number is an integer value)

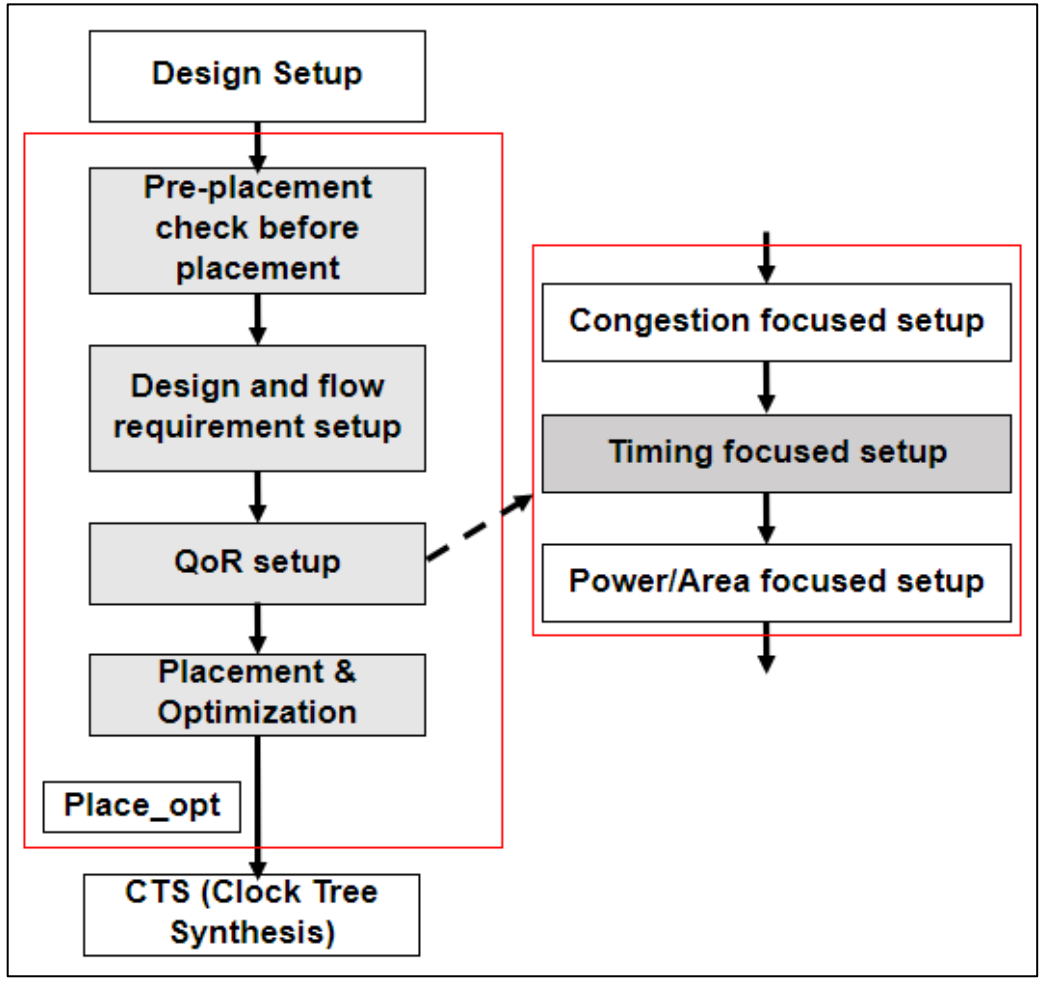
- **Limitations**

  The pin density map displays the number of pins that are within the grid or overlap the grid



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Introduction to Placement : Timing-Focused Setup Steps



- Once congestion is acceptable, analyze timing and logical DRCs
- Perform the following setup steps, as applicable, to improve Timing
  - Auto Timing Control
  - Buffering Aware Placement
  - Control cell max density
  - Enable ICG optimization

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Introduction to Placement : Timing/DRC QoR summary Report

```
Timing
------------------------------------------------------------------------
Context                                 WNS           TNS           NVE
------------------------------------------------------------------------
func.ff_125c       (Setup)             0.83          0.00             0
func.ff_n40c       (Setup)             1.03          0.00             0
func.ss_125c       (Setup)            -0.06         -0.17             6
func.ss_n40c       (Setup)            -0.14         -4.35           347
scan_capture.ff_125c
                   (Setup)             9.73          0.00             0
scan_capture.ff_n40c
                   (Setup)             9.65          0.00             0
scan_capture.ss_125c
                   (Setup)             9.46          0.00             0
scan_capture.ss_n40c
                   (Setup)             9.23          0.00             0
scan_shift.ff_125c (Setup)             9.76          0.00             0
scan_shift.ff_n40c (Setup)             9.65          0.00             0
scan_shift.ss_125c (Setup)             9.36          0.00             0
scan_shift.ss_n40c (Setup)             8.97          0.00             0
Design             (Setup)            -0.14         -4.40           351

func.ff_125c       (Hold)             -2.51      -1178.56          2972
func.ff_n40c       (Hold)             -2.45       -990.09          2984
func.ss_125c       (Hold)             -3.04      -1909.11          2016
func.ss_n40c       (Hold)             -2.99      -1773.16          1446
scan_capture.ff_125c
                   (Hold)             -0.10         -60.71          1452
scan_capture.ff_n40c
                   (Hold)             -0.10         -67.87          1569
scan_capture.ss_125c
                   (Hold)             -0.09          -1.51            86
scan_capture.ss_n40c
                   (Hold)             -0.10          -1.18            78
scan_shift.ff_125c (Hold)             -0.09         -37.10          1360
scan_shift.ff_n40c (Hold)             -0.08         -45.50          1400
scan_shift.ss_125c (Hold)             -0.03          -0.11             5
scan_shift.ss_n40c (Hold)             -0.03          -0.11             6
Design             (Hold)             -3.04      -2214.93          4453
------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------
Cell Area (netlist):                          766940.70
Cell Area (netlist and physical only):       1190940.70
Nets with DRC Violations:       95
1
```

Hold is not addressed during place_opt

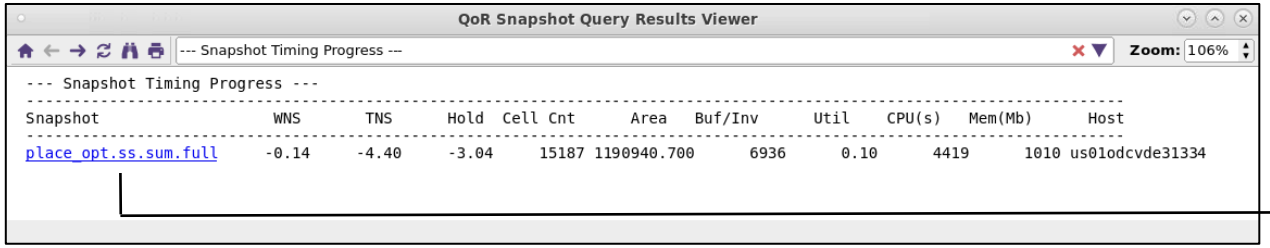
- Use **report_qor –summary** to obtain a summary of timing violations across all active scenarios, listed by type (setup & hold), and number of logical DRC violations
- Use **report_timing** to violating timing path details
- Use **report_constraints –all –max_tran | -max_cap** for violating DRC details

- **WNS** = Worst Negative Slack
- **TNS** = Total Negative Slack
- **NVE** = Number of violating Endpoints
- **report_qor –summary** = This command gives the summary of timing violations across all active scenarios (setup, hold and logical DRC violations)
- **report_timing** = This command gives the violating path details
- **report_constraints** = This command gives the details of max transition and capacitance violation

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Introduction to Placement ：Capture Comprehensive Design Metrics

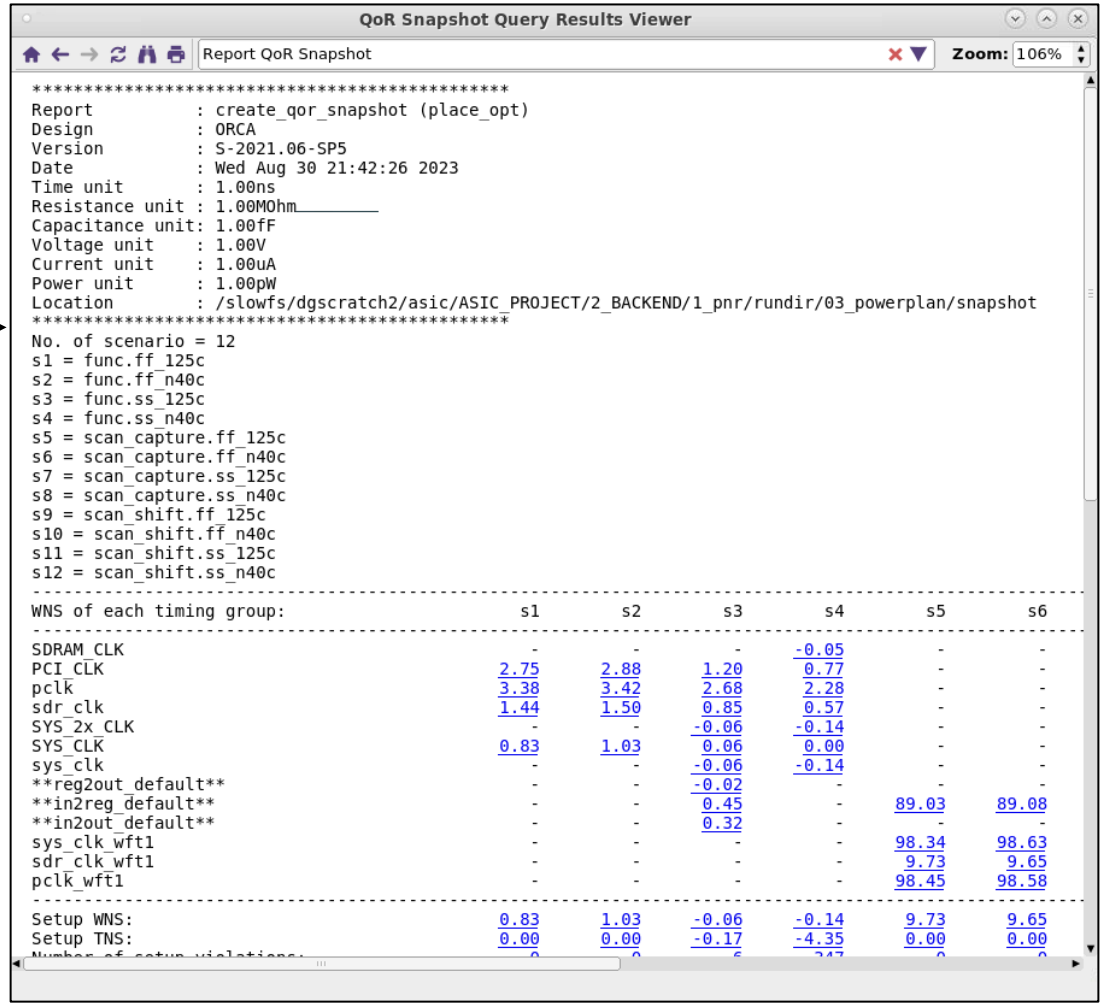
```
query_qor_snapshot -name place_opt -display
```



- **Setup/Hold Timing across all scenarios**
- **Numbers of violations (max tran/cap/etc ...)**
- **Cell/buffer/inverter counts, Memory/CPU status**
- **Timing histograms**
- **Links to detailed reports**

- **create_qor_snapshot**

  Generates a quality-of-results report for specified modes/corners/scenarios and stores the report into a set of files under location designated by application option "time.snapshot_storage_location"



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Introduction to Placement ：Capture Comprehensive Design Metrics

```
create_qor_snapshot -name place_opt
report_qor_snapshot -display
```

QoR Snapshot Query Results Viewer

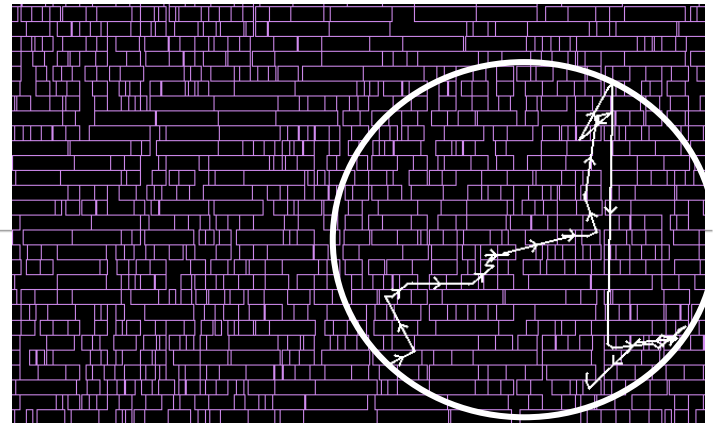
Create Exceptions

Zoom: 106%

Create Exceptions
[X]Append [ ]Overwrite

Exception Type: [X]False Paths [ ]Multicycle Paths [ ]Max Delay [ ]Min Delay [ ]Input Delay [ ]Output Delay [ ]Path Margin

Setup/Hold: [X]Setup and Hold [ ]Setup [ ]Hold
Delay Rise/Fall: [X]Rising and Falling [ ]Rising [ ]Falling
From/Through/To: [X]From-To Only [ ]From Only [ ]To Only [ ]From-To-Through [ ]Through Only
Path Point Rise/Fall: [X]Rise and Fall [ ]Rise [ ]Fall
Reset Path: [ ]



Input file: /slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/03_powerplan/snapshot/place_opt.tim.max.rpt
Filters: -wns , 0.000000 -zero_path , 0.000000 -fanout 40 -transition_degradation 2.000000 -slew_degradation 2.000000 -logic_levels 50
And Columns: none
Sort Column: wns (ascending)
Number of Paths: 61

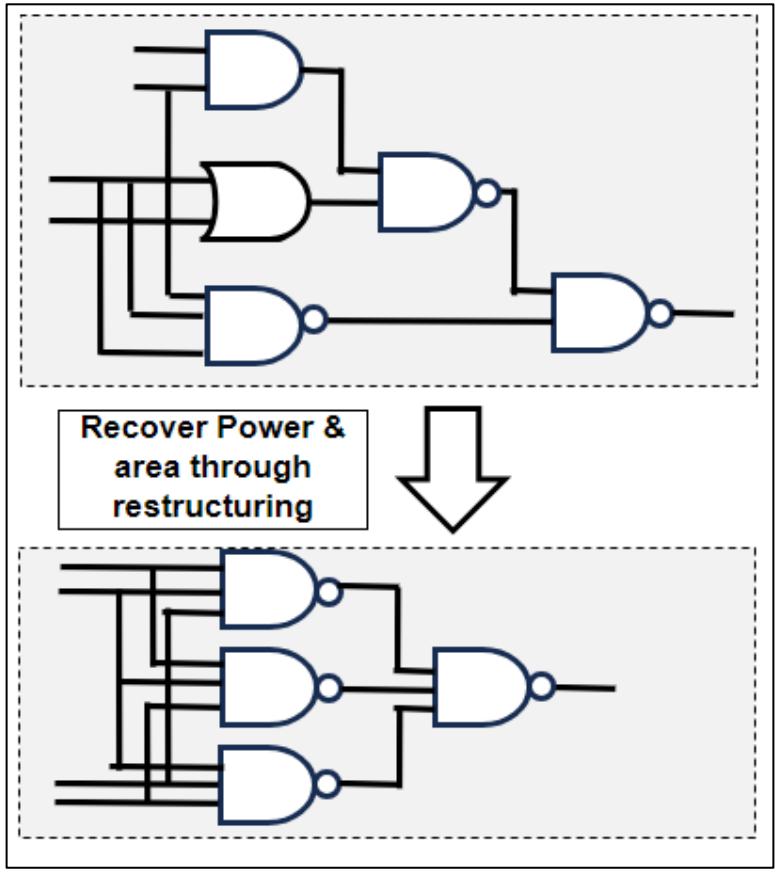
[ ]Exception Path ?	[ ]Highlight Path ?	Display Path ?	Path Group ? ↓	Start Point ? ↓	End Point ? ↓	WNS ? [↑]	Worst Incremental ? ↓	Logic Levels ? ↓	Zero Path ? ↓	Largest Fanout ? ↓	Slew Degradation ? ↓	Transition Degradation ? ↓
[ ]	[ ]	P0	**in2out default**	sdr_clk	sd_DQ[0]	-0.950424	1.380711	5	0.950001	1197	1.000000	1.000000
[ ]	[ ]	P1	**reg2out default**	I_ORCA_TOP/I_SDRAM_IF/sd_DQ_en_reg_0_/CLK	sd_DQ[0]	-0.885269	1.642094	3	0.950001	1197	1.000000	1.000000
[ ]	[X]	P2	SYS_2x_CLK	I_ORCA_TOP/I_RISC_CORE/Oprnd_B_reg_1_/CLK	I_ORCA_TOP/I_RISC_CORE/Neg_Flag_reg/D	0.000057	0.225506	18	1.730156	140	1.000000	1.539150
[ ]	[ ]	P3	sys_clk	I_ORCA_TOP/I_BLENDER/s2_op1_reg_1_/CLK	I_ORCA_TOP/I_BLENDER/s3_op1_reg_31_/D	0.001185	0.295887	32	3.738501	326	1.000000	1.935031
[ ]	[ ]	P4	SYS_CLK	I_ORCA_TOP/I_BLENDER/s2_op1_reg_1_/CLK	I_ORCA_TOP/I_BLENDER/s3_op1_reg_31_/D	0.001185	0.295887	32	3.738501	326	1.000000	1.935031
[ ]	[ ]	P5	SYS_2x_CLK	I_ORCA_TOP/I_RISC_CORE/Crnt_Instrn_1_reg_27_/CLK	I_ORCA_TOP/I_RISC_CORE/Neg_Flag_reg/D	0.083532	0.350285	15	2.095556	140	1.004629	1.495352

- **Generates table from the QoR snapshot**
- **Click on the entries for more information**
- **Apply Filtering (false paths) - ignore outliers**
- **Select paths to highlight in the layout**

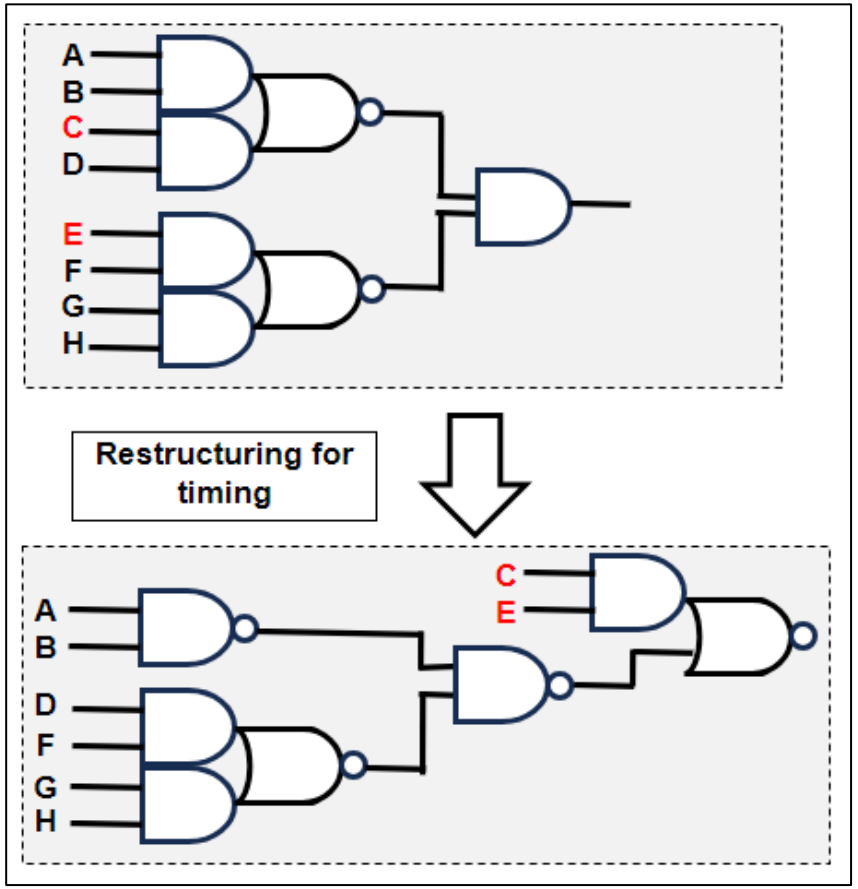
- **query_qor_snapshot**
  Analyzes timing report files from existing QoR snapshots, applies any specified filters, and displays the results in an appropriate format

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Introduction to Placement ：Logic Restructuring for Area and Timing



Area restructuring finds functionally equivalent cones of logic that improve area/leakage without degrading timing

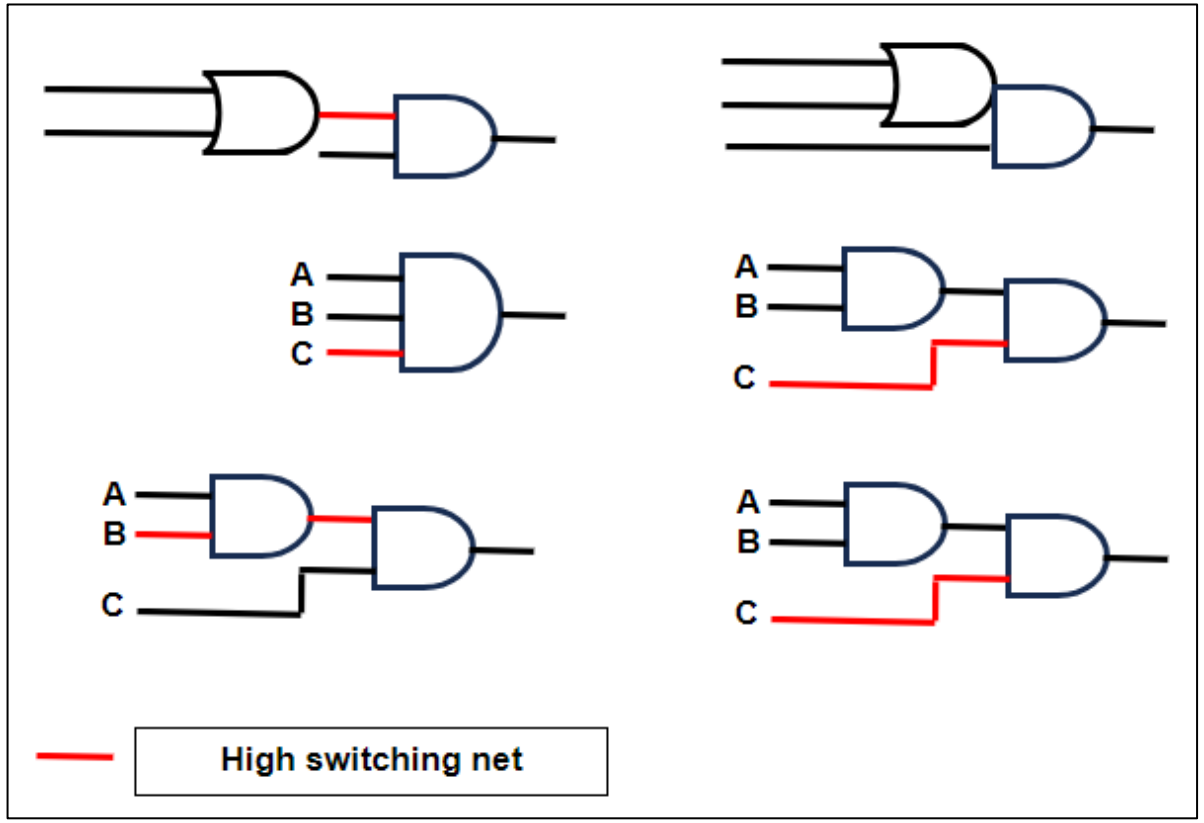


Timing-based restructuring solutions are evaluated in parallel with other transforms like sizing and buffering, to find the best overall timing solution

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Introduction to Placement : Logic Restructuring for Power

- **Logic cones of combinational gates are identified and remapped with equivalent implementations with better powe r and similar timing**



Composition: High switching net can be absorbed in a complex gate

Decomposition: High switching nets driving smaller gets is better for internal power

Rewiring: Feeding the high switching net to the last stage

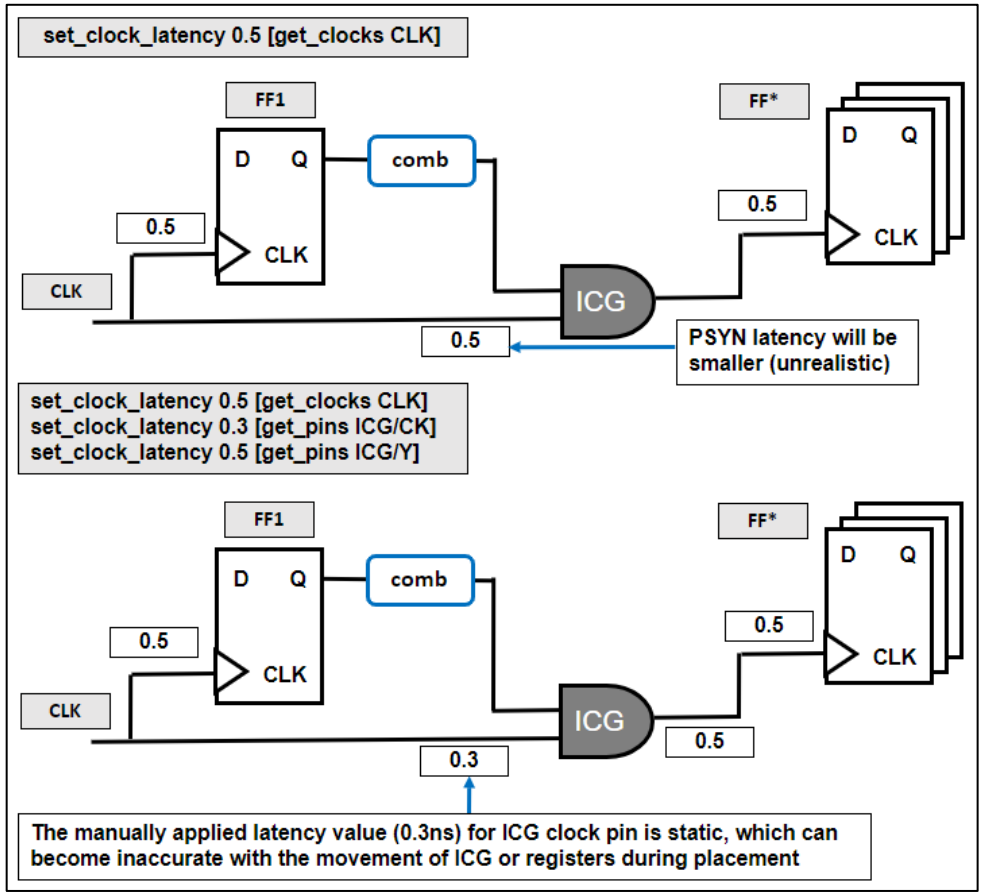
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Introduction to Placement ：Optimizing the Clock Gate Enable Logic

- **Pre-CTS, by default, ICG clock pins are assigned the same latency as that of register clock pins**
  - Enable-logic optimization will be optimistic
  - May lead to ICG timing violation after CTS

- **Common solutions:**
  - Apply annotated latencies on ICG clock pins
  - Apply path margins on ICG enable pins

- **Disadvantages with these approaches**
  - Time consuming and error prone
  - Applied values are static
    - ICG placement is not driven to improve (increase) ICG clock pin latencies
    - Optimization of enable logic does not see ICG latency changes from ICG placement changes

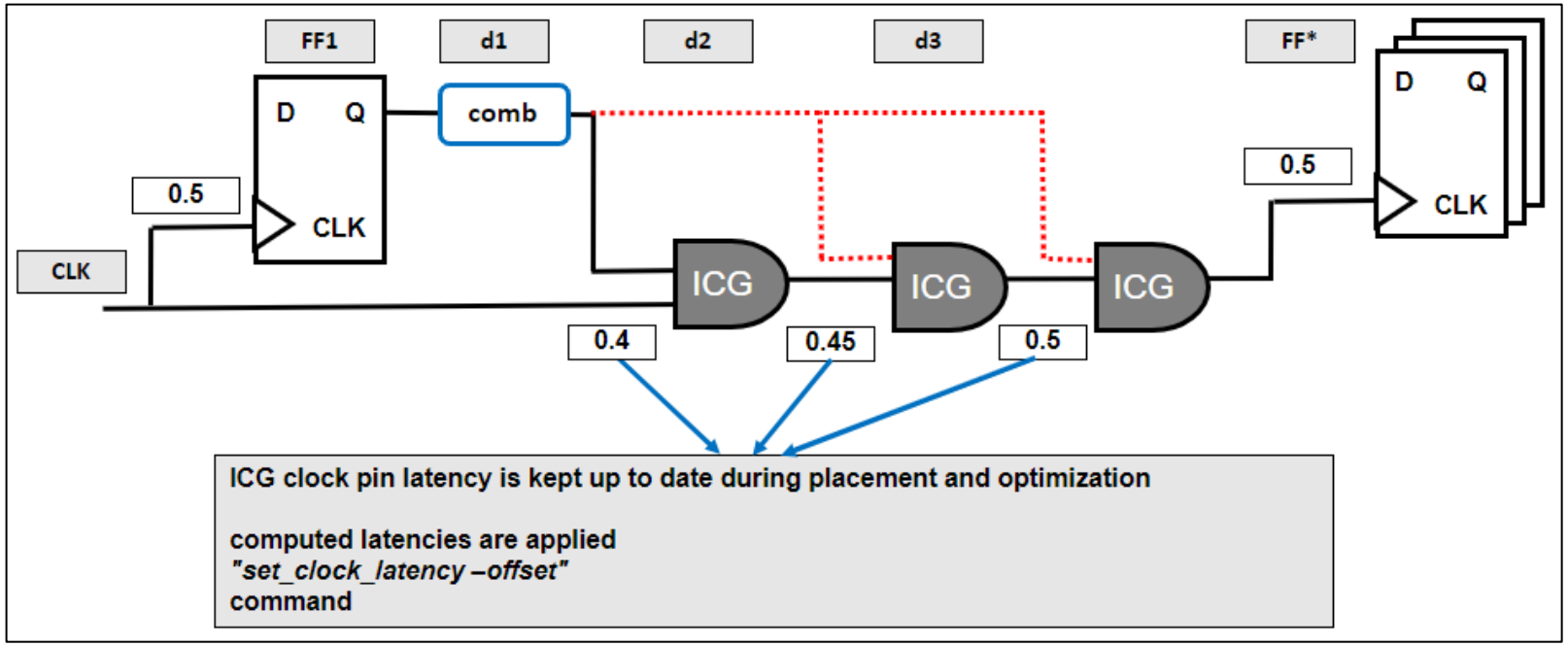


- **Applying a positive path margin use "set_path_margin" command to ICG enable pins reduces the required path del ay time to these enable pins by the margin amount**
- **This may allow optimization to see and try to fix the violating enable timing paths**

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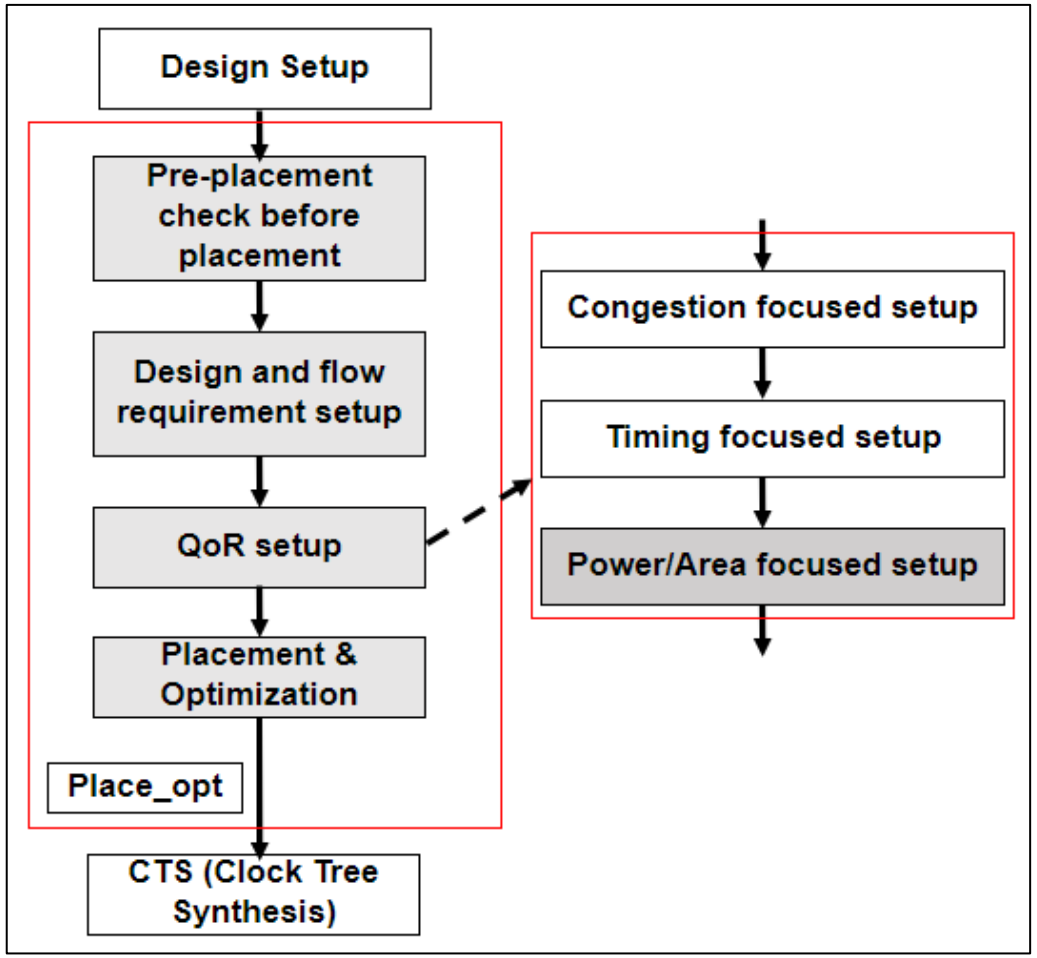
Introduction to Placement ：Optimizing the Clock Gate Enable Logic

- **Integrated Clock Gate Latency Estimation is on by default during place_opt**
- **Clock gate latency estimation is integrated into the pre-CTS optimization flow**
  - Ensures that the clock gate latencies are accurate throughout the pre-CTS flow by updating latencies after necessary steps such as
    - Placement
    - Integrated multibit banking
    - Clock gate optimization steps (merging, splitting ...)
- **Allows more accurate data path and useful skew optimization of enable control registers and combinational logic**



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Introduction to Placement : Congestion-Focused Setup Steps



- Analyze power dissipation and area
- Perform the following setup steps, as applicable, to reduce power dissipation and area
  - Increase power and area optimization effort levels
  - Enable pre-route layer optimization
  - Enable Route-Driven Extraction (RDE)
  - Enable low-power placement

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Introduction to Placement : Logic Restructuring for Power

- **Report power dissipation**
- You can use "**report_power"** command in icc2

```
 Attributes
 ----------
     u  -  User defined power group

Power Group         Internal Power        Switching Power          Leakage Power            Total Power    (   %  )    Attrs
--------------------------------------------------------------------------------------------------------------------------
io_pad                   6.64e+08              3.37e+06              6.55e+08              1.32e+09    ( 24.6%)
memory                   3.13e+07              1.18e+05              0.00e+00              3.14e+07    (  0.6%)
black_box                0.00e+00              0.00e+00              0.00e+00              0.00e+00    (  0.0%)
clock_network            1.31e+08              6.42e+07              6.72e+05              1.96e+08    (  3.6%)
register                -5.73e+06              1.23e+06              2.53e+09              2.53e+09    ( 47.1%)
sequential               6.28e+04              3.36e+04              1.23e+09              1.23e+09    ( 22.9%)
combinational            2.11e+07              1.94e+07              2.24e+07              6.29e+07    (  1.2%)
--------------------------------------------------------------------------------------------------------------------------
Total                    8.41e+08 pW           8.84e+07 pW           4.44e+09 pW           5.37e+09 pW
1
```

report_power

Calculates and reports dynamic and static power for the design or instance.

You can use the **report_power –groups sequential –cell –flat** command option to get a report of all the sequential leaf cells in the design and their power consumption information

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Introduction to Placement : Logic Restructuring for Power

- **Report Area details**
- You can use "**report_design"** command in icc2

```
icc2_shell> report_design
****************************************
Report : design
Design : ORCA
Version: S-2021.06-SP5
Date   : Fri Oct 13 10:44:10 2023
****************************************

Total number of std cells in library : 863
Total number of dont_use lib cells   : 26
Total number of dont_touch lib cells : 20
Total number of buffers              : 33
Total number of inverters            : 45
Total number of flip-flops           : 318
Total number of latches              : 36
Total number of ICGs                 : 36

Cell Instance Type  Count         Area
--------------------------------------
TOTAL LEAF CELLS    16195  1189426.764
Standard cells      16075    69308.627
Hard macro cells       12   128118.137
Soft macro cells        0        0.000
Always on cells         0        0.000
Physical only          37   424000.000
Fixed cells           689  1121273.984
Moveable cells      15506    68152.780
Sequential           3154   155904.209
Buffer/inverter      7695    23250.618
ICG cells              27      198.995
```

report_design

Reports netlist, floorplan, routing, and library information for the current block. By default, the command reports the number of standard cells, macro cells, and other elements in the design.

Unit of area determined by "unit length" in the tech file

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Introduction to Placement : RDE for Preroute Optimization

- **What is RDE (Route-Driven Extraction)**
  - RDE can reduce the impact of miscorrelation between virtual-router-based preroute timing and postroute timing.
  - The RDE flow runs global routing inside preroute optimization (starting at the place_opt final_opto stage) and the tool uses the global routin g data to build an RDE extraction table (This extraction engine is used subsequently for all virtual net extraction)

- **RDE uses global routing on the placed design to capture:**
  - Detailed layer occupancy beyond min/max/layer usage
  - GR topology and length
  - Via resistance
  - Cell pin geometries

- **Once data is captured, all virtual routes will use the GR RC model for RC extraction**
  - Ensures close parasitic and timing correlation to actual routing
  - Tends to result in better area and power with similar post-route timing QoR
  - Does not increase full flow run time

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Introduction to Placement : RDE for Preroute Optimization

- **Automatically assigns timing-critical nets to higher layers using min layer constraints during final_opto (feature is always on)**
  - Reduces the number of buffers needed for the same path
  - May also "demote" nets to lower layers that were promoted previously

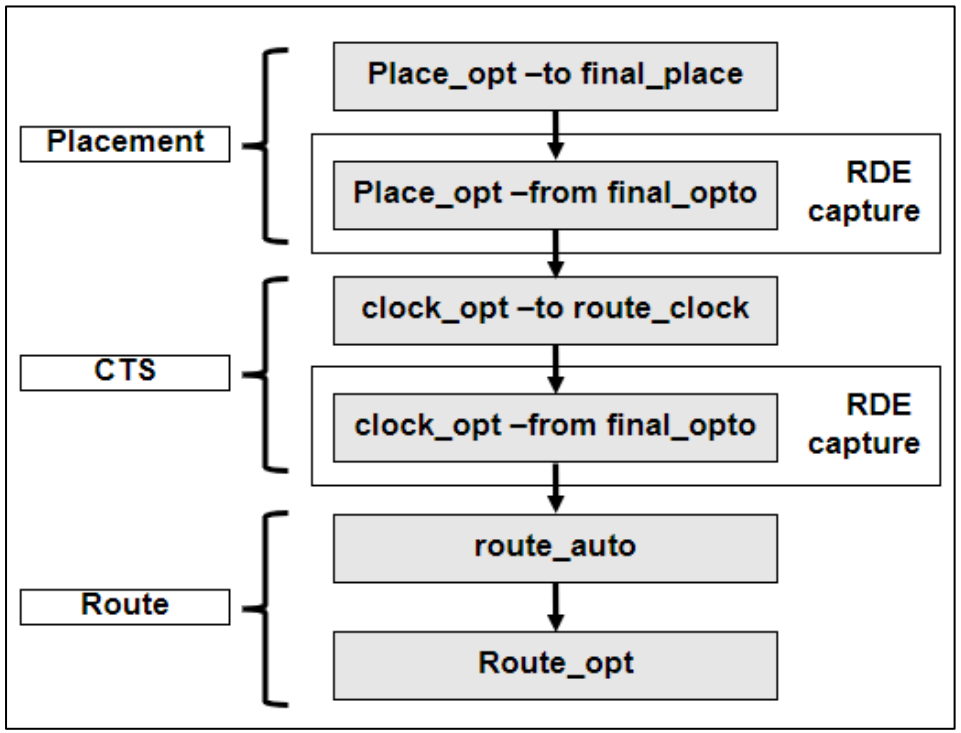
- **Nets will always be rebuffered once promoted / demoted to achieve the best buffering solution**

- **Minimum layer constraints are carried all the way to post-route optimization**

- **The layer optimization also looks for long repeater chains that have been buffered previously, assigns the net to higher layers and rebuffers, thereby reducing the number of buffers and possibly improving its timing (This optimization occurs in the final_opto stage)**

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Introduction to Placement : RDE in the Pre-Route Flow



- **RDE capture occurs twice in the flow**
  - **At the beginning of place_opt and clock_opt final_opto**
- **RDE data is stored with the design, and is reused for any subse quent optimizations**
- **RDE is enabled by default for 16nm and below**

opt.common.enable_rde

Controls whether route-driven extraction (RDE) is enabled during preroute optimization to handle preroute to postroute miscorrelation.

- To enable for other technologies, set the application option to true
- By default (auto), the tool infers whether the design uses advanced technolo gy nodes (<=16nm) and, if so, automatically enables the flow
- If set to true, the RDE flow is always triggered
- If this flow is enabled during pre-route optimization, you should continue to r un all steps in pre-route optimization with the flow enabled

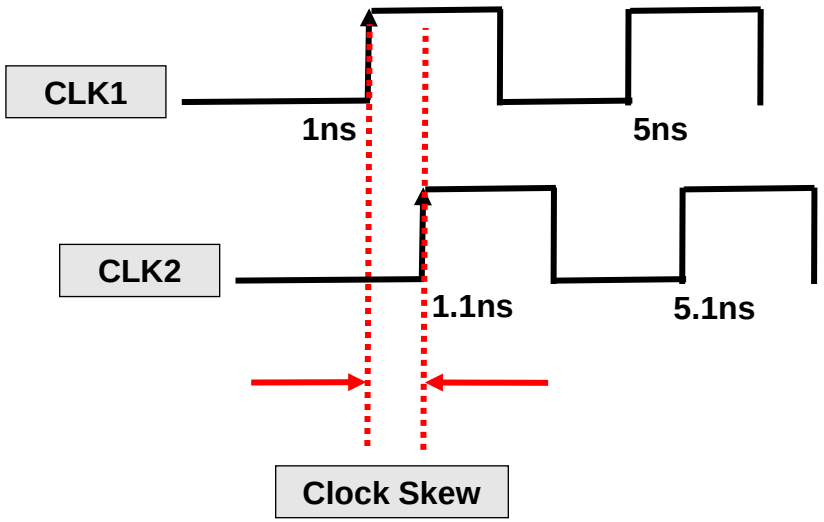
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Introduction to Placement : setup (Data Arrival Time & Data Required Time)

- **What is Latency?**

  Clock latency is the time taken by a clock signal to move from the clock source to the clock pin of a particular flip flop

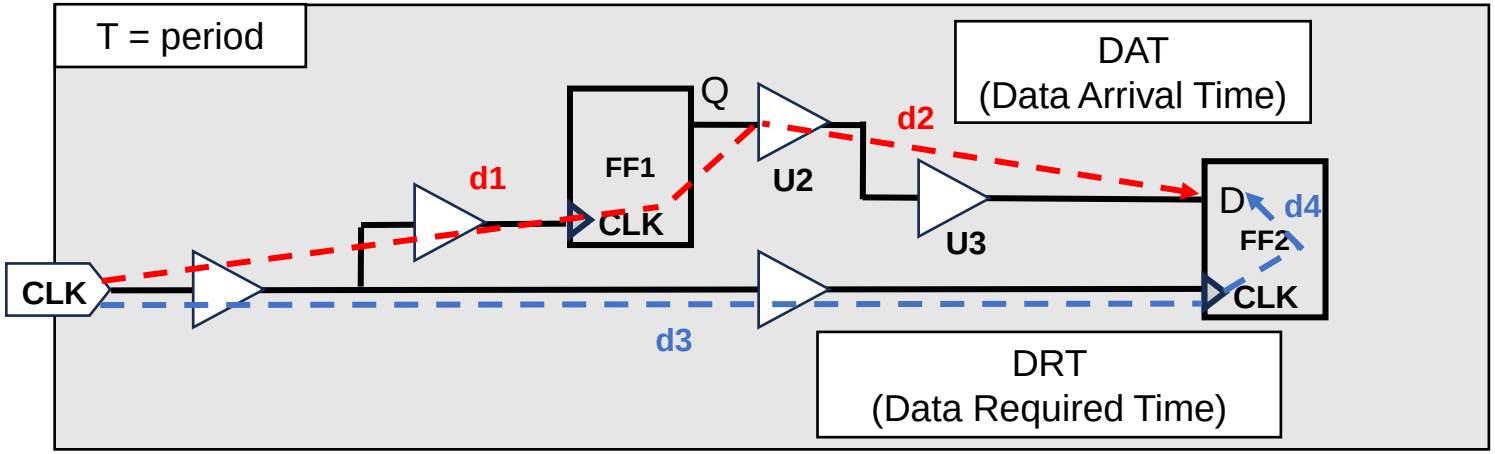
- **What is Skew?**



- Clock Skew is the timing differences between signals in a clock distribution system
- For example, you can see positive edge of both the clock signal (CLK1 and CLK2) has some time delay even when there is same clock source (CLK_S) - This Time delay is known as CLOCK Skew
- Clock Skew = 1.1ns - 1ns = 0.1ns

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Introduction to Placement : setup (Data Arrival Time & Data Required Time)

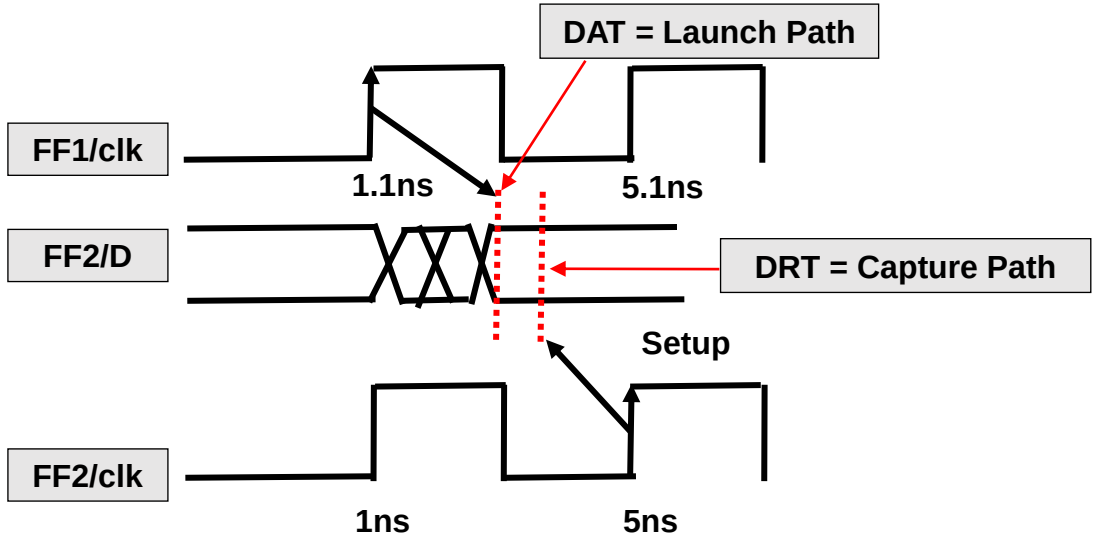


Data Arrival Time FF2:D = d1 + d2

- The arrival time represents the time at which the data arrives at the input of the receiving sequential element
- In this example, the arrival time is considered from the setup launch edge at CLK, taken as a time reference
- It follows the clock network along the red line until the clock pin on FF1 (delay d1), and it continues along the data path always following the red line until the data pin D on FF2

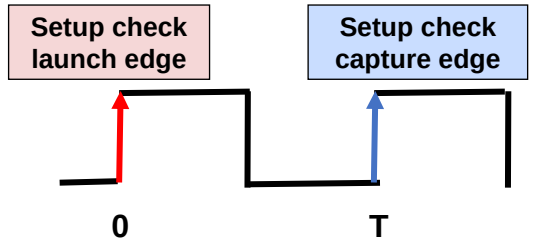
Data Required Time FF2:D = T + d3 - d4

- The required time represents when the data is required to be present at the same pin FF2:D
- Using the period T provided to the tool through the clock constraint, the event gets propagated through the clock network along the blue line until the clock pin of FF2 (delay d3)
- Taking FF2 setup (delay d4) into account, this means the clock constraint requires the data to be present d4 time before the capturing clock edge on FF2



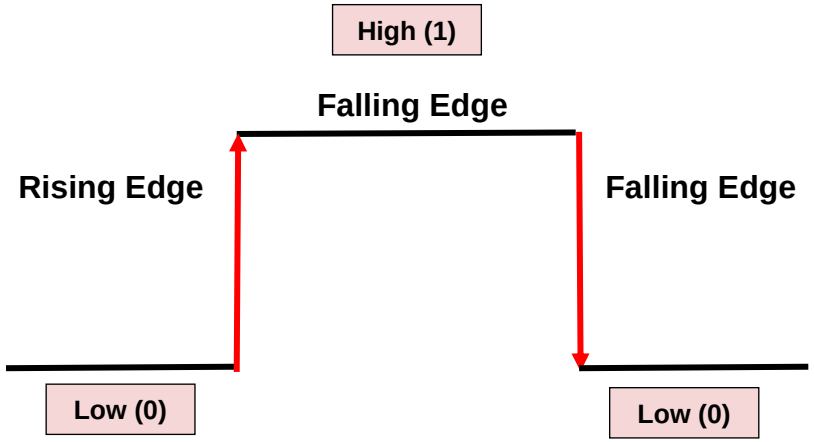
Slack = Data Required Time – Data Arrival Time

- If the slack is negative, the path is violating the setup relationship between the two sequential elements



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Introduction to Placement : Rising & Falling Edge



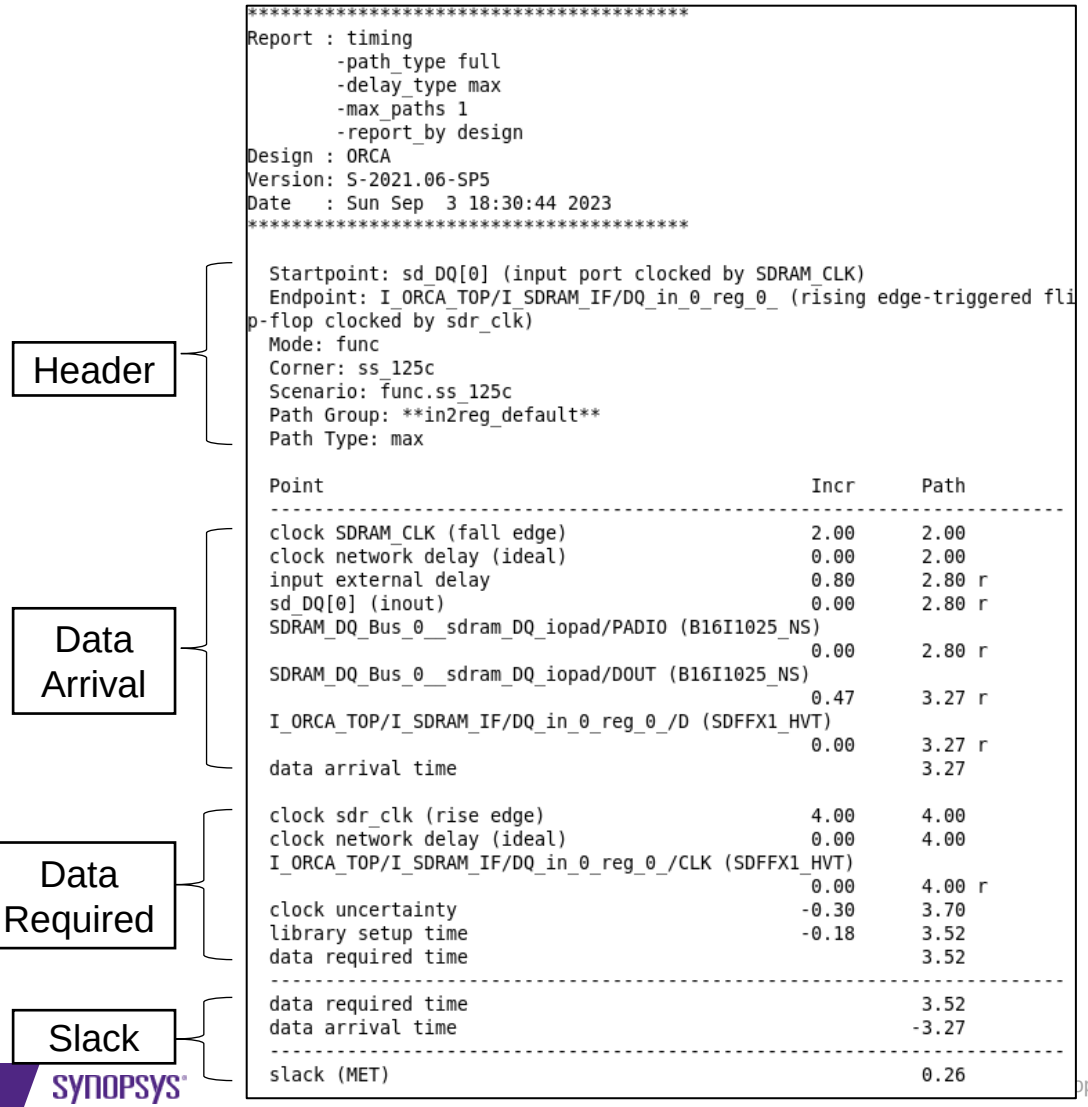
- **Rising edge**
  When the input signal is transitioning from a low stage to a high stage
- **Falling edge**
  When the input signal is transitioning grom a high stage to a low stage
- **Either edge**
  When the input signal is changing state, from high to low or from low to high
  - ❖ "r" in the path column for the rising edge of the signal in timing report
  - ❖ "f" in the path column for the falling edge of the signal in timing report

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Introduction to Placement : report_timing (ideal clock)

- **report_timing**

Displays timing information about a design




```
****************************************
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -report_by design
Design : ORCA
Version: S-2021.06-SP5
Date   : Sun Sep  3 18:30:44 2023
****************************************

  Startpoint: sd_DQ[0] (input port clocked by SDRAM_CLK)
  Endpoint: I_ORCA_TOP/I_SDRAM_IF/DQ_in_0_reg_0_ (rising edge-triggered fli
p-flop clocked by sdr_clk)
  Mode: func
  Corner: ss_125c
  Scenario: func.ss_125c
  Path Group: **in2reg_default**
  Path Type: max

  Point                                          Incr       Path
  -----------------------------------------------------------------------
  clock SDRAM_CLK (fall edge)                    2.00       2.00
  clock network delay (ideal)                    0.00       2.00
  input external delay                           0.80       2.80 r
  sd_DQ[0] (inout)                               0.00       2.80 r
  SDRAM_DQ_Bus_0__sdram_DQ_iopad/PADIO (B16I1025_NS)
                                                 0.00       2.80 r
  SDRAM_DQ_Bus_0__sdram_DQ_iopad/DOUT (B16I1025_NS)
                                                 0.47       3.27 r
  I_ORCA_TOP/I_SDRAM_IF/DQ_in_0_reg_0_/D (SDFFX1_HVT)
                                                 0.00       3.27 r
  data arrival time                                         3.27

  clock sdr_clk (rise edge)                      4.00       4.00
  clock network delay (ideal)                    0.00       4.00
  I_ORCA_TOP/I_SDRAM_IF/DQ_in_0_reg_0_/CLK (SDFFX1_HVT)
                                                 0.00       4.00 r
  clock uncertainty                             -0.30       3.70
  library setup time                            -0.18       3.52
  data required time                                        3.52
  -----------------------------------------------------------------------
  data required time                                        3.52
  data arrival time                                        -3.27
  -----------------------------------------------------------------------
  slack (MET)                                               0.26
```

- **At any stage of the design you will be report_timing**
- **You can use ICC2 tool to report the timing after placement, after CTS**
- **When report_timing make sure use "full_path" reporting for easy analysis**
- **The –delay determines whether hold or setup is reported**

  -delay min = hold

  -delay max = setup

- **Use –scenario option if you have created multiple scenarios in P&R flow**
- **It consists of start point (FF1) and end point (FF2)**
- **Path type**

  - max = setup

  - min = hold

We analyze ideal network (setup) at placement stage, because we didn't run CTS (clock tree synthesis) not yet

❖ **Ideally, clock signals should have minimum rise and fall times, specified duty cycles, and zero skew**

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Introduction to Placement : report_timing (ideal clock)

- **report_timing - Header**

Displays timing information about a design

```
****************************************
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -report_by design
Design : ORCA
Version: S-2021.06-SP5
Date   : Sun Sep  3 18:30:44 2023
****************************************

  Startpoint: sd_DQ[0] (input port clocked by SDRAM_CLK)
  Endpoint: I_ORCA_TOP/I_SDRAM_IF/DQ_in_0_reg_0_ (rising edge-triggered fli
p-flop clocked by sdr_clk)
  Mode: func
  Corner: ss_125c
  Scenario: func.ss_125c
  Path Group: **in2reg_default**
  Path Type: max
```

- **The header of the timing report is given below (This gives the options have used while running report_timing)**
- **"-delay_type max" is a setup violation report**
- **"Path Type: max" is a setup violation report**
- **-scenario option is specified and timing is reported for the scenario "func.ss_125c "**

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Introduction to Placement : report_timing (ideal clock)

- **report_timing – Data Arrival Time & Data Required Time**

Displays timing information about a design

```
****************************************
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -report_by design
Design : ORCA
Version: S-2021.06-SP5
Date   : Sun Sep  3 18:30:44 2023
****************************************

  Startpoint: sd_DQ[0] (input port clocked by SDRAM_CLK)
  Endpoint: I_ORCA_TOP/I_SDRAM_IF/DQ_in_0_reg_0_ (rising edge-triggered fli
p-flop clocked by sdr_clk)
  Mode: func
  Corner: ss_125c
  Scenario: func.ss_125c
  Path Group: **in2reg_default**
  Path Type: max

  Point                                              Incr       Path
  -----------------------------------------------------------------------
  clock SDRAM_CLK (fall edge)                        2.00       2.00
  clock network delay (ideal)                        0.00       2.00
  input external delay                               0.80       2.80 r
  sd_DQ[0] (inout)                                   0.00       2.80 r
  SDRAM_DQ_Bus_0__sdram_DQ_iopad/PADIO (B16I1025_NS)
                                                     0.00       2.80 r
  SDRAM_DQ_Bus_0__sdram_DQ_iopad/DOUT (B16I1025_NS)
                                                     0.47       3.27 r
  I_ORCA_TOP/I_SDRAM_IF/DQ_in_0_reg_0_/D (SDFFX1_HVT)
                                                     0.00       3.27 r
  data arrival time                                             3.27

  clock sdr_clk (rise edge)                          4.00       4.00
  clock network delay (ideal)                        0.00       4.00
  I_ORCA_TOP/I_SDRAM_IF/DQ_in_0_reg_0_/CLK (SDFFX1_HVT)
                                                     0.00       4.00 r
  clock uncertainty                                 -0.30       3.70
  library setup time                                -0.18       3.52
  data required time                                            3.52
  -----------------------------------------------------------------------
  data required time                                            3.52
  data arrival time                                            -3.27
  -----------------------------------------------------------------------
  slack (MET)                                                   0.26
```

Data Arrival

Data Required

Slack

Rising edge (max delay)

- **This report is taken after placement, but before completing CTS step**
- **The data path if from port sdr_clk to the sd_DQ[0] port to the D pin of th e I_ORCA_TOP/I_SDRAM_IF/DQ_in_0_reg_0_**
- **The clock at both launch and the capture edges are ideal (clock network is reported after the line "data arrival time")**
- **With an ideal clock network delay of 0.00, the setup time is MET in this r eport**

- **Data Arrival section**

  Reports the total time taken to arrive at D pin of FF2

- **Data Required section**

  Reports the total time taken to arrive the clock pin of FF1 minus the setup time of FF2

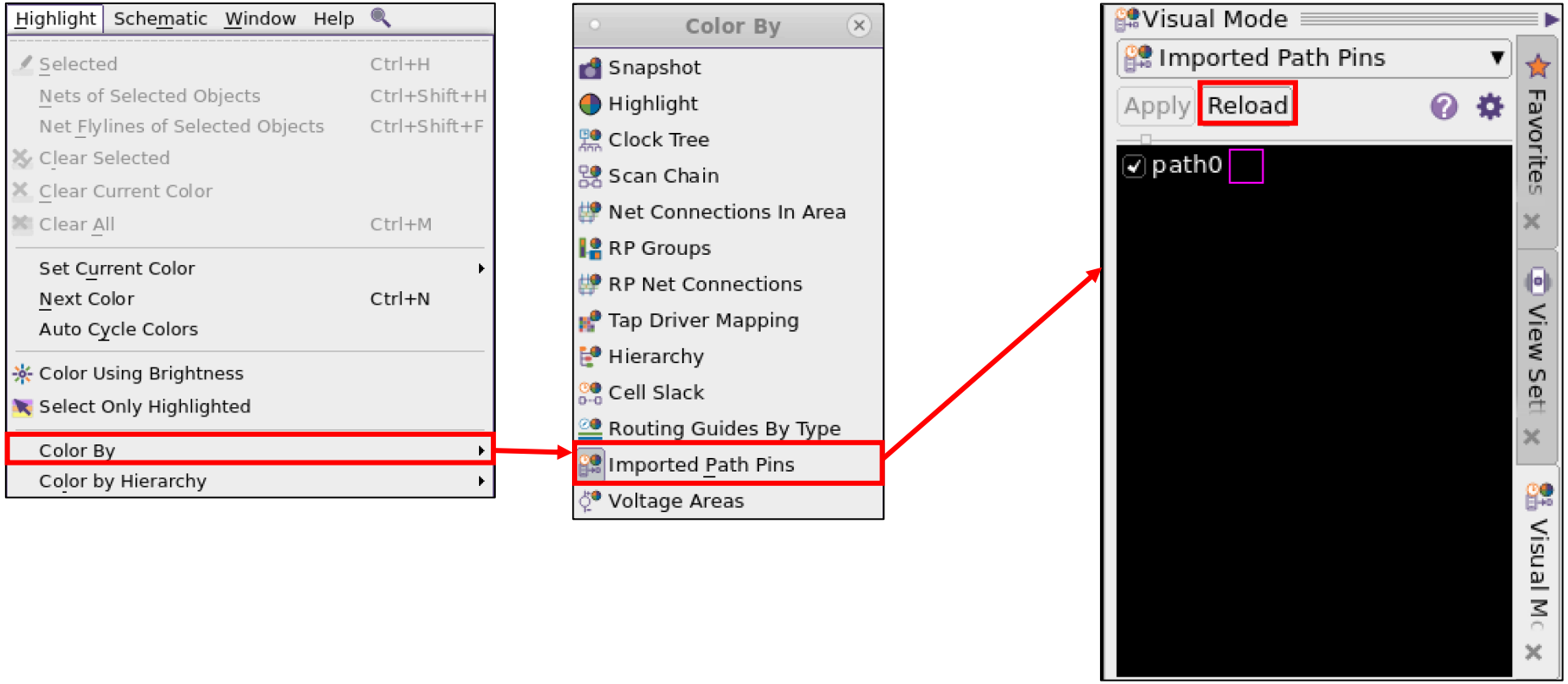
- **Slack**

  Timing difference between required and arrival time

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Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **The "imported Path Pins" visual mode takes timing path data from a timing report and highlights the timing path in the Layout**
- **Paths can be viewed one-by-one**
- **Menu Location**
  - Highlight > Color By > Imported Path Pins



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Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Load Timing Report**

1. Choose Highlight > color By > imported Path Pins to open the imported path pins visual Mode
2. Update the timing for your design with the update_timing command (**update_timing -full**)
3. Report the timing for a path with the report_timing command and save the result to a fil

```
report_timing  \
        -from   I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg \
        -to   I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/syncff_reg_6_ \
        -physical -nets -nosplit -input_pins -delay_type min > ./test_timing.rpt
```

4. Click the Reload button
5. Select Import Path Pins from Timing Report and click the file chooser button
6. Select the timing.rpt file you wrote out in step 3 and click ok
7. Choose the show path as option to specify the path type to show: Flylines, routes or both (default option is Flylines) and apply

❖ The Imported Path Pins Visual Mode panel displays a histogram bar for the path with the number of pins in the path

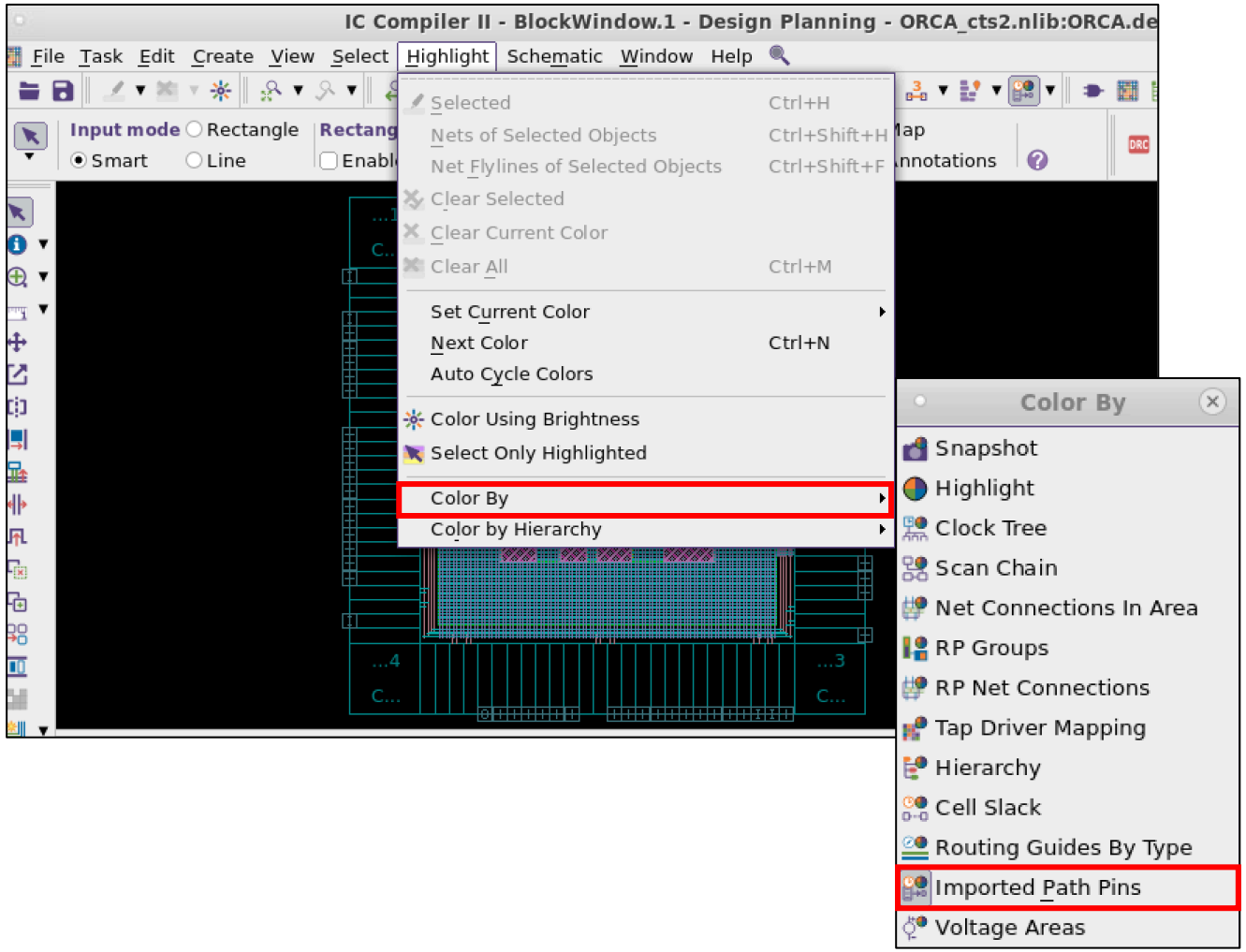
❖ Repeat steps 3 through 6 to import and display additional paths (Choose the Clear all buckets before importing new path pins option to delete the histogram before importing the new path)

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Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Load Timing Report**

1. Choose Highlight > color By > imported Path Pins to open the imported path pins visual Mode





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Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Load Timing Report**

2. Update the timing for your design with the update_timing command (**update_timing -full**)

```
icc2_shell> update_timing -full
```

3. Report the timing for a path with the report_timing command and save the result to a file

```
report_timing  \
        -from   I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg \
        -to   I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/syncff_reg_6_  \
        -physical -nets -nosplit -input_pins -delay_type min > ./test_timing.rpt
```

4. Click the Reload button

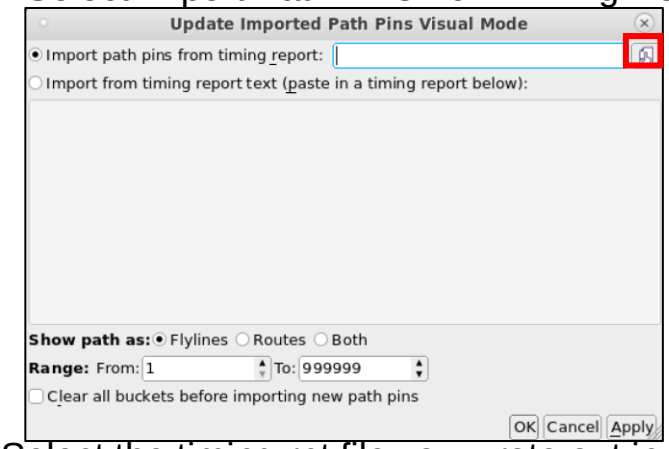


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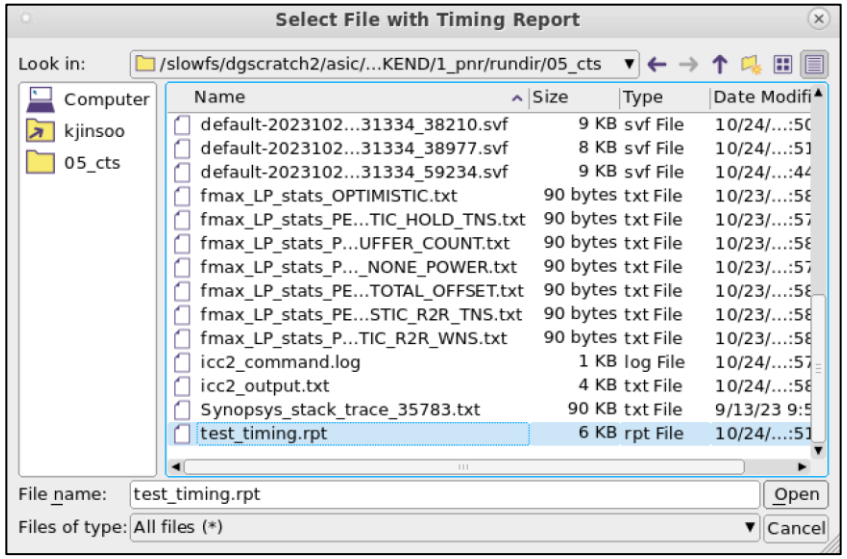
Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

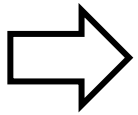
- **Load Timing Report**

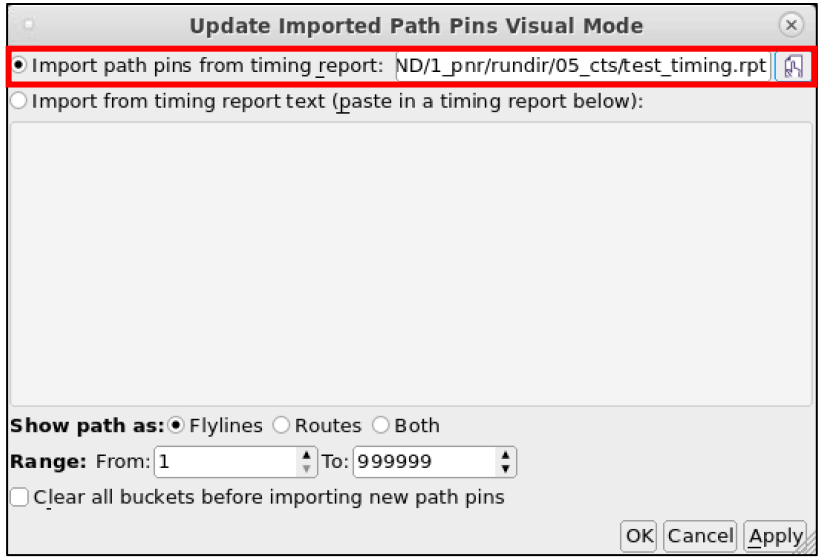
5. Select Import Path Pins from Timing Report and click the file chooser button



6. Select the timing.rpt file you wrote out in step 3 and click ok (test_timing.rpt)





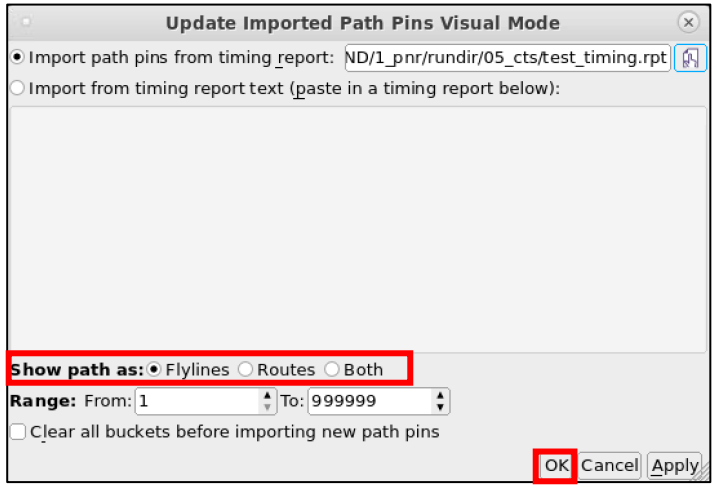


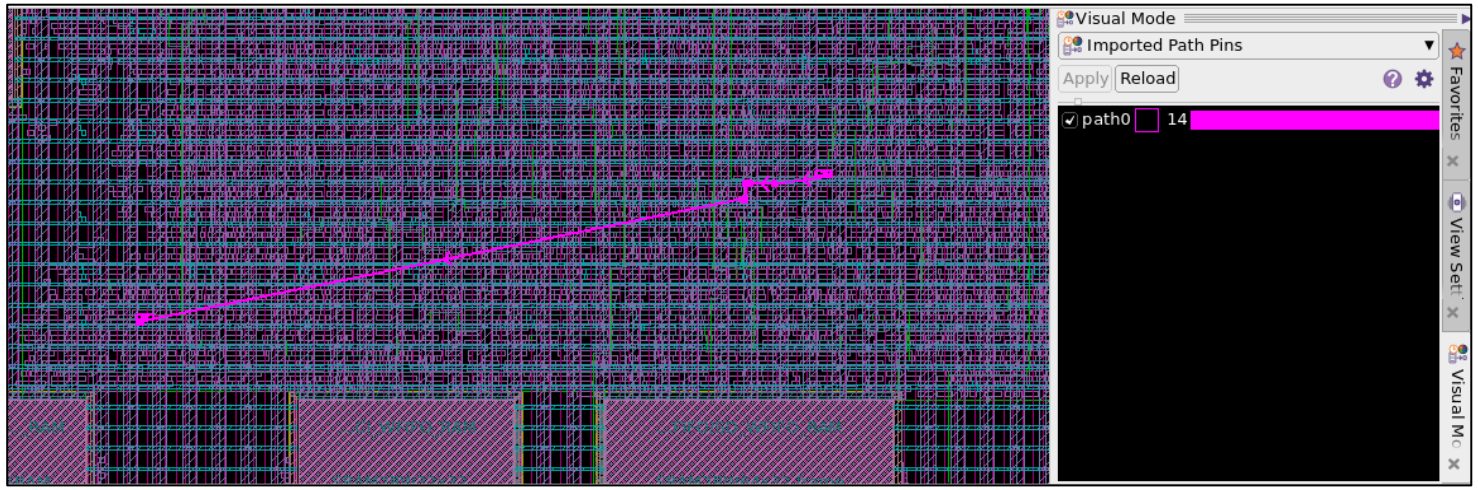
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Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Load Timing Report**

7. Choose the show path as option to specify the path type to show: Flylines, routes or both (default option is Flylines)





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WEEK 7 : LAB for Placement

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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Go to the Script Directory**

```
  < user@linux script >> pwd
/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/04_placement/script
  < user@linux script >> ls
01_design_setup_and_copy.tcl    04_read_tluplus.tcl  07_set_tiecell_and_ideal_network.tcl   10_refine_opt.tcl         13_report_timing_and_constraints.tcl
02_create_routing_blockage.tcl  05_mcmm.tcl          08_set_false_path_and_app_options.tcl  11_insert_spare_cell.tcl  14_save_and_exit.tcl
03_insert_guide_buf.tcl         06_read_sdc.tcl      09_place_opt.tcl                       12_check_placement.tcl    full_script
```

  - **Review 01_design_setup_and_copy.tcl**

```
# module load
#module load icc2/2021.06-SP5
# open icc2_shell
#icc2_shell -gui | tee -i ./04_placement.log

#set current step & before step
set step placement
set before_step powerplan

#source design_setup.tcl
source -e ./icc2_common_env/design_setup.tcl

#source timer.rpt script - report runtime for placement
source ./icc2_common_env/sec_runtime.tcl

sec_StartTimer copy_lib

#open lib
open_lib ../03_${before_step}/nlib/ORCA_${before_step}.nlib
#copy lib
copy_lib -from ../03_${before_step}/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
#close & open current lib
close_lib -all
open_lib ./nlib/ORCA_${step}.nlib
#open block
open_block ./nlib/ORCA_${step}.nlib:ORCA.design
#link_block
link_block -rebind

sec_ReportResourceUtil copy_lib
```


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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Review 02_create_routing_blockage.tcl**

```
#create routing blockages for all Macros & PLL

source -e ./user_script/create_routing_blockage.tcl
```

  - **Open & Review the create_routing_blockage.tcl**

```
set only_macros "I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM0 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM1 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM2 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM3 I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_RAM I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_RAM I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM I_ORCA_TOP/I_RISC_CORE/I_REG_FILE/REG_FILE_A_RAM I_ORCA_TOP/I_RISC_CORE/I_REG_FILE/REG_FILE_B_RAM"
set pll "I_CLOCK_GEN/I_PLL_PCI I_CLOCK_GEN/I_PLL_SD"


####################################
#create_routing_blockage_for macros
####################################
set macro_bbox [get_attr [get_cells $only_macros] bbox]
set i 0

foreach x $macro_bbox {
set llx [expr [lindex $x 0 0] + 1]
set urx [expr [lindex $x 1 0] - 1]
set lly [expr [lindex $x 0 1] + 1]
set ury [expr [lindex $x 1 1] - 1]
set new_bboxes "{$llx $lly} {$urx $ury}"
create_routing_blockage -name "mc_routing_blk_${i}" -layers "M2 M4 M7 M8 M9 MRDL" -boundary $new_bboxes

incr i
}
```

```
####################################
#create_routing_blockage_for PLL
####################################
set pll_bbox [get_attr [get_cells $pll] bbox]
set i 0

foreach x $pll_bbox {
set llx [expr [lindex $x 0 0] + 2]
set urx [expr [lindex $x 1 0] - 2]
set lly [expr [lindex $x 0 1] + 2]
set ury [expr [lindex $x 1 1] - 2]
set new_bboxes "{$llx $lly} {$urx $ury}"
create_routing_blockage -name "pll_blkg_hard_${i}" -layers "M2 M3 M4 M5 M7 M8 M9 MRDL" -boundary $new_bboxes

incr i
}
```

Prevents Metal layers from being routed over the Macro when routing
(we can use only M1~M6 for routing layers)

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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Review 03_insert_guide_buf.tcl**

```
#insert guide buffer (all Macros & PLL signal inout port)
#insert buffer (scan_en_iopad/DOUT) - purpose of high_fanout_synthesis

source ./user_script/insert_guide_buf.tcl
```

  - **Open & Review the insert_guide_buf.tcl**

```
# Insert Guide buffer at all signal input & output pins (Macros & PLL)
set all_macros [get_cells -hierarchical -filter "is_hard_macro && !is_physical_only"]

set sig_pins [get_pins -of $all_macros -filter "port_type == signal && net_name !~ *VDD*  && net_name !~ *VSS* && net_name !~ *UNCONNEC
TED*"]


set buf "saed32rvt/NBUFFX2_RVT"


foreach x $sig_pins {
add_buffer $x -lib_cell $buf -new_cell_names macro_guide_buf
incr i
}

# Insert Guide buffer in front of scan_en_iopad/DOUT pin for HFS (High Fanout Synthesis)

add_buffer [get_pins -filter "full_name == scan_en_iopad/DOUT"] -inverter_pair -new_cell_names scan_en_HFS -lib_cell saed32rvt/INVX2_RV
T

#legalize guide_buffer

legalize_placement

#set fixed guide buffers & scan_en_HFS buffer

set guide_buf [get_cells -physical_context -design [current_block] -quiet {*guide_buf*}]
change_selection $guide_buf
set_fixed_objects [get_selection]

set scan_en_inv [get_cells -physical_context -design [current_block] -quiet {*scan_en_HFS*}]
change_selection $scan_en_inv
set_fixed_objects [get_selection]
```

Set all Macros signal pins & add buffer in front of signal pins (except clock & power pins)

Insert inverter_pair in front of scan_en_iopad/DOUT pin for HFS during place_opt
Inverter_pair = buffer

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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Review 04_read_tluplus.tcl**

```
#remove scenarios modes corners before setting

remove_scenarios -all
remove_modes -all
remove_corners -all


# Read TLU+ files
# TLU+ file is given by Foundary Company
# This file contains Metal capacitance at different spacing and width in the form of a lookup -
# talbe which provides high accuracy and runtime (This file can be used to extract RC value from interconnects)

read_parasitic_tech -name {minTLU} -tlup $data(tech_tlupbc)
read_parasitic_tech -name {maxTLU} -tlup $data(tech_tlupwc)
```

  - **Review 05_mcmm.tcl**

```
# source mcmm(multi corners multi modes).tcl

source -e ./icc2_common_env/mcmm.tcl
#set scenario_status -- report

# all about ss corner
set_scenario_status {*.ss*} -setup true -hold true \
                            -active true -max_transition true \
                            -max_capacitance true \
                            -min_capacitance false \
                            -leakage_power true \
                            -dynamic_power true
# all about ff corner
set_scenario_status {*.ff*} -setup true -hold true \
                            -active true -max_transition true \
                            -max_capacitance true \
                            -min_capacitance false \
                            -leakage_power true \
                            -dynamic_power true
```


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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Open & Review the mcmm.tcl**

```
#set scenarios modes corners

#mode
set m_constr(func) func
set m_constr(scan_shift) scan_shift
set m_constr(scan_capture) capture

#corners
set c_constr(ss_125c) ss_125c
set c_constr(ss_n40c) ss_n40c
set c_constr(ff_125c) ff_125c
set c_constr(ff_n40c) ff_n40c

#scenarios
set s_constr(func.ss_125c) func.ss_125c
set s_constr(func.ss_n40c) func.ss_n40c
set s_constr(func.ff_125c) func.ff_125c
set s_constr(func.ff_n40c) func.ff_n40c

set s_constr(scan_shift.ss_125c) scan_shift.ss_125c
set s_constr(scan_shift.ss_n40c) scan_shift.ss_n40c
set s_constr(scan_shift.ff_125c) scan_shift.ff_125c
set s_constr(scan_shift.ff_n40c) scan_shift.ff_n40c

set s_constr(scan_capture.ss_125c) scan_capture.ss_125c
set s_constr(scan_capture.ss_n40c) scan_capture.ss_n40c
set s_constr(scan_capture.ff_125c) scan_capture.ff_125c
set s_constr(scan_capture.ff_n40c) scan_capture.ff_n40c
```

```
# scenario mode corner creation

foreach m [array names m_constr] {
create_mode $m
}

foreach c [array names c_constr] {
create_corner $c
}

foreach s [array names s_constr] {
lassign [split $s "."] m c
create_scenario -name $s  -mode $m  -corner $c
}
```

```
# set parasitic_parameters after read TLU+ files

#set ff_n40c corner
current_corner ff_n40c
set_parasitic_parameters -corners ff_n40c -library ORCA_${step}.nlib -late_s
pec minTLU -early_spec minTLU -late_temperature -40 -early_temperature -40

#set ss_n40c corner
current_corner ss_n40c
set_parasitic_parameters -corners ss_n40c -library ORCA_${step}.nlib -late_s
pec maxTLU -early_spec maxTLU -late_temperature -40 -early_temperature -40

#set ff_125c corner
current_corner ff_125c
set_parasitic_parameters -corners ff_125c -library ORCA_${step}.nlib -late_s
pec minTLU -early_spec minTLU -late_temperature 125 -early_temperature 125

#set ss_125c corner
current_corner ss_125c
set_parasitic_parameters -corners ss_125c -library ORCA_${step}.nlib -late_s
pec maxTLU -early_spec maxTLU -late_temperature 125 -early_temperature 125
```

Sets parasitic parameters such as parasitic spec (TLU+) and temperature after read TLU+ files

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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Open & Review the mcmm.tcl**

```
#set pvt value for all corners

set_process_number 1.01 -corners ff_n40c
set_temperature -40 -min -40 -corners ff_n40c
set_voltage 1.16 -corner ff_n40c -object_list [get_supply_nets VDD]
set_voltage 2.75 -corner ff_n40c -object_list [get_supply_nets AVDD1]
set_voltage 2.75 -corner ff_n40c -object_list [get_supply_nets AVDD2]
set_voltage 1.16 -corner ff_n40c -object_list [get_supply_nets DVDD1]
set_voltage 1.16 -corner ff_n40c -object_list [get_supply_nets DVDD2]
set_voltage 2.75 -corner ff_n40c -object_list [get_supply_nets VDDIO]
set_voltage 0.00 -corner ff_n40c -object_list [get_supply_nets VSS]
set_voltage 0.00 -corner ff_n40c -object_list [get_supply_nets VSSIO]
set_voltage 0.00 -corner ff_n40c -object_list [get_supply_nets AVSS]

set_process_number 0.99 -corners ss_n40c
set_temperature -40 -min -40 -corners ss_n40c
set_voltage 0.95 -corner ss_n40c -object_list [get_supply_nets VDD]
set_voltage 2.25 -corner ss_n40c -object_list [get_supply_nets AVDD1]
set_voltage 2.25 -corner ss_n40c -object_list [get_supply_nets AVDD2]
set_voltage 0.95 -corner ss_n40c -object_list [get_supply_nets DVDD1]
set_voltage 0.95 -corner ss_n40c -object_list [get_supply_nets DVDD2]
set_voltage 2.25 -corner ss_n40c -object_list [get_supply_nets VDDIO]
set_voltage 0.00 -corner ss_n40c -object_list [get_supply_nets VSS]
set_voltage 0.00 -corner ss_n40c -object_list [get_supply_nets VSSIO]
set_voltage 0.00 -corner ss_n40c -object_list [get_supply_nets AVSS]
```

```
set_process_number 1.01 -corners ff_125c
set_temperature 125 -min 125 -corners ff_125c
set_voltage 1.16 -corner ff_125c -object_list [get_supply_nets VDD]
set_voltage 2.75 -corner ff_125c -object_list [get_supply_nets AVDD1]
set_voltage 2.75 -corner ff_125c -object_list [get_supply_nets AVDD2]
set_voltage 1.16 -corner ff_125c -object_list [get_supply_nets DVDD1]
set_voltage 1.16 -corner ff_125c -object_list [get_supply_nets DVDD2]
set_voltage 2.75 -corner ff_125c -object_list [get_supply_nets VDDIO]
set_voltage 0.00 -corner ff_125c -object_list [get_supply_nets VSS]
set_voltage 0.00 -corner ff_125c -object_list [get_supply_nets VSSIO]
set_voltage 0.00 -corner ff_125c -object_list [get_supply_nets AVSS]

set_process_number 0.99 -corners ss_125c
set_temperature 125 -min 125 -corners ss_125c
set_voltage 0.95 -corner ss_125c -object_list [get_supply_nets VDD]
set_voltage 2.25 -corner ss_125c -object_list [get_supply_nets AVDD1]
set_voltage 2.25 -corner ss_125c -object_list [get_supply_nets AVDD2]
set_voltage 0.95 -corner ss_125c -object_list [get_supply_nets DVDD1]
set_voltage 0.95 -corner ss_125c -object_list [get_supply_nets DVDD2]
set_voltage 2.25 -corner ss_125c -object_list [get_supply_nets VDDIO]
set_voltage 0.00 -corner ss_125c -object_list [get_supply_nets VSS]
set_voltage 0.00 -corner ss_125c -object_list [get_supply_nets VSSIO]
set_voltage 0.00 -corner ss_125c -object_list [get_supply_nets AVSS]
```

- **set_process_number**
  Sets the process number for one or more corners, for part or all of the current design, or only for cells from specific reference libraries (match .lib file)
- **set_temperature**
  Sets the operating temperature for one or more corners, for part or all of the current design (match corners temperature)
- **set_voltage**
  Applies an operating voltage to a list of supply nets, supply ports, cells or ports, or to the current design (match upf & .lib files)

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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Review 06_read_sdc.tcl**

```
#read_sdc(synopsys design constraints)
#SDC contain information such as clock definitions, input and output delays, false paths, and timing exeptions

#scan_shift mode
current_mode scan_shift
read_sdc $data(shift_sdc) ;# ${libdir}/SDC/ORCA_shift.sdc

#func mode
current_mode func
read_sdc $data(func_sdc) ;# ${libdir}/SDC/ORCA_func.sdc

#scan_capture mode
current_mode scan_capture
read_sdc $data(capture_sdc) ;# ${libdir}/SDC/ORCA_capture.sdc
```

  - **Review 07_remove_propagated_clocks.tcl**

```
#set TIE cell
#A tie cell is a special type of standard cell that provides a constant high or low signal to the input of any logic
 gate

set_lib_cell_purpose -include optimization [get_lib_cells */TIE*] ;# set TIE cell used in optimizaiton
set_dont_touch [get_lib_cells */TIE*] false ;# set TIE cells "dont_touch" value false
```


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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Review 07_remove_propagated_clocks.tcl**

```
#remove_propagated clocks before placement
#set ideal network for all clock
#we can analyze only ideal network in placement stage (because you didn't run CTS not yet)

set mode "func scan_shift scan_capture"

foreach MODE $mode {
current_mode $MODE

echo "current mode is $MODE"
        switch $MODE {
                func {
                        remove_propagated_clocks [all_clocks]
                        set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        remove_ideal_network $clock_sink
                        set_ideal_network [get_pins $clock_sink]
                        set_ideal_network [get_nets buf_sdram_clk]
                }
                scan_shift {
                        remove_propagated_clocks [all_clocks]
                        set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        remove_ideal_network $clock_sink
                        set_ideal_network [get_pins $clock_sink]
                        set_ideal_network [get_nets buf_sdram_clk]
                }
                scan_capture {
                        remove_propagated_clocks [all_clocks]
                        set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        remove_ideal_network $clock_sink
                        set_ideal_network [get_pins $clock_sink]
                        set_ideal_network [get_nets buf_sdram_clk]
                }
        }
}
```


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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Review 08_set_false_path_and_app_options.tcl**

```
#set a false path constraint from the ScanEnable port (top-level)

#set current mode func
current_mode "func"

#set false path
set_false_path -from scan_en -to [get_clocks "PCI* SD* SYS*"] ;#Identifies paths in a design to mark as false, so that they are not considered during timing analysis

#report high fanout
report_net_fanout -high_fanout

#set false path about buf_sdram_clk
current_mode "func"
set_false_path  -through [get_pins DFTC_70/Y]

#source set_disable_timing.tcl

source -e ./user_script/set_disable_timing.tcl ;#Disables timing through the specified cells, pins, ports, or timing arcs in the current_mode, if library cells or pins are specified,
 it disables the timing arcs for all modes that utilize the library

#set_app_options for placement

set_app_options -list {opt.tie_cell.max_fanout {8}} ;#Specifies the maximum fanout a tie-cell can drive
set_app_options -list {cts.common.max_fanout {16}} ;#Fanout constraint for clock tree synthesis
set_app_options -list {opt.common.max_fanout {16}} ;#Fanout constraint for data path optimization
set_app_options -list {place.coarse.congestion_layer_aware {true}} ;#Controls whether the tool considers the congestion of each layer separately during coarse placement
set_app_options -list {place.coarse.continue_on_missing_scandef {true}} ;#When this option is true scandef checking prior to coarse placement is disabled
set_app_options -list {opt.common.drc_mode_buffering {true}} ;#Buffering mode for drc stages of place_opt and clock_opt
set_app_options -list {place_opt.initial_drc.global_route_based {1}} ;#Run global route based buffering (GRopto flow) during the initial_drc place_opt stage
set_app_options -list {place_opt.initial_place.buffering_aware {true}} ;#Runs buffering-aware timing-driven placement for the initial placement step inside the place_opt command

set_app_options -list {place.coarse.pin_density_aware {true}} ;#Enables pin density-aware coarse placement
set_app_options -list {place_opt.congestion.effort {high}} ;#set congestion effort during placement
set_app_options -list {place_opt.final_place.effort {high}} ;#Specifies the CPU effort level for the final coarse placement invoked in place_opt
```


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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Open & Review the set_disable_timing.tcl**

```
#set current_scenarios

current_scenario "func.ss_125c"

#set_disable_timing SDRAM_DQ_Bus_1__sdram_DQ_iopad ~ SDRAM_DQ_Bus_15__sdram_DQ_iopad from EN pin to PADIO pin

set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_1__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_1__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_2__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_2__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_3__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_3__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_4__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_4__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_5__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_5__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_6__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_6__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_7__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_7__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_8__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_8__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_9__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_9__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_10__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_10__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_11__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_11__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_12__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_12__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_13__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_13__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_14__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_14__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_15__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_15__sdram_DQ_iopad/PADIO } ]
```

This command is used when it is not necessary to calculate the Timing Arc

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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Review 09_place_opt.tcl**

```
#place_opt - place and optimize the current design
#< 5 stages of place_opt >
#initial_place - initial_drc - initial_opto - final_place - final_opto

sec_StartTimer place_opt

#place_opt
place_opt

sec_ReportResourceUtil place_opt

#congestion
report_congestion -rerun_global_router ;#Reports the congestion statistics

#timing
report_timing -significant_digits 5 ;#Displays timing information about a design

#Qor summary
report_qor -summary ;#Displays QoR information and statistics for the current design

############################################################
#report Timing & constraints for placement after place_opt
############################################################

#update timing
update_timing -full ;#Updates timing information on the current design

sec_StartTimer report

#write "reports files" of placement at ${rundir}/reports/placement/* directory
source ./user_script/placement_rpt.tcl

sec_ReportResourceUtil report

#save_lib

save_lib -all
```


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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Review 10_refine_opt.tcl**

```
#refine_opt
#By running refine_opt command, you can optimize the place_opt completed db once again

set_app_options -list {refine_opt.place.congestion_effort {high}} ;#Specifies the effort level for the congestion alleviation in refine_opt.
set_app_options -list {refine_opt.place.effort {high}} ;#Specifies the CPU effort level for coarse placements invoked in refine_opt
set_app_options -list {place_opt.final_place.effort {high}} ;#Specifies the CPU effort level for the final coarse placement invoked in place_opt

sec_StartTimer refine_opt

refine_opt ;#By running refine_opt command, you can optimize the place_opt completed db once again

sec_ReportResourceUtil refine_opt
```

  - **Review 11_insert_spare_cell.tcl**

```
#source Insert_spare cell script
#Spare cells enable us to modify/improve the functionality of a chip with minimal changes in the mask

source -e ./user_script/spare_cell_insertion.tcl
```


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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Open & Review the spare_cell_insertion.tcl**

```
set core_bdry "{425.543 425.846} {1014.459 1014.152}"

#add NBUFFX2_RVT
add_spare_cells -num_instances 100 \
        -boundary $core_bdry \
        -cell_name ORCA_spare_cell_NBUF \
        -lib_cell "saed32rvt/NBUFFX2_RVT"

#add INV_RVT
add_spare_cells -num_instances 100 \
        -boundary $core_bdry \
        -cell_name ORCA_spare_cell_INV \
        -lib_cell "saed32rvt/INVX2_RVT"

#add AND_RVT
add_spare_cells -num_instances 50 \
        -boundary $core_bdry \
        -cell_name ORCA_spare_cell_AND \
        -lib_cell "saed32rvt/AND2X2_RVT"

#add FF_RVT
add_spare_cells -num_instances 50 \
        -boundary $core_bdry \
        -cell_name ORCA_spare_cell_FF \
        -lib_cell "saed32rvt/RSDFFX2_RVT"

#add XOR2x2_RVT
add_spare_cells -num_instances 50 \
        -boundary $core_bdry \
        -cell_name ORCA_spare_cell_XOR \
        -lib_cell "saed32rvt/XOR2X2_RVT"
```

```
#spread & legalize spare cells after adding

spread_spare_cells -cells [get_cells -phy -filter "full_name =~ ORCA_spare_cell*"] \
                   -boundary $core_bdry

legalize_placement -incremental
set_placement_status legalize_only \
        [get_flat_cells -filter is_spare_cell]
```

- **We can use (BUF, INV, AND, Flip Flop, XOR) 5 types of spare cells in this design**
- **Spread_spare_cells**
  Spreads spare cells evenly throughout a rectilinear region
  (The spare cells usually come through Verilog ECO or manual netlist editing Commands)

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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Review 12_check_placement.tcl**

```
####################
#check placement
####################

#legality
check_legality ;#Checks the legality of the current placement

#mv_design
check_mv_design ;#Checks for violations in a multi-voltage design

#utilization of core area (use -region option)
report_utilization -region {{{425.5430 425.8460} {1014.4590 1014.1520}}} ;#Reports utilization of the current block

#congestion
report_congestion -rerun_global_router;#Reports the congestion statistics

#timing
report_timing -significant_digits 5 ;#Displays timing information about a design

#Qor summary
report_qor -summary ;#Displays QoR information and statistics for the current design
```


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Placement : Introduction to placement.tcl

- Review and check the variables in each scripts
  - **Review 13_report_timing_and_constraints.tcl & 14_save_and_exit.tcl**

```
#report Timing & constraints for placement after refine_opt

#update timing
update_timing -full ;#Updates timing information on the current design

sec_StartTimer report

#write "reports files" of placement at ${rundir}/reports/placement/* directory
source ./user_script/refine_opt_rpt.tcl

sec_ReportResourceUtil report
```

```
#save_lib
save_lib -as ./nlib/ORCA_refine_opt.nlib

#exit
exit
```

  - **Open & Review the placement_rpt.tcl**

```
#write report files at ${WOWRK_DIR}/2_BACKEND/1_pnr/rundir/04_placement/reports/placement/* directory
set rpt "reports"
set step "placement"

switch $step {
placement {
redirect -file ./$rpt/${step}/${step}.qor.snap.rpt {report_qor -summary -nosplit -significant_digits 3 } ;# report_qor -summary
redirect -file ./$rpt/${step}/${step}.util.rpt {report_utilization} ;# report utilization
redirect -file ./$rpt/${step}/${step}.congestion.rpt {report_congestion} ;# report congestion
redirect -file ./$rpt/${step}/${step}.legality.check.rpt {check_legality} ;# check legality
redirect -file ./$rpt/${step}/${step}.fanout.rpt {report_net_fanout -high_fanout} ;# report high fanout
redirect -file ./$rpt/${step}/${step}.max_tran.rpt {report_constraints -max_transition -all_violators -corners *} ;# report max_transition violat
ion

#report timing (set, hold) about all scenarios
set all_sce [get_attribute -objects [get_scenarios -filter "active == true" ] -name name]
foreach x $all_sce {
redirect -file ./$rpt/${step}/timing_report_max_${step}_${x} \
{report_timing -scenarios $x -significant_digits 3 -delay_type max -transition_time -max_paths 100 -nets -input_pins -report_by group -physical -
attributes -nosplit -derate -crosstalk_delta -path_type full_clock}
redirect -file ./$rpt/${step}/timing_report_min_${step}_${x} \
{report_timing -scenarios $x -significant_digits 3 -delay_type min -transition_time -max_paths 100 -nets -input_pins -report_by group -physical -
attributes -nosplit -derate -crosstalk_delta -path_type full_clock}
        }
    }
}
```


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Placement : Use ICC2 (IC Compiler II tool)

It is also good to Read Design by source 02_floorplan.tcl, but when you proceed for the first time, it is recommended that you open the script and apply it one phrase at a time

1. Invoke tool icc2_shell

- **Recommend the icc2/2021.06-SP5 version**

>> icc2_shell –gui | tee -i ./04_placement.log ;# (| tee -I ./01_read_design.log) is the command to output log

```
< user@linux 04_placement >> icc2_shell -gui | tee -i ./04_placement.log


                            IC Compiler II (TM)

                Version S-2021.06-SP5 for linux64 - Jan 14, 2022 -SLE

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                           https://solvnetplus.synopsys.com)

  Loading user preference file /u/kjinsoo/.synopsys_icc2_gui/preferences.tcl
  icc2_shell>
```

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/01_design_setup_and_copy.tcl

- **set step & source design_setup.tcl**

```
icc2_shell> set step placement
placement
icc2_shell> set before_step powerplan
powerplan
icc2_shell> source -e ./icc2_common_env/design_setup.tcl
```


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Placement : Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/01_design_setup_and_copy.tcl

- **open lib**

```
icc2_shell> open_lib ../03_${before_step}/nlib/ORCA_${before_step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir
/03_powerplan/nlib/ORCA_powerplan.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir
/00_create_ndm/ndm/saed32sram.ndm' (FILE-007)
{ORCA_powerplan.nlib}
```

Check that ndm is properly apllied each time you open the nlib

- **Copy ORCA_powerplan.nlib to ORCA_placement.nlib & open ORCA_placement.nlib**

```
icc2_shell> copy_lib -from ../03_${before_step}/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_
${step}.nlib
Information: Copying from library "ORCA_powerplan.nlib" to "./nlib/ORCA_placement.nlib".
{ORCA_placement.nlib}
icc2_shell> close_lib -all
Closing all libraries...
1
icc2_shell> open_lib ./nlib/ORCA_${step}.nlib
```


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Placement : Use ICC2 (IC Compiler II tool)

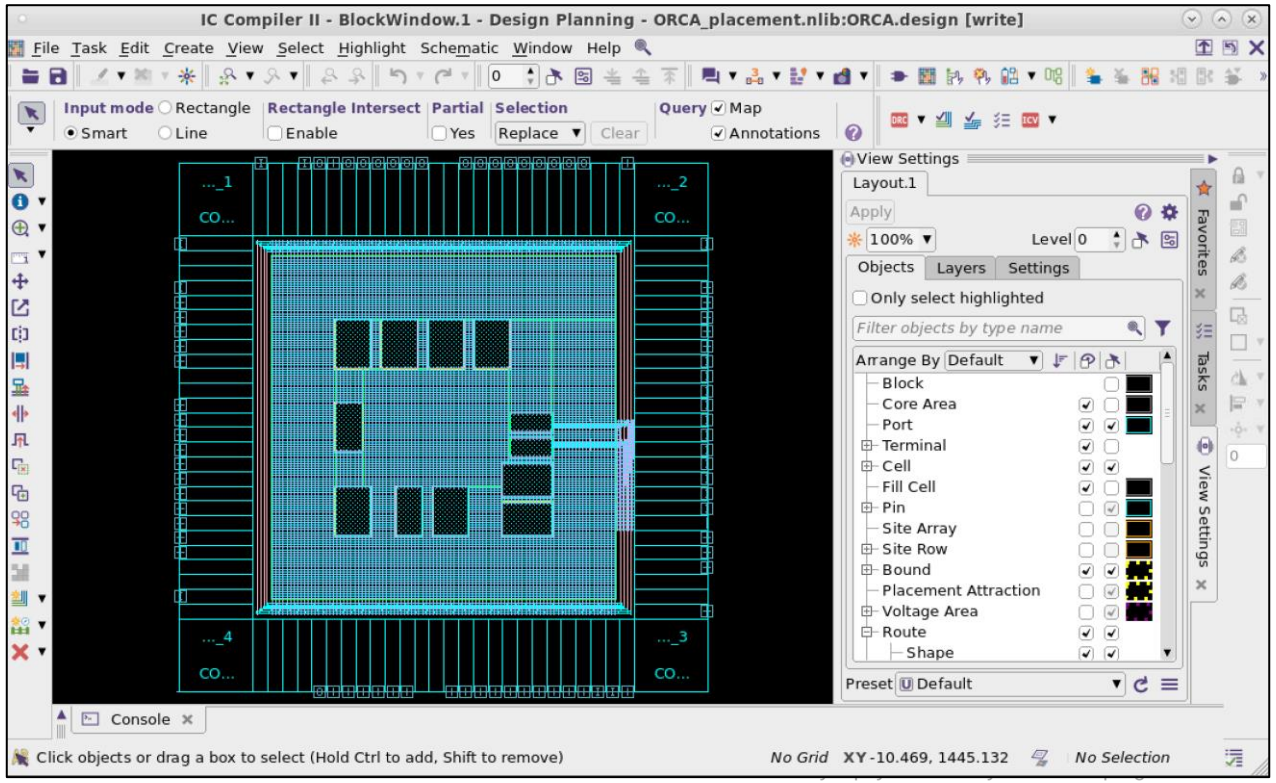
2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/01_design_setup_and_copy.tcl

- **open block**

```
icc2_shell> open_block ./nlib/ORCA_${step}.nlib:ORCA.design
Information: User units loaded from library 'saed32hvt' (LNK-040)
Opening block 'ORCA_placement.nlib:ORCA.design' in edit mode
{ORCA_placement.nlib:ORCA.design}
icc2_shell> link_block
Using libraries: ORCA_placement.nlib EXPLORE_physical_only saed32hvt saed32io_fc saed32lvt saed32pll
 saed32rvt saed32sram
Visiting block ORCA_placement.nlib:ORCA.design
```

Check using libraries

- **Check the copied block in the gui window**



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Placement : Use ICC2 (IC Compiler II tool)

3. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/02_create_routing_blockage.tcl

- **Create routing blockage for all Macros (M9 MRDL)**

```
icc2_shell> source -e ./user_script/create_routing_blockage.tcl
set only_macros "I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM0 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM1 I_ORCA_
TOP/I_CONTEXT_MEM/CONTEXT_RAM2 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM3 I_ORCA_TOP/I_RISC_CORE/REG_FILE
_B_RAM I_ORCA_TOP/I_RISC_CORE/REG_FILE_A_RAM I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM I_ORCA_TOP/I_P
CI_WRITE_FIFO/PCI_WFIFO_RAM I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_RAM I_ORCA_TOP/I_SDRAM_WRITE_FIFO/
SD_WFIFO_RAM"
set pll "I_CLOCK_GEN/I_PLL_PCI I_CLOCK_GEN/I_PLL_SD"
####################################
#create_routing_blockage_for macros
####################################
set macro_bbox [get_attr [get_cells $only_macros] bbox]
set i 0
foreach x $macro_bbox {
set llx [expr [lindex $x 0 0] + 1]
set urx [expr [lindex $x 1 0] - 1]
set lly [expr [lindex $x 0 1] + 1]
set ury [expr [lindex $x 1 1] - 1]
set new_bboxes "{$llx $lly} {$urx $ury}"
create_routing_blockage -name "mc_routing_blk_${i}" -layers "M6 M9 MRDL" -boundary $new_bboxes

incr i
}
####################################
#create_routing_blockage_for PLL
####################################
set pll_bbox [get_attr [get_cells $pll] bbox]
set i 0
foreach x $pll_bbox {
set llx [expr [lindex $x 0 0] + 2]
set urx [expr [lindex $x 1 0] - 2]
set lly [expr [lindex $x 0 1] + 2]
set ury [expr [lindex $x 1 1] - 2]
set new_bboxes "{$llx $lly} {$urx $ury}"
create_routing_blockage -name "pll_blkg_hard_${i}" -layers "M9 MRDL" -boundary $new_bboxes

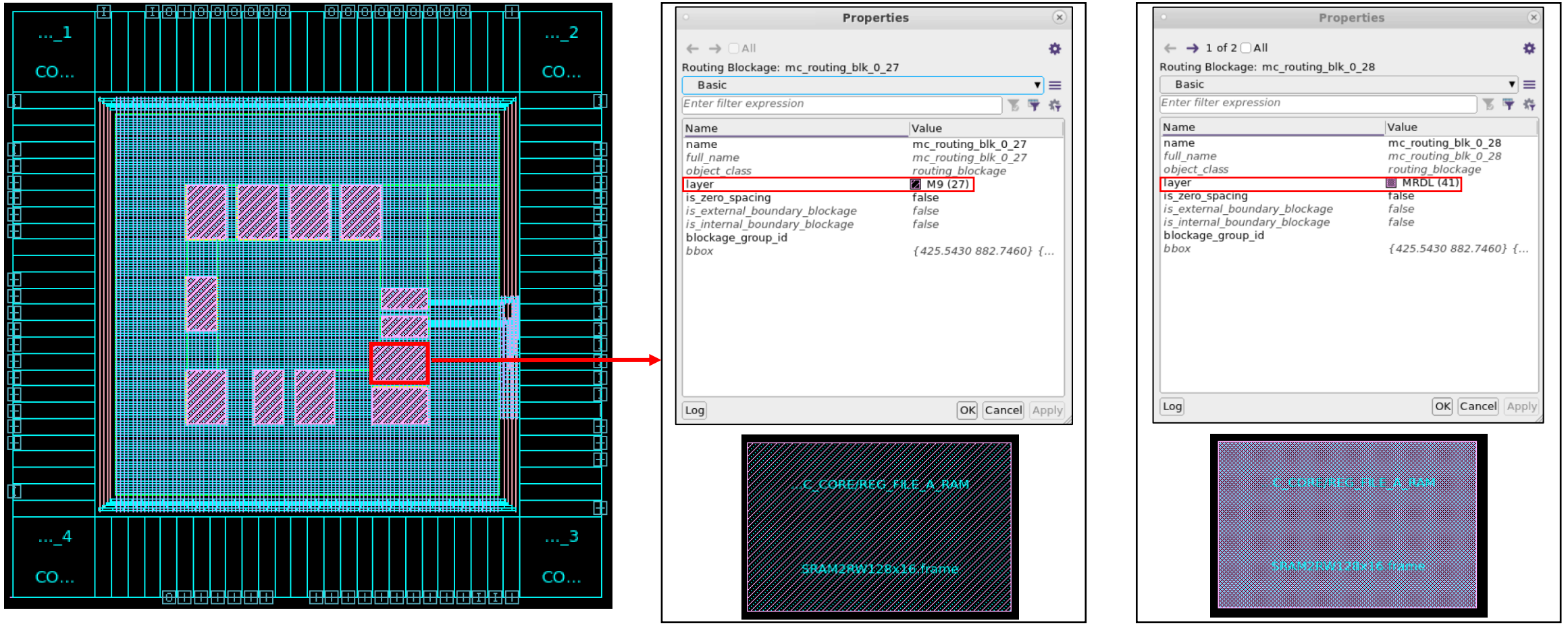
incr i
}
```


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Placement : Use ICC2 (IC Compiler II tool)

3. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/02_create_routing_blockage.tcl

- **Check M9 MRDL Routing blockages (Macro & PLL)**

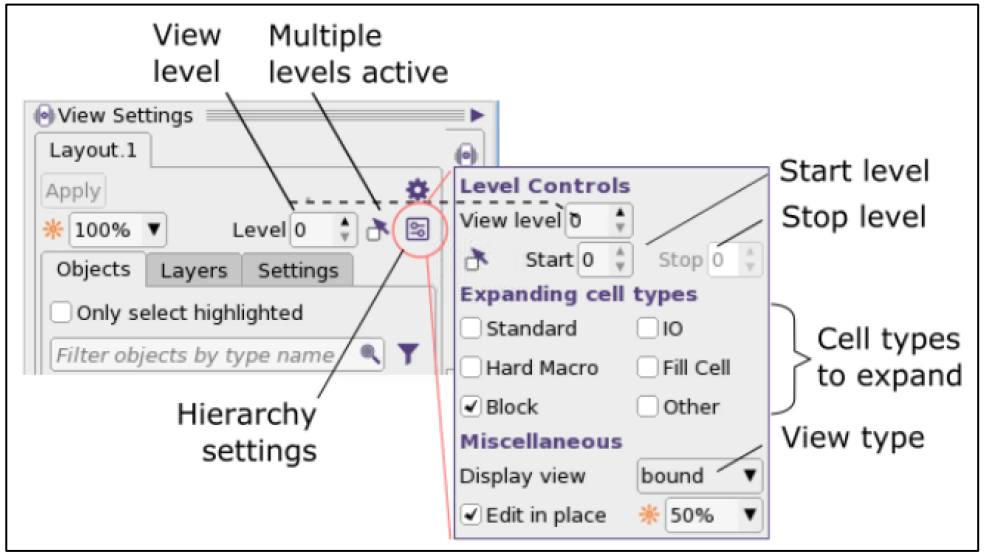


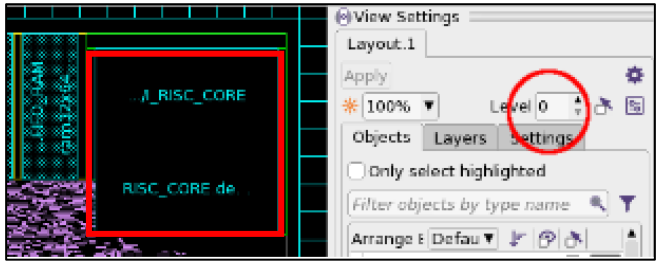
- **(M9, MRDL) Two routing blockages are created by overlapping on the Macros**
- **You can select the required layers in the (View Settings – Layers) menu and you can check each routing blockage**

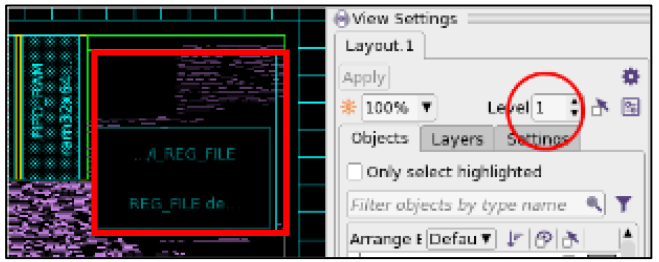
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Placement : Use ICC2 (IC Compiler II tool)

- **Hierarchy Display Controls**
  - When a design is loaded into the tool, the hierarchy view level is 0 and the layout view displays the top block and its contents (default value is 0)
  - Use the Level setting, "Multiple Levels Active" option, and Level Controls options to control how design objects in the physical hierarchy are displayed







Show or hide objects chosen in the Expanding cell types section by increasing or decreasing the view level in the View Settings panel

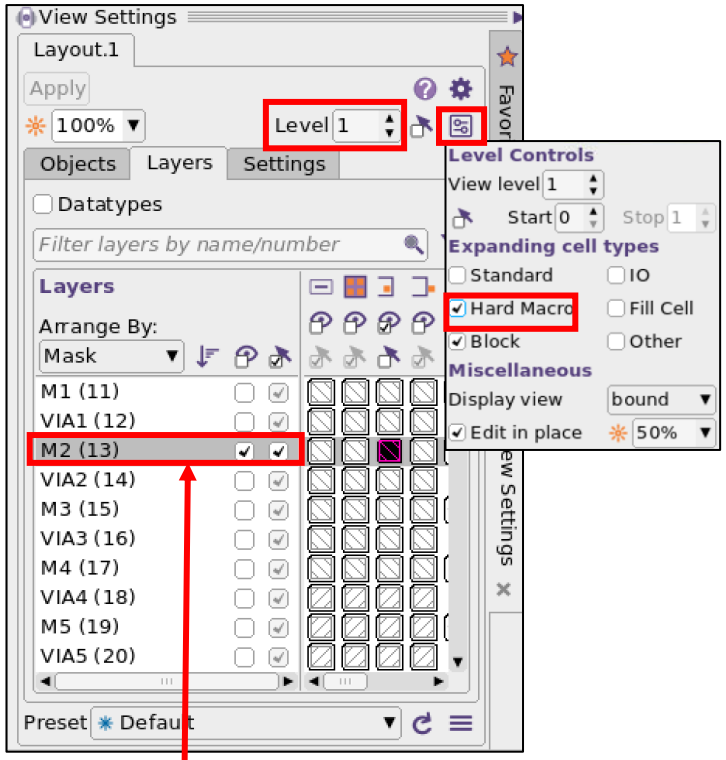
- **Limit the hierarchy levels at which objects can be selected by setting the start and stop level values in the "Level Controls section"**
- **Enable editing of the objects within the hierarchy by selecting the "Multiple Levels Active" option**
- **Specify the cell types to expand by selecting them in the Expanding cell types section**
- **Specify the view type (abstract, bound, design, frame, layout, or outline) to Display view option in the "Miscellaneous" section**

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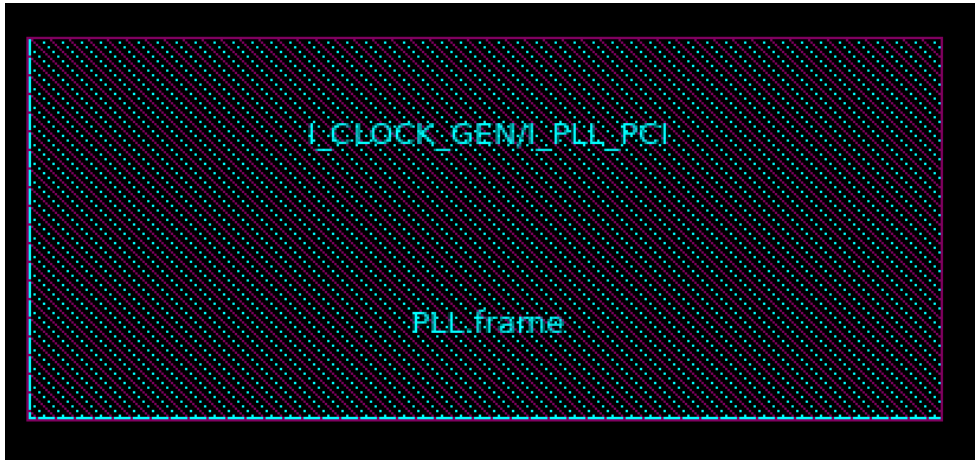
Placement : Use ICC2 (IC Compiler II tool)

3. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/02_create_routing_blockage.tcl

- **Check M1~M5 routing blockages inside Macro & PLL**



Select the layer of routing blockage you want to check



Use "Hierarchy Display Controls" panels to check cell routing blockages inside Macro & PLL
(check M1 to M6 routing blockages inside Macro & PLL)

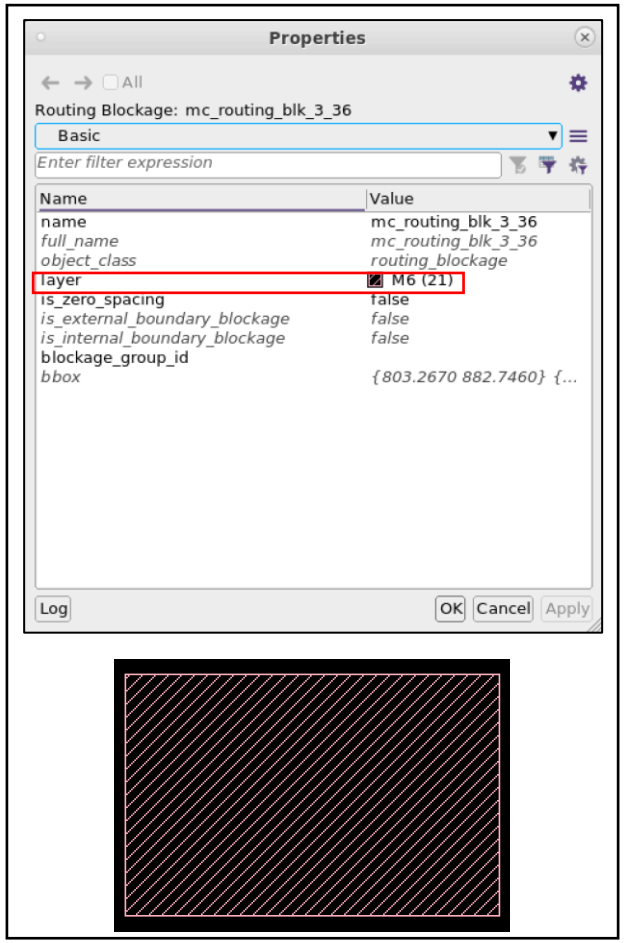
There are routing blockages inside the cell from M1 to M5, so there is no need to create M1 to M5 routing blockage on Macro & PLL

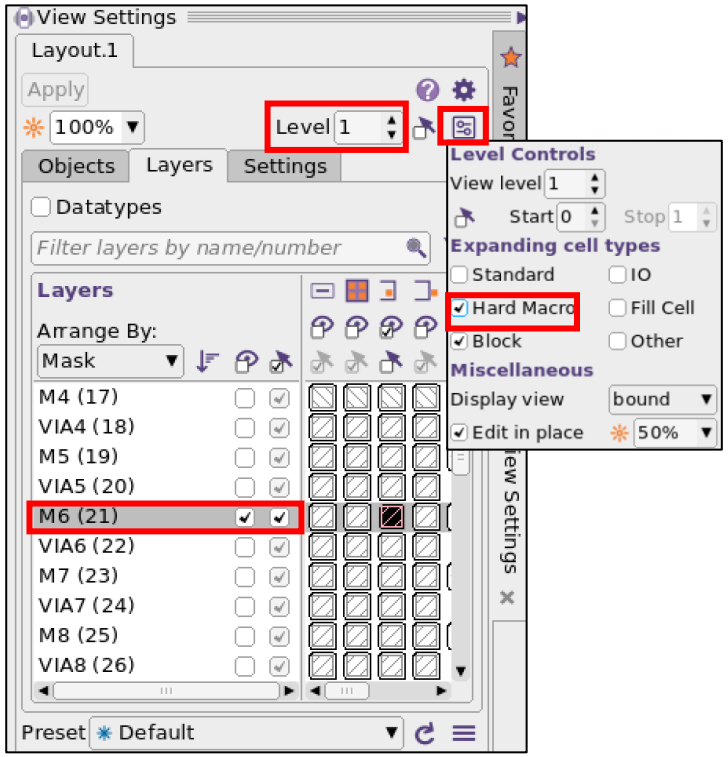
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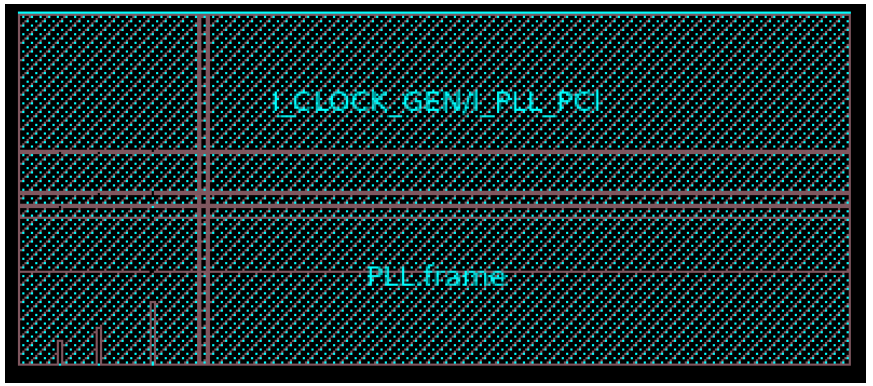
Placement : Use ICC2 (IC Compiler II tool)

3. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/02_create_routing_blockage.tcl

- **Check M6 Routing blockages (only PLL)**







M6 Routing blockage exists inside the PLL, so we will create routing blockages only Macros

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Placement : Use ICC2 (IC Compiler II tool)

4. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/03_insert_guide_buf.tcl

- **Insert guide buffer (Macro & PLL)**

```
icc2_shell> set all_macros [get_cells -hierarchical -filter "is_hard_macro && !is_physical_only"]
{I_CLOCK_GEN/I_PLL_PCI I_CLOCK_GEN/I_PLL_SD I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM0 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM1 I_ORCA_TOP/I_
CONTEXT_MEM/CONTEXT_RAM2 I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM3 I_ORCA_TOP/I_RISC_CORE/REG_FILE_B_RAM I_ORCA_TOP/I_RISC_CORE/REG_FILE_
A_RAM I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_RAM I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_RAM I_
ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM}
icc2_shell> set sig_pins [get_pins -of $all_macros -filter "port_type == signal && net_name !~ *VDD*  && net_name !~ *VSS* && net_nam
e !~ *UNCONNECTED*"]

icc2_shell> set buf "saed32rvt/NBUFFX2_RVT"
saed32rvt/NBUFFX2_RVT
icc2_shell> foreach x $sig_pins {
? add_buffer $x -lib_cell $buf -new_cell_names macro_guide_buf -net_name guide_buf
? incr i
? }
Cell is added at (927.0780 709.9870).
Cell is added at (901.3030 722.6680).
Cell is added at (901.3040 733.5020).
Cell is added at (927.0780 640.8790).
Cell is added at (901.3030 653.5600).
```

- **Insert inverter pair (scan_en_iopad/DOUT)**

```
add_buffer [get_pins -filter "full_name == scan_en_iopad/DOUT"] -inverter_pair \
?      -new_cell_names scan_en_HFS \
?      -new_net_names scan_en_guide_inv \
?      -lib_cell saed32rvt/INVX2_RVT
Cell is added at (1240.4240 896.5490).
Cell is added at (1240.4240 896.5490).
{scan_en_HFS scan_en_HFS_1}
```


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Placement : Use ICC2 (IC Compiler II tool)

4. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/03_insert_guide_buf.tcl

- **Legalize guide buffer & inverter**

```
icc2_shell> set guide_buf [get_attr [get_cells -physical_context -quiet {*macro_guide_buf*}] full_name]

icc2_shell> set scan_en_inv [get_attr [get_cells -physical_context -quiet {*scan_en_HFS*}] full_name]
scan_en_HFS scan_en_HFS_1
icc2_shell> legalize_placement -cells "$guide_buf $scan_en_inv"
```

- **Fixed guide buffer & inverter**

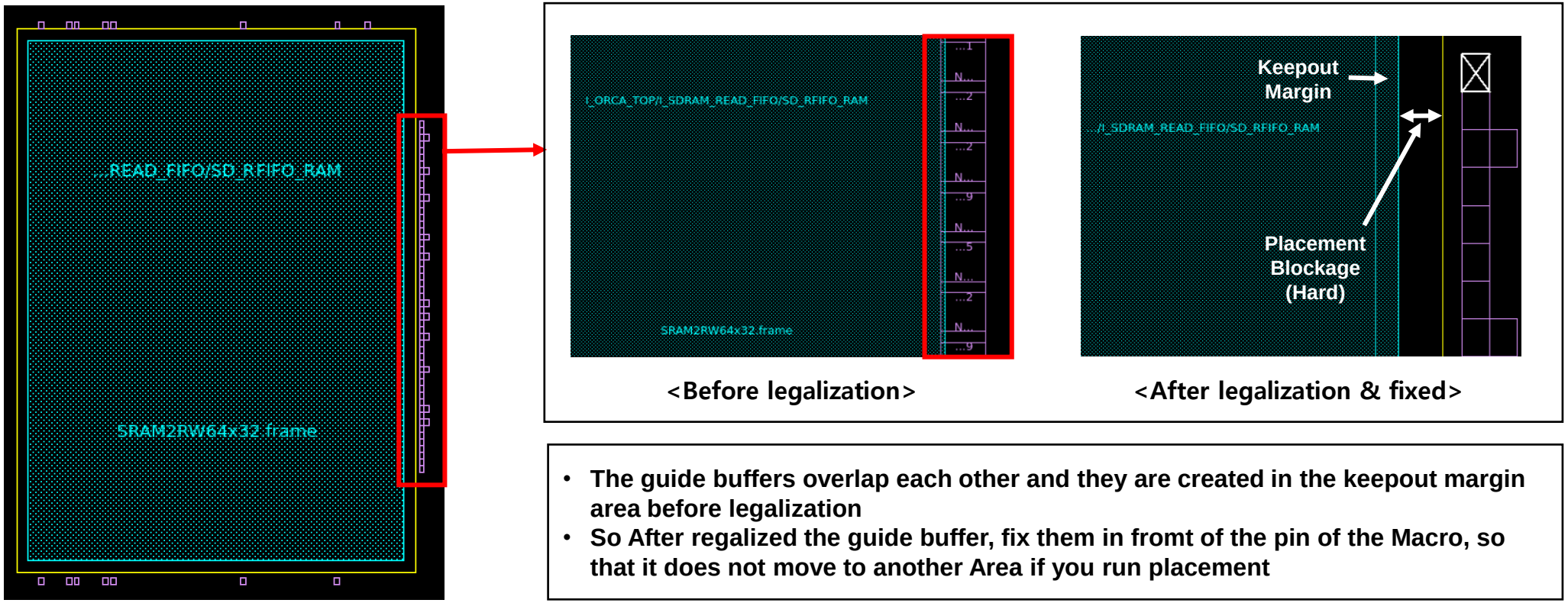
```
icc2_shell> set guide_buf [get_cells -physical_context -design [current_block] -quiet {*guide_buf*}]
icc2_shell> change_selection $guide_buf
icc2_shell> set_fixed_objects [get_selection]
1
icc2_shell>
icc2_shell> set scan_en_inv [get_cells -physical_context -design [current_block] -quiet {*scan_en_HFS*}]
{scan_en_HFS scan_en_HFS_1}
icc2_shell> change_selection $scan_en_inv
icc2_shell> set_fixed_objects [get_selection]
1
```


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Placement : Use ICC2 (IC Compiler II tool)

4. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/03_insert_guide_buf.tcl

- **Check Macro & PLL guide BUF**



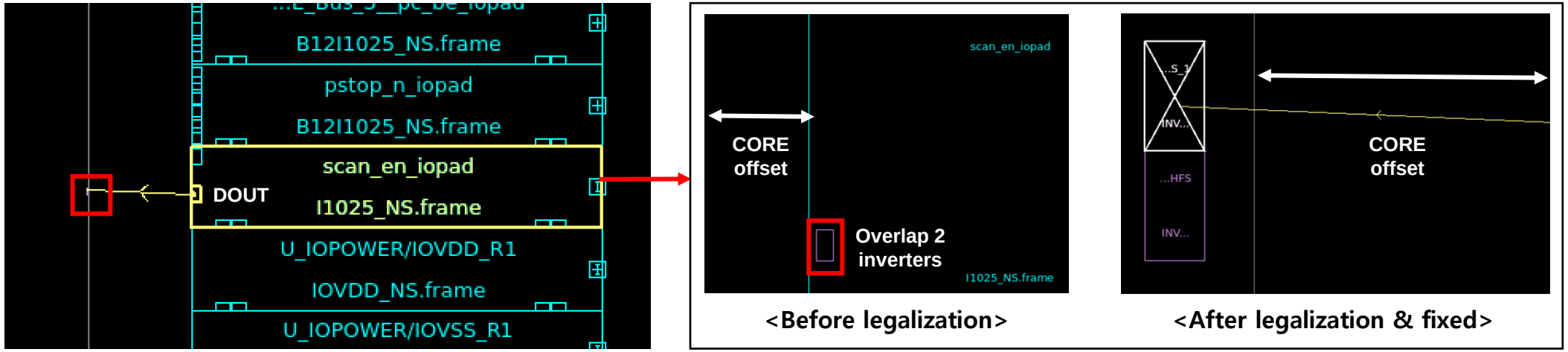
- **The guide buffers overlap each other and they are created in the keepout margin area before legalization**
- **So After regalized the guide buffer, fix them in fromt of the pin of the Macro, so that it does not move to another Area if you run placement**

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Placement : Use ICC2 (IC Compiler II tool)

4. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/03_insert_guide_buf.tcl

- **Check inverter pair (scan_en_iopad/DOUT)**



- **You can display net fanout use report_net_fanout –high fanout command**
- **The nets connected to theses inverters perform High Fanout Synthesis(HFS) in the optimization step to prevent max transition violation**
- **This net proceeds HFS with inverter during optimization, so we create inverter (pair) in front of scan_en_iopad/DOUT pin (Inverter pair = 1buffer)**

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Placement : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/04_read_tluplus.tcl

- **Remove scenarios, modes, corners before setting**

```
icc2_shell>  remove_scenarios -all
Information: The command 'remove_scenarios' cleared the undo history. (UNDO-016)
1
icc2_shell> remove_modes -all
1
icc2_shell> remove_corners -all
1
```

- **Read TLU+ files (best case & worst case)**

```
icc2_shell> read_parasitic_tech -name {minTLU} -tlup $data(tech_tlupbc)
1
icc2_shell> read_parasitic_tech -name {maxTLU} -tlup $data(tech_tlupwc)
1
```

- **TLU+(Table Look Up)** file contains advanced process effect that can be used to extract RC value interconnects
- The file contains Metal Capacitance at different spacing and width in the form of a lookup table which provides high accuracy and run-time

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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Set mode, corners, scenarios**

```
icc2_shell> #mode
icc2_shell> set m_constr(func) func
func
icc2_shell> set m_constr(scan_shift) scan_shift
scan_shift
icc2_shell> set m_constr(scan_capture) capture
capture
icc2_shell>
icc2_shell> #corners
icc2_shell> set c_constr(ss_125c) ss_125c
ss_125c
icc2_shell> set c_constr(ss_n40c) ss_n40c
ss_n40c
icc2_shell> set c_constr(ff_125c) ff_125c
ff_125c
icc2_shell> set c_constr(ff_n40c) ff_n40c
ff_n40c
icc2_shell>
icc2_shell> #scenarios
icc2_shell> set s_constr(func.ss_125c) func.ss_125c
func.ss_125c
icc2_shell> set s_constr(func.ss_n40c) func.ss_n40c
func.ss_n40c
icc2_shell> set s_constr(func.ff_125c) func.ff_125c
func.ff_125c
icc2_shell> set s_constr(func.ff_n40c) func.ff_n40c
func.ff_n40c
icc2_shell>
icc2_shell> set s_constr(scan_shift.ss_125c) scan_shift.ss_125c
scan_shift.ss_125c
icc2_shell> set s_constr(scan_shift.ss_n40c) scan_shift.ss_n40c
scan_shift.ss_n40c
icc2_shell> set s_constr(scan_shift.ff_125c) scan_shift.ff_125c
scan_shift.ff_125c
icc2_shell> set s_constr(scan_shift.ff_n40c) scan_shift.ff_n40c
scan_shift.ff_n40c
icc2_shell>
icc2_shell> set s_constr(scan_capture.ss_125c) scan_capture.ss_125c
scan_capture.ss_125c
icc2_shell> set s_constr(scan_capture.ss_n40c) scan_capture.ss_n40c
scan_capture.ss_n40c
icc2_shell> set s_constr(scan_capture.ff_125c) scan_capture.ff_125c
scan_capture.ff_125c
icc2_shell> set s_constr(scan_capture.ff_n40c) scan_capture.ff_n40c
scan_capture.ff_n40c
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Create mode, corners, scenarios**

```
icc2_shell> foreach m [array names m_constr] {
? create_mode $m
? }
icc2_shell>
icc2_shell> foreach c [array names c_constr] {
? create_corner $c
? }
icc2_shell>
icc2_shell> foreach s [array names s_constr] {
? lassign [split $s "."] m c
? create_scenario -name $s  -mode $m  -corner $c
? }
Created scenario func.ff_125c for mode func and corner ff_125c
All analysis types are activated.
Created scenario scan_shift.ss_125c for mode scan_shift and corner ss_125c
All analysis types are activated.
Created scenario func.ss_125c for mode func and corner ss_125c
All analysis types are activated.
Created scenario scan_capture.ff_n40c for mode scan_capture and corner ff_n40c
All analysis types are activated.
Created scenario scan_shift.ff_n40c for mode scan_shift and corner ff_n40c
All analysis types are activated.
Created scenario scan_capture.ss_n40c for mode scan_capture and corner ss_n40c
All analysis types are activated.
Created scenario func.ff_n40c for mode func and corner ff_n40c
All analysis types are activated.
Created scenario scan_shift.ss_n40c for mode scan_shift and corner ss_n40c
All analysis types are activated.
Created scenario func.ss_n40c for mode func and corner ss_n40c
All analysis types are activated.
Created scenario scan_capture.ff_125c for mode scan_capture and corner ff_125c
All analysis types are activated.
Created scenario scan_shift.ff_125c for mode scan_shift and corner ff_125c
All analysis types are activated.
Created scenario scan_capture.ss_125c for mode scan_capture and corner ss_125c
All analysis types are activated.
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Check modes after create modes**

  Check modes are created correctly and associated with scenarios

```
--------------------------------------------------------------------------------
Mode scan_capture
--------------------------------------------------------------------------------

Current: true  Default: false  Empty: true

Scenarios associated with this mode:

                                          Leakage   Dynamic
Scenario            Corner          Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
--------------------------------------------------------------------------------------------------------------------------
scan_capture.ff_125c
                    ff_125c         true    true   true  true      true      true      true     true     false    false
scan_capture.ff_n40c
                    ff_n40c         true    true   true  true      true      true      true     true     false    false
scan_capture.ss_125c
                    ss_125c         true    true   true  true      true      true      true     true     false    false
scan_capture.ss_n40c
                    ss_n40c         true    true   true  true      true      true      true     true     false    false


--------------------------------------------------------------------------------
Mode scan_shift
--------------------------------------------------------------------------------

Current: false  Default: false  Empty: true

Scenarios associated with this mode:

                                          Leakage   Dynamic
Scenario            Corner          Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
--------------------------------------------------------------------------------------------------------------------------
scan_shift.ff_125c  ff_125c         true    true   true  true      true      true      true     true     false    false
scan_shift.ff_n40c  ff_n40c         true    true   true  true      true      true      true     true     false    false
scan_shift.ss_125c  ss_125c         true    true   true  true      true      true      true     true     false    false
scan_shift.ss_n40c  ss_n40c         true    true   true  true      true      true      true     true     false    false
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Check modes after create modes**

  Check modes are created correctly and associated with scenarios

```
icc2_shell> report_modes
****************************************
Report : mode
Design : ORCA
Version: S-2021.06-SP5
Date   : Sun Sep 10 17:57:00 2023
****************************************

--------------------------------------------------------------------------------
Mode func
--------------------------------------------------------------------------------

Current: false  Default: false  Empty: true

Scenarios associated with this mode:

                                                    Leakage   Dynamic
Scenario            Corner         Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
--------------------------------------------------------------------------------------------------------------------------
func.ff_125c        ff_125c        true    true   true  true      true      true      true     true     false    false
func.ff_n40c        ff_n40c        true    true   true  true      true      true      true     true     false    false
func.ss_125c        ss_125c        true    true   true  true      true      true      true     true     false    false
func.ss_n40c        ss_n40c        true    true   true  true      true      true      true     true     false    false
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Check corners after create Scenarios**

Check corners information & PVT Settings (ff_125c) default

```
icc2_shell> report_corners
****************************************
Report : corner
Design : ORCA
Version: S-2021.06-SP5
Date   : Sun Sep 10 18:01:24 2023
****************************************

--------------------------------------------------------------------------------
Corner ff_125c
--------------------------------------------------------------------------------

Current: false  Default: false  Empty: true

Scenarios associated with this corner:

                                                 Leakage   Dynamic
Scenario            Mode            Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
-------------------------------------------------------------------------------------------------------------------------
func.ff_125c        func            true    true   true  true      true      true      true     true     false    false
scan_capture.ff_125c scan_capture   true    true   true  true      true      true      true     false    false
scan_shift.ff_125c  scan_shift      true    true   true  true      true      true      true     true     false    false


Top-Level PVT Settings:
  early process label             (none)
  early process number            1.01
  early voltage                   1.16
    source                        default
    file/line                     --
  early temperature               -40.00
    source                        default
    file/line                     --

  late process label              (none)
  late process number             1.01
  late voltage                    1.16
    source                        default
    file/line                     --
  late temperature                -40.00
    source                        default
    file/line                     --
```

```
0 objects with explicit PVT settings.


Top-level cell timing derates:
 data_cell derate early:   rise: 1.00    fall: 1.00
 clk_cell derate early:   rise: 1.00    fall: 1.00
 cell_check derate early:   rise: 1.00    fall: 1.00
 data_cell derate  late:   rise: 1.00    fall: 1.00
 clk_cell derate  late:   rise: 1.00    fall: 1.00
 cell_check derate  late:   rise: 1.00    fall: 1.00

Global net timing derates:
 net_data, early:   rise: 1.00    fall: 1.00
 net_data, late:   rise: 1.00    fall: 1.00
 net_clock, early:   rise: 1.00    fall: 1.00
 net_clock, late:   rise: 1.00    fall: 1.00
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_mcmm.tcl

- **Check corners after create Scenarios**

  Check corners information & PVT Settings (ss_125c) default

```
--------------------------------------------------------------------------------
Corner ss_125c
--------------------------------------------------------------------------------

Current: true  Default: false  Empty: true

Scenarios associated with this corner:

                                                 Leakage   Dynamic
Scenario            Mode            Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
-------------------------------------------------------------------------------------------------------------------------
func.ss_125c        func            true    true   true  true      true      true      true     true     false    false
scan_capture.ss_125c scan_capture   true    true   true  true      true      true      true     false    false    false
scan_shift.ss_125c  scan_shift      true    true   true  true      true      true      true     true     false    false


Top-Level PVT Settings:
  early process label             (none)
  early process number            1.01
  early voltage                   1.16
    source                        default
    file/line                     --
  early temperature               -40.00
    source                        default
    file/line                     --

  late process label              (none)
  late process number             1.01
  late voltage                    1.16
    source                        default
    file/line                     --
  late temperature                -40.00
    source                        default
    file/line                     --
```

```
0 objects with explicit PVT settings.


Top-level cell timing derates:
 data_cell derate early:   rise: 1.00    fall: 1.00
 clk_cell derate early:   rise: 1.00    fall: 1.00
 cell_check derate early:   rise: 1.00    fall: 1.00
 data_cell derate  late:   rise: 1.00    fall: 1.00
 clk_cell derate  late:   rise: 1.00    fall: 1.00
 cell_check derate  late:   rise: 1.00    fall: 1.00

Global net timing derates:
 net_data, early:   rise: 1.00    fall: 1.00
 net_data, late:   rise: 1.00    fall: 1.00
 net_clock, early:   rise: 1.00    fall: 1.00
 net_clock, late:   rise: 1.00    fall: 1.00
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Check corners after create Scenarios**

  Check corners information & PVT Settings (ff_n40c) default

```
--------------------------------------------------------------------------------
Corner ff_n40c
--------------------------------------------------------------------------------

Current: false  Default: false  Empty: true

Scenarios associated with this corner:

                                                  Leakage   Dynamic
Scenario            Mode            Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
-------------------------------------------------------------------------------------------------------------------------
func.ff_n40c        func            true    true   true  true      true      true      true     true     false    false
scan_capture.ff_n40c scan_capture   true    true   true  true      true      true      true     true     false    false
scan_shift.ff_n40c  scan_shift      true    true   true  true      true      true      true     true     false    false

Top-Level PVT Settings:
  early process label             (none)
  early process number            1.01
  early voltage                   1.16
    source                        default
    file/line                     --
  early temperature               -40.00
    source                        default
    file/line                     --

  late process label              (none)
  late process number             1.01
  late voltage                    1.16
    source                        default
    file/line                     --
  late temperature                -40.00
    source                        default
    file/line                     --
```

```
0 objects with explicit PVT settings.


Top-level cell timing derates:
 data_cell derate early:   rise: 1.00    fall: 1.00
 clk_cell derate early:   rise: 1.00    fall: 1.00
 cell_check derate early:   rise: 1.00    fall: 1.00
 data_cell derate  late:   rise: 1.00    fall: 1.00
 clk_cell derate  late:   rise: 1.00    fall: 1.00
 cell_check derate  late:   rise: 1.00    fall: 1.00

Global net timing derates:
 net_data, early:   rise: 1.00    fall: 1.00
 net_data, late:   rise: 1.00    fall: 1.00
 net_clock, early:   rise: 1.00    fall: 1.00
 net_clock, late:   rise: 1.00    fall: 1.00
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Check corners after create Scenarios**

Check corners information & PVT Settings (ss_n40c) default

```
--------------------------------------------------------------------------------
Corner ss_n40c
--------------------------------------------------------------------------------

Current: false  Default: false  Empty: true

Scenarios associated with this corner:

                                                 Leakage   Dynamic
Scenario            Mode            Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
--------------------------------------------------------------------------------------------------------------------------
func.ss_n40c        func            true    true   true  true      true      true      true     true     false    false
scan_capture.ss_n40c scan_capture   true    true   true  true      true      true      true     true     false    false
scan_shift.ss_n40c  scan_shift      true    true   true  true      true      true      true     true     false    false


Top-Level PVT Settings:
  early process label             (none)
  early process number            1.01
  early voltage                   1.16
    source                        default
    file/line                     --
  early temperature               -40.00
    source                        default
    file/line                     --

  late process label              (none)
  late process number             1.01
  late voltage                    1.16
    source                        default
    file/line                     --
  late temperature                -40.00
    source                        default
    file/line                     --
```

```
0 objects with explicit PVT settings.


Top-level cell timing derates:
 data_cell derate early:   rise: 1.00    fall: 1.00
 clk_cell derate early:   rise: 1.00    fall: 1.00
 cell_check derate early:   rise: 1.00    fall: 1.00
 data_cell derate  late:   rise: 1.00    fall: 1.00
 clk_cell derate  late:   rise: 1.00    fall: 1.00
 cell_check derate  late:   rise: 1.00    fall: 1.00

Global net timing derates:
 net_data, early:   rise: 1.00    fall: 1.00
 net_data, late:   rise: 1.00    fall: 1.00
 net_clock, early:   rise: 1.00    fall: 1.00
 net_clock, late:   rise: 1.00    fall: 1.00
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Check scenarios after create Scenarios**

  Check Scenarios are created correctly with the previously created modes and corners

```
icc2_shell> report_scenarios
****************************************
Report : scenario
Design : ORCA
Version: S-2021.06-SP5
Date   : Thu Sep  7 21:50:07 2023
****************************************

                                                                                                                                                  Leakage  Dynamic                                                  Cell  Signa
Name            Mode            Corner          Active  Setup  Hold  Power    Power    Max_tran  Max_cap  Min_cap  EM    EM
-------------------------------------------------------------------------------------------------------------------------------------------------------
func.ff_125c *  func            ff_125c         true    true   true  true     true     true      true     true     false false
func.ff_n40c *  func            ff_n40c         true    true   true  true     true     true      true     true     false false
func.ss_125c *  func            ss_125c         true    true   true  true     true     true      true     true     false false
func.ss_n40c *  func            ss_n40c         true    true   true  true     true     true      true     true     false false
scan_capture.ff_125c *
                scan_capture    ff_125c         true    true   true  true     true     true      true     true     false false
scan_capture.ff_n40c *
                scan_capture    ff_n40c         true    true   true  true     true     true      true     true     false false
scan_capture.ss_125c *
                scan_capture    ss_125c         true    true   true  true     true     true      true     true     false false
scan_capture.ss_n40c *
                scan_capture    ss_n40c         true    true   true  true     true     true      true     true     false false
scan_shift.ff_125c *
                scan_shift      ff_125c         true    true   true  true     true     true      true     true     false false
scan_shift.ff_n40c *
                scan_shift      ff_n40c         true    true   true  true     true     true      true     true     false false
scan_shift.ss_125c *
                scan_shift      ss_125c         true    true   true  true     true     true      true     true     false false
scan_shift.ss_n40c *
                scan_shift      ss_n40c         true    true   true  true     true     true      true     true     false false

1
```


SYNOPSYS
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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Set parasitic parameters after read TLU+ files**

  Sets parasitic parameters such as parasitic spec (TLU+) and temperature

```
icc2_shell> #set ff_n40c corner
icc2_shell> current_corner ff_n40c
{ff_n40c}
icc2_shell> set_parasitic_parameters -corners ff_n40c -library ORCA_${step}.nlib -late_spec minTLU -early_spec minTLU
-late_temperature -40 -early_temperature -40
1
icc2_shell>
icc2_shell> #set ss_n40c corner
icc2_shell> current_corner ss_n40c
{ss_n40c}
icc2_shell> set_parasitic_parameters -corners ss_n40c -library ORCA_${step}.nlib -late_spec maxTLU -early_spec maxTLU
-late_temperature -40 -early_temperature -40
1
icc2_shell>
icc2_shell> #set ff_125c corner
icc2_shell> current_corner ff_125c
{ff_125c}
icc2_shell> set_parasitic_parameters -corners ff_125c -library ORCA_${step}.nlib -late_spec minTLU -early_spec minTLU
-late_temperature 125 -early_temperature 125
1
icc2_shell>
icc2_shell> #set ss_125c corner
icc2_shell> current_corner ss_125c
{ss_125c}
icc2_shell> set_parasitic_parameters -corners ss_125c -library ORCA_${step}.nlib -late_spec maxTLU -early_spec maxTLU
-late_temperature 125 -early_temperature 125
1
```

set_parasitic_parameters option

- **-corners**: sets the parasitic parameter to the constraint corner
- **-library** : Specifies the library name that contains the specified parasitic specification (ORCA_placement.nlib)
- **-early_spec** : Specifies the early parasitic tech specification name (minTLU)
- **-late_spec** : Specifies the late parasitic tech specification name (maxTLU)
- **-early_temperature** : Specifies the early temperature value
- **-late_temperature** : Specifies the late temperature value

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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Set PVT value for all corners**
  - **set_process_number**

    Sets the process number for one or more corners, for part or all of the current design, or only for cells from specific reference libraries

    -number : Specifies the process number to be used for both early and late analysis

    -corners : Specifies the corners to which to apply the process number

  - **If corner is ff_n40c, you can check process number at *ff1p16vb40c.lib file**

    Ex) open **$WORK_DIR/2_BACKEND/1_pnr/libdir/LIB/saed32rvt_ff1p16vn40c.lib** & search nom_process

```
nom_voltage : 1.160000;
nom_temperature : -40.000000;
nom_process : 1.01;
default_wire_load_capacitance : 0.024051;
default_wire_load_resistance : 0.001860;
default_wire_load_area : 0.010000;
default_wire_load_mode : "enclosed";
default_threshold_voltage_group : "saed32cell_svt";
```

- **You can find the process number that matches the corner in the corner's lib file**

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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Set PVT value for all corners**
  - **set_temperatrue**

    Sets the operating temperature for one or more corners, for part or all of the current design

    -min : The early temperature, if different than the late temperature

    -corners : Specifies the corners to which to apply the temperature
  - **set_voltage**

    Applies an operating voltage to a list of supply nets, supply ports, cells or ports, or to the current design

    -object_list : Specifies the supply nets, supply ports, cells, or ports that have the operating voltages specified in this command

    -corners : Specifies the corners to which to apply the temperature

    ❖ **We will set the max and min top-level default primary voltage, specified in .lib file**

    Open $WORK_DIR/2_BACKEND/1_pnr/libdir/LIB/*.lib files & search voltage_map

    - **(VDD & VSS) for STD cells & Macros (ff* corners)**

      Ex) open saed32rvt_ff1p16nv40c.lib

```
simulator : "HSPICE -- C-2009.03-SP1 32-BIT (May 25 2009)";
voltage_map(VDD,1.160000);
voltage_map(VDDG,1.160000);
voltage_map(VSS,0.000000);
output_voltage (DC_0) {
        vol : 0.058;
        voh : 1.102;
        vomin : 0;
        vomax : VDD;
}
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **We will set the max and min top-level default primary voltage, specified in .lib file**

  Open $WORK_DIR/2_BACKEND/1_pnr/libdir/LIB/*.lib files & search voltage_map

  - **(VDD & VSS) for STD cells & Macros(ss* corners)**

    Ex) open saed32rvt_ff1p16nv40c.lib

```
simulator : "HSPICE -- C-2009.03-SP1 32-BIT (May 25 2009)";
voltage_map(VDD,0.950000);
voltage_map(VDDG,0.950000);
voltage_map(VSS,0.000000);
output_voltage (DC_0) {
        vol : 0.0475;
        voh : 0.9025;
        vomin : 0;
        vomax : VDD;
}
```

  - **(DVDD, AVDD) for PLL (ff* corners)**

    Ex) open saed32pll_ff1p16vn40c_2p75v.lib

```
simulator : "HSPICE -- C-2009.03-SP1 32-BIT (May 25 2009)";
voltage_map("DVDD",1.160000);
voltage_map("AVDD",2.750000);
voltage_map("VSS",0.000000);
```

  - **(DVDD, AVDD) for PLL (ff* corners)**

    Ex) open saed32pll_ff1p16vn40c_2p75v.lib

```
simulator : "HSPICE -- C-2009.03-SP1 32-BIT (May 25 2009)";
voltage_map("DVDD",0.950000);
voltage_map("AVDD",2.250000);
voltage_map("VSS",0.000000);
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **We will set the max and min top-level default primary voltage, specified in .lib file**

  Open $WORK_DIR/2_BACKEND/1_pnr/libdir/LIB/*.lib files & search voltage_map

  - **(AVSS, VDDIO, VSSIO) for IO PAD (ff* corners)**

    Ex) open saed32io_fc_ff1p16vn40c_2p75v.lib

```
simulator : "HSPICE -- C-2009.03-SP1 32-BIT (May 25 2009)";
voltage_map("VDD",1.1600000);
voltage_map("AVDD",1.1600000);
voltage_map("AVSS",0.000000);
voltage_map("VDDIO",2.7500000);
voltage_map("VSS",0.000000);
voltage_map("VSSIO",0.000000);
input_voltage (COREIN) {
        vil : 0.05800000;
        vih : 1.10200000;
        vimin : 0.000000;
        vimax : 1.1600000;
```

  - **(AVSS, VDDIO, VSSIO) for IO PAD (ss* corners)**

    Ex) open saed32io_fc_ss0p95vb40c_2p25v.lib

```
simulator : "HSPICE -- C-2009.03-SP1 32-BIT (May 25 2009)";
voltage_map("VDD",0.9500000);
voltage_map("AVDD",0.9500000);
voltage_map("AVSS",0.000000);
voltage_map("VDDIO",2.2500000);
voltage_map("VSS",0.000000);
voltage_map("VSSIO",0.000000);
input_voltage (COREIN) {
        vil : 0.047500000;
        vih : 0.902500000;
        vimin : 0.000000;
        vimax : 0.9500000;
}
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **ORCA max and min top-level default primary voltage**

<create_supply_net in upf file>

```
# supply net creation for domain TOP
create_supply_net VDD            -domain TOP
create_supply_net VSS            -domain TOP
create_supply_net VDDIO          -domain TOP
create_supply_net VSSIO          -domain TOP
create_supply_net AVDD1          -domain TOP
create_supply_net AVDD2          -domain TOP
create_supply_net DVDD1          -domain TOP
create_supply_net DVDD2          -domain TOP
create_supply_net AVSS           -domain TOP
```

- **Check Power net used for design in upf file**

<*ff corners>

Supply_net	Primary voltage
VDD	1.16
AVDD1	2.75
AVDD2	2.75
DVDD1	1.16
DVDD2	1.16
VDDIO	2.75
VSS	0.00
VSSIO	0.00
AVSS	0.00

<*ss corners>

Supply_net	Primary voltage
VDD	0.95
AVDD1	2.25
AVDD2	2.25
DVDD1	0.95
DVDD2	0.95
VDDIO	2.25
VSS	0.00
VSSIO	0.00
AVSS	0.00

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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Set PVT for ff_n40c corners**

```
icc2_shell> set_process_number 1.01 -corners ff_n40c
1
icc2_shell> set_temperature -40 -min -40 -corners ff_n40c
1
icc2_shell> set_voltage 1.16 -corner ff_n40c -object_list [get_supply_nets VDD]
1
icc2_shell> set_voltage 2.75 -corner ff_n40c -object_list [get_supply_nets AVDD1]
1
icc2_shell> set_voltage 2.75 -corner ff_n40c -object_list [get_supply_nets AVDD2]
1
icc2_shell> set_voltage 1.16 -corner ff_n40c -object_list [get_supply_nets DVDD1]
1
icc2_shell> set_voltage 1.16 -corner ff_n40c -object_list [get_supply_nets DVDD2]
1
icc2_shell> set_voltage 2.75 -corner ff_n40c -object_list [get_supply_nets VDDIO]
1
icc2_shell> set_voltage 0.00 -corner ff_n40c -object_list [get_supply_nets VSS]
1
icc2_shell> set_voltage 0.00 -corner ff_n40c -object_list [get_supply_nets VSSIO]
1
icc2_shell> set_voltage 0.00 -corner ff_n40c -object_list [get_supply_nets AVSS]
1
```

- **Set PVT for ss_n40c corners**

```
icc2_shell> set_process_number 0.99 -corners ss_n40c
1
icc2_shell> set_temperature -40 -min -40 -corners ss_n40c
1
icc2_shell> set_voltage 0.95 -corner ss_n40c -object_list [get_supply_nets VDD]
1
icc2_shell> set_voltage 2.25 -corner ss_n40c -object_list [get_supply_nets AVDD1]
1
icc2_shell> set_voltage 2.25 -corner ss_n40c -object_list [get_supply_nets AVDD2]
1
icc2_shell> set_voltage 0.95 -corner ss_n40c -object_list [get_supply_nets DVDD1]
1
icc2_shell> set_voltage 0.95 -corner ss_n40c -object_list [get_supply_nets DVDD2]
1
icc2_shell> set_voltage 2.25 -corner ss_n40c -object_list [get_supply_nets VDDIO]
1
icc2_shell> set_voltage 0.00 -corner ss_n40c -object_list [get_supply_nets VSS]
1
icc2_shell> set_voltage 0.00 -corner ss_n40c -object_list [get_supply_nets VSSIO]
1
icc2_shell> set_voltage 0.00 -corner ss_n40c -object_list [get_supply_nets AVSS]
1
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Set PVT for ff_125c corners**

```
icc2_shell> set_process_number 1.01 -corners ff_125c
1
icc2_shell> set_temperature 125 -min 125 -corners ff_125c
1
icc2_shell> set_voltage 1.16 -corner ff_125c -object_list [get_supply_nets VDD]
1
icc2_shell> set_voltage 2.75 -corner ff_125c -object_list [get_supply_nets AVDD1]
1
icc2_shell> set_voltage 2.75 -corner ff_125c -object_list [get_supply_nets AVDD2]
1
icc2_shell> set_voltage 1.16 -corner ff_125c -object_list [get_supply_nets DVDD1]
1
icc2_shell> set_voltage 1.16 -corner ff_125c -object_list [get_supply_nets DVDD2]
1
icc2_shell> set_voltage 2.75 -corner ff_125c -object_list [get_supply_nets VDDIO]
1
icc2_shell> set_voltage 0.00 -corner ff_125c -object_list [get_supply_nets VSS]
1
icc2_shell> set_voltage 0.00 -corner ff_125c -object_list [get_supply_nets VSSIO]
1
icc2_shell> set_voltage 0.00 -corner ff_125c -object_list [get_supply_nets AVSS]
1
```

- **Set PVT for ss_125c corners**

```
icc2_shell> set_process_number 0.99 -corners ss_125c
1
icc2_shell> set_temperature 125 -min 125 -corners ss_125c
1
icc2_shell> set_voltage 0.95 -corner ss_125c -object_list [get_supply_nets VDD]
1
icc2_shell> set_voltage 2.25 -corner ss_125c -object_list [get_supply_nets AVDD1]
1
icc2_shell> set_voltage 2.25 -corner ss_125c -object_list [get_supply_nets AVDD2]
1
icc2_shell> set_voltage 0.95 -corner ss_125c -object_list [get_supply_nets DVDD1]
1
icc2_shell> set_voltage 0.95 -corner ss_125c -object_list [get_supply_nets DVDD2]
1
icc2_shell> set_voltage 2.25 -corner ss_125c -object_list [get_supply_nets VDDIO]
1
icc2_shell> set_voltage 0.00 -corner ss_125c -object_list [get_supply_nets VSS]
1
icc2_shell> set_voltage 0.00 -corner ss_125c -object_list [get_supply_nets VSSIO]
1
icc2_shell> set_voltage 0.00 -corner ss_125c -object_list [get_supply_nets AVSS]
1
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **report_corners after set PVT values**
- **Corner ff_125c**

```
icc2_shell> report_corners
****************************************
Report : corner
Design : ORCA
Version: S-2021.06-SP5
Date   : Fri Sep  8 00:46:50 2023
****************************************

--------------------------------------------------------------------------------
Corner ff_125c
--------------------------------------------------------------------------------

Current: false  Default: false  Empty: false

Scenarios associated with this corner:

                                                    Leakage   Dynamic
Scenario            Mode            Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
-------------------------------------------------------------------------------------------------------------------------
func.ff_125c        func            true    true   true  true      true      true      true     true     false    false
scan_capture.ff_125c scan_capture   true    true   true  true      true      true      true     true     false    false
scan_shift.ff_125c  scan_shift      true    true   true  true      true      true      true     true     false    false


Top-Level PVT Settings:
  early process label             (none)
  early process number            1.01
  early voltage                   1.16
    source                        UPF
    file/line                     --
  early temperature               125.00
    source                        hierarchical setting
    file/line                     --

  late process label              (none)
  late process number             1.01
  late voltage                    1.16
    source                        UPF
    file/line                     --
  late temperature                125.00
    source                        hierarchical setting
    file/line                     --
```

```
0 objects with explicit PVT settings.


Top-level cell timing derates:
 data_cell derate early:   rise: 1.00    fall: 1.00
 clk_cell derate early:   rise: 1.00    fall: 1.00
 cell_check derate early:   rise: 1.00    fall: 1.00
 data_cell derate  late:   rise: 1.00    fall: 1.00
 clk_cell derate  late:   rise: 1.00    fall: 1.00
 cell_check derate  late:   rise: 1.00    fall: 1.00

Global net timing derates:
 net_data, early:   rise: 1.00    fall: 1.00
 net_data, late:   rise: 1.00    fall: 1.00
 net_clock, early:   rise: 1.00    fall: 1.00
 net_clock, late:   rise: 1.00    fall: 1.00
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **report_corners after set PVT values**
- **Corner ff_n40c**

```
--------------------------------------------------------------------------------
Corner ff_n40c
--------------------------------------------------------------------------------

Current: false  Default: false  Empty: false

Scenarios associated with this corner:

                                                  Leakage   Dynamic
Scenario            Mode            Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
--------------------------------------------------------------------------------------------------------------------------
func.ff_n40c        func            true    true   true  true      true      true      true     true     false    false
scan_capture.ff_n40c scan_capture   true    true   true  true      true      true      true     true     false    false
scan_shift.ff_n40c  scan_shift      true    true   true  true      true      true      true     true     false    false


Top-Level PVT Settings:
  early process label             (none)
  early process number            1.01
  early voltage                   1.16
    source                        UPF
    file/line                     --
  early temperature               -40.00
    source                        hierarchical setting
    file/line                     --

  late process label              (none)
  late process number             1.01
  late voltage                    1.16
    source                        UPF
    file/line                     --
  late temperature                -40.00
    source                        hierarchical setting
    file/line                     --
```

```
0 objects with explicit PVT settings.


Top-level cell timing derates:
 data_cell derate early:   rise: 1.00    fall: 1.00
 clk_cell derate early:   rise: 1.00    fall: 1.00
 cell_check derate early:   rise: 1.00    fall: 1.00
 data_cell derate  late:   rise: 1.00    fall: 1.00
 clk_cell derate  late:   rise: 1.00    fall: 1.00
 cell_check derate  late:   rise: 1.00    fall: 1.00

Global net timing derates:
 net_data, early:   rise: 1.00    fall: 1.00
 net_data, late:   rise: 1.00    fall: 1.00
 net_clock, early:   rise: 1.00    fall: 1.00
 net_clock, late:   rise: 1.00    fall: 1.00
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **report_corners after set PVT values**
- **Corner ss_125c**

```
--------------------------------------------------------------------------------
Corner ss_125c
--------------------------------------------------------------------------------

Current: true  Default: false  Empty: false

Scenarios associated with this corner:

                                                 Leakage   Dynamic
Scenario            Mode            Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
-------------------------------------------------------------------------------------------------------------------------
func.ss_125c        func            true    true   true  true      true      true      true     true     false    false
scan_capture.ss_125c scan_capture   true    true   true  true      true      true      true     true     false    false
scan_shift.ss_125c  scan_shift      true    true   true  true      true      true      true     true     false    false


Top-Level PVT Settings:
  early process label             (none)
  early process number            0.99
  early voltage                   0.95
    source                        UPF
    file/line                     --
  early temperature               125.00
    source                        hierarchical setting
    file/line                     --

  late process label              (none)
  late process number             0.99
  late voltage                    0.95
    source                        UPF
    file/line                     --
  late temperature                125.00
    source                        hierarchical setting
    file/line                     --
```

```
Top-Level PVT Settings:
  early process label             (none)
  early process number            0.99
  early voltage                   0.95
    source                        UPF
    file/line                     --
  early temperature               125.00
    source                        hierarchical setting
    file/line                     --

  late process label              (none)
  late process number             0.99
  late voltage                    0.95
    source                        UPF
    file/line                     --
  late temperature                125.00
    source                        hierarchical setting
    file/line                     --
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **report_corners after set PVT values**
- **Corner ss_n40c**

```
--------------------------------------------------------------------------------
Corner ss_n40c
--------------------------------------------------------------------------------

Current: false  Default: false  Empty: false

Scenarios associated with this corner:

                                                  Leakage   Dynamic
Scenario            Mode            Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
--------------------------------------------------------------------------------------------------------------------------
func.ss_n40c        func            true    true   true  true      true      true      true     true     false    false
scan_capture.ss_n40c scan_capture   true    true   true  true      true      true      true     true     false    false
scan_shift.ss_n40c  scan_shift      true    true   true  true      true      true      true     true     false    false


Top-Level PVT Settings:
  early process label             (none)
  early process number            0.99
  early voltage                   0.95
    source                        UPF
    file/line                     --
  early temperature               -40.00
    source                        hierarchical setting
    file/line                     --

  late process label              (none)
  late process number             0.99
  late voltage                    0.95
    source                        UPF
    file/line                     --
  late temperature                -40.00
    source                        hierarchical setting
    file/line                     --
```

```
0 objects with explicit PVT settings.


Top-level cell timing derates:
 data_cell derate early:   rise: 1.00    fall: 1.00
 clk_cell derate early:   rise: 1.00    fall: 1.00
 cell_check derate early:   rise: 1.00    fall: 1.00
 data_cell derate  late:   rise: 1.00    fall: 1.00
 clk_cell derate  late:   rise: 1.00    fall: 1.00
 cell_check derate  late:   rise: 1.00    fall: 1.00

Global net timing derates:
 net_data, early:   rise: 1.00    fall: 1.00
 net_data, late:   rise: 1.00    fall: 1.00
 net_clock, early:   rise: 1.00    fall: 1.00
 net_clock, late:   rise: 1.00    fall: 1.00
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **set_scenario_status (**Configures analysis settings for scenarios**)**
  - **Set_scenario_status for all *ss scenarios**

```
icc2_shell> set_scenario_status {*.ss*} -setup true -hold true \
?     -active true -max_transition true \
?     -max_capacitance true \
?     -min_capacitance false \
?             -leakage_power true \
?             -dynamic_power true
Scenario func.ss_125c (mode func corner ss_125c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario func.ss_n40c (mode func corner ss_n40c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario scan_capture.ss_125c (mode scan_capture corner ss_125c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario scan_capture.ss_n40c (mode scan_capture corner ss_n40c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario scan_shift.ss_125c (mode scan_shift corner ss_125c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario scan_shift.ss_n40c (mode scan_shift corner ss_n40c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
1
```

  - **Set_scenario_status for all *ff scenarios**

```
set_scenario_status {*.ff*} -setup true -hold true \
?                             -active true -max_transition true \
?             -max_capacitance true \
?     -min_capacitance false \
?      -leakage_power true \
?                     -dynamic_power true
Scenario func.ff_125c (mode func corner ff_125c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario func.ff_n40c (mode func corner ff_n40c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario scan_capture.ff_125c (mode scan_capture corner ff_125c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario scan_capture.ff_n40c (mode scan_capture corner ff_n40c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario scan_shift.ff_125c (mode scan_shift corner ff_125c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario scan_shift.ff_n40c (mode scan_shift corner ff_n40c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
1
```


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Placement : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/05_mcmm.tcl

- **Report_scenarios after set_scenarios_status**

  Check the list to analyze in the placement step

```
icc2_shell> report_scenarios
****************************************
Report : scenario
Design : ORCA
Version: S-2021.06-SP5
Date   : Sun Sep 10 17:43:43 2023
****************************************

                                                                                                                  Leakage  Dynamic                                Cell  Signal
Name            Mode            Corner          Active  Setup  Hold  Power    Power    Max_tran  Max_cap  Min_cap  EM    EM
-------------------------------------------------------------------------------------------------------------------------------
func.ff_125c *  func            ff_125c         true    true   true  true     true     true      true     false    false false
func.ff_n40c *  func            ff_n40c         true    true   true  true     true     true      true     false    false false
func.ss_125c *  func            ss_125c         true    true   true  true     true     true      true     false    false false
func.ss_n40c *  func            ss_n40c         true    true   true  true     true     true      true     false    false false
scan_capture.ff_125c *
                scan_capture    ff_125c         true    true   true  true     true     true      true     false    false false
scan_capture.ff_n40c *
                scan_capture    ff_n40c         true    true   true  true     true     true      true     false    false false
scan_capture.ss_125c *
                scan_capture    ss_125c         true    true   true  true     true     true      true     false    false false
scan_capture.ss_n40c *
                scan_capture    ss_n40c         true    true   true  true     true     true      true     false    false false
scan_shift.ff_125c *
                scan_shift      ff_125c         true    true   true  true     true     true      true     false    false false
scan_shift.ff_n40c *
                scan_shift      ff_n40c         true    true   true  true     true     true      true     false    false false
scan_shift.ss_125c *
                scan_shift      ss_125c         true    true   true  true     true     true      true     false    false false
scan_shift.ss_n40c *
                scan_shift      ss_n40c         true    true   true  true     true     true      true     false    false false

1
```


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Placement : Use ICC2 (IC Compiler II tool)

7. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/06_read_sdc.tcl

- **read_sdc (scan_shift, func, scan_capture)**

```
icc2_shell> current_mode scan_shift
{scan_shift}
icc2_shell> read_sdc $data(shift_sdc)
Information: Loading SDC version 2.1 file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_shift.sdc' (FILE-007)
Warning: Redefining clock 'pclk_wft1'.
        Previously defined at: /slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_shift.sdc, line 22 (UIC-034)
Warning: Redefining clock 'sys_clk_wft1'.
        Previously defined at: /slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_shift.sdc, line 23 (UIC-034)
Warning: Redefining clock 'sdr_clk_wft1'.
        Previously defined at: /slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_shift.sdc, line 24 (UIC-034)
Warning: Creating virtual clock named 'forcePI_wft1' with no sources. (UIC-021)
Warning: Redefining clock 'forcePI_wft1'.
        Previously defined at: /slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_shift.sdc, line 52 (UIC-034)
Warning: Creating virtual clock named 'measurePO_wft1' with no sources. (UIC-021)
Warning: Redefining clock 'measurePO_wft1'.
        Previously defined at: /slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_shift.sdc, line 53 (UIC-034)
1
icc2_shell> current_mode func
{func}
icc2_shell> read_sdc $data(func_sdc)
Information: Loading SDC version 2.1 file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_func.sdc' (FILE-007)
INFO: set_units command temporarily sets input units and restores back when read_sdc is done.
Warning: Redefining clock 'pclk'.
        Previously defined at: /slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_func.sdc, line 105 (UIC-034)
Warning: Redefining clock 'sys_clk'.
        Previously defined at: /slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_func.sdc, line 107 (UIC-034)
Warning: Redefining clock 'sdr_clk'.
        Previously defined at: /slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_func.sdc, line 109 (UIC-034)
Warning: Creating a clock on internal pin 'I_CLOCK_GEN/I_PLL_PCI/CLK_1X'. (UIC-019)
Warning: Redefining clock 'PCI_CLK'.
        Previously defined at: /slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_func.sdc, line 111 (UIC-034)
Warning: Creating a clock on internal pin 'I_CLOCK_GEN/I_PLL_SD/CLK_1X'. (UIC-019)
Warning: Redefining clock 'SDRAM_CLK'.
        Previously defined at: /slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_func.sdc, line 113 (UIC-034)
Warning: Creating 'clock' on a hierarchical pin 'I_CLOCK_GEN/I_CLKMUL/CLK_1X'. (UIC-018)
Warning: Creating 'clock' on a hierarchical pin 'I_CLOCK_GEN/I_CLKMUL/CLK_2X'. (UIC-018)
1
icc2_shell> current_mode scan_capture
{scan_capture}
icc2_shell> read_sdc $data(capture_sdc)
Information: Loading SDC version 2.1 file '/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/libdir/SDC/ORCA_capture.sdc' (FILE-007)
Warning: Creating virtual clock named 'forcePI_wft1' with no sources. (UIC-021)
Warning: Creating virtual clock named 'measurePO_wft1' with no sources. (UIC-021)
1
```


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Placement : Use ICC2 (IC Compiler II tool)

8. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/07_set_tiecell_and_ideal_network.tcl

- **Set TIE cell used in optimization**
  - **set_lib_cell_purpose (specifies valid purposes for library cells)**

-include *include_list*

Specifies the purposes for which the library cells specified in the **lib_cell_list** argument can be used. Valid values are

- **cts** for clock tree synthesis
- **optimization** and **hold** delay, electrical design rule and power optimization, as well as hold optimization when opt.common.honor_lib_cell_purpose false (default)
- **hold** for hold fixing
- **power** for power optimization
- **all** for no restrictions on inserting the specified cells
- **none** to prevent insertion of the specified cells for any purpose

This option is mutually exclusive with the **-exclude** option; you must specify one of these options.

```
icc2_shell> set_lib_cell_purpose -include optimization [get_lib_cells */TIE*]
Information: The design specific attribute override for lib_cell 'saed32hvt:TIEH_HVT.timing' is set in the current block 'ORCA', because the actual library setting may not be ove
rwritten. (ATTR-12)
Information: The design specific attribute override for lib_cell 'saed32hvt:TIEL_HVT.timing' is set in the current block 'ORCA', because the actual library setting may not be ove
rwritten. (ATTR-12)
Information: The design specific attribute override for lib_cell 'saed32lvt:TIEH_LVT.timing' is set in the current block 'ORCA', because the actual library setting may not be ove
rwritten. (ATTR-12)
Information: The design specific attribute override for lib_cell 'saed32lvt:TIEL_LVT.timing' is set in the current block 'ORCA', because the actual library setting may not be ove
rwritten. (ATTR-12)
Information: The design specific attribute override for lib_cell 'saed32rvt:TIEH_RVT.timing' is set in the current block 'ORCA', because the actual library setting may not be ove
rwritten. (ATTR-12)
Information: The design specific attribute override for lib_cell 'saed32rvt:TIEL_RVT.timing' is set in the current block 'ORCA', because the actual library setting may not be ove
rwritten. (ATTR-12)
1
icc2_shell> set_dont_touch [get_lib_cells */TIE*] false
1
```

The dont_touch attribute prevents modification or replacement of these objects during optimization

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Placement : Use ICC2 (IC Compiler II tool)

8. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/07_set_tiecell_and_ideal_network.tcl

- **remove_propagated_clocks & set_ideal_network (func, scan_shift, scan_capture)**

```
icc2_shell> set mode "func scan_shift scan_capture"
func scan_shift scan_capture
icc2_shell>
icc2_shell> foreach MODE $mode {
? current_mode $MODE
?
? echo "current mode is $MODE"
? switch $MODE {
? func {
? remove_propagated_clocks [all_clocks]
? set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
? remove_ideal_network $clock_sink
? set_ideal_network [get_pins $clock_sink]
? set_ideal_network [get_nets buf_sdram_clk]
? }
? scan_shift {
? remove_propagated_clocks [all_clocks]
? set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
? remove_ideal_network $clock_sink
? set_ideal_network [get_pins $clock_sink]
? set_ideal_network [get_nets buf_sdram_clk]
? }
? scan_capture {
? remove_propagated_clocks [all_clocks]
? set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
? remove_ideal_network $clock_sink
? set_ideal_network [get_pins $clock_sink]
? set_ideal_network [get_nets buf_sdram_clk]
? }
? }
? }
current mode is func
current mode is scan_shift
Warning: Virtual clock 'forcePI_wft1' cannot be made propagated. (UIC-025)
Warning: Virtual clock 'measurePO_wft1' cannot be made propagated. (UIC-025)
current mode is scan_capture
Warning: Virtual clock 'forcePI_wft1' cannot be made propagated. (UIC-025)
Warning: Virtual clock 'measurePO_wft1' cannot be made propagated. (UIC-025)
```

You can check ideal network to use **report_ideal_network** command

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Placement : Use ICC2 (IC Compiler II tool)

9. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/08_set_false_path_and_app_options.tcl

- **set false path about scan_en at "func" mode**
  - **Specify as false path because the constraint value in func.sdc is 0**

```
set_case_analysis 0 [get_ports scan_en]
```

```
icc2_shell> current_mode "func"
{func}
icc2_shell> set_false_path -from scan_en -to [get_clocks "PCI* SD* SYS*"]
1
```

- **Report high fanout**

```
icc2_shell> report_net_fanout -high_fanout
****************************************
Report : net_fanout
         -high_fanout
Design : ORCA
Version: S-2021.06-SP5
Date   : Sun Sep 10 19:34:10 2023
****************************************

NetName                 Fanout Driver
-------                 ------ ------
buf_pclk                  1090 I_CLK_SOURCE_PCLK/Y
buf_sdram_clk             1197 DFTC_70/Y
scan_en_guide_inv         2917 scan_en_HFS/Y
1
```

This net should be HFS(High Fanout Synthesis) during place_opt

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Placement : Use ICC2 (IC Compiler II tool)

9. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/08_set_false_path_and_app_options.tcl

- **set false path about DFTC_70/Y pin (because this Driver connected clk net)**

```
icc2_shell> current_mode "func"
{func}
icc2_shell> set_false_path  -through [get_pins DFTC_70/Y]
1
```

- **Source set_disable_timing.tcl**
  - **set_disable_timing (Disables timing through the specified cells, pins, ports, or timing arcs in the current_mode)**
    - **Timing ARC**

      Timing Arc represents the timing relationship between 2 pins of any element or Block or any boundaries

      (Basically it represents the timing characteristic of the element or block or boundaries)

```
icc2_shell> source -e ./user_script/set_disable_timing.tcl
#set current_scenarios
current_scenario "func.ss_125c"
#set_disable_timing SDRAM_DQ_Bus_1__sdram_DQ_iopad ~ SDRAM_DQ_Bus_15__sdram_DQ_iopad from EN pin to PADIO pin
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_1__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_1__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_2__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_2__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_3__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_3__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_4__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_4__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_5__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_5__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_6__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_6__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_7__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_7__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_8__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_8__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_9__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_9__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_10__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_10__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_11__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_11__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_12__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_12__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_13__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_13__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_14__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_14__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_15__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_15__sdram_DQ_iopad/PADIO } ]
1
```


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Placement : Use ICC2 (IC Compiler II tool)

9. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/08_set_false_path_and_app_options.tcl

- **set_app_options for placement**

```
icc2_shell> set_app_options -list {opt.tie_cell.max_fanout {8}}
opt.tie_cell.max_fanout 8
icc2_shell> set_app_options -list {cts.common.max_fanout {16}}
cts.common.max_fanout 16
icc2_shell> set_app_options -list {opt.common.max_fanout {16}}
opt.common.max_fanout 16
icc2_shell> set_app_options -list {place.coarse.congestion_layer_aware {true}}
place.coarse.congestion_layer_aware true
icc2_shell> set_app_options -list {place.coarse.continue_on_missing_scandef {true}}
place.coarse.continue_on_missing_scandef true
icc2_shell> set_app_options -list {opt.common.drc_mode_buffering {true}}
opt.common.drc_mode_buffering true
icc2_shell> set_app_options -list {place_opt.initial_drc.global_route_based {1}}
place_opt.initial_drc.global_route_based 1
icc2_shell> set_app_options -list {place_opt.initial_place.buffering_aware {true}}
place_opt.initial_place.buffering_aware true
icc2_shell>
icc2_shell> set_app_options -list {place.coarse.pin_density_aware {true}}
place.coarse.pin_density_aware true
icc2_shell> set_app_options -list {place_opt.congestion.effort {high}}
Warning: application option <place_opt.congestion.effort> is deprecated and scheduled for removal in a future release. See the man page for this  application option for migration
 assistance. (NDMUI-441)
place_opt.place.congestion_effort high
icc2_shell> set_app_options -list {place_opt.final_place.effort {high}}
place_opt.final_place.effort high
```

10. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/09_place_opt_and_refine_opt.tcl

- **Run place_opt**

```
icc2_shell> sec_StartTimer place_opt
1
icc2_shell>
icc2_shell> #place_opt
icc2_shell> place_opt
icc2_shell> sec_ReportResourceUtil place_opt
```


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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Check placement after place_opt**
  - **report_utilization about Core area after placement**

```
icc2_shell> report_utilization -region {{{425.5430 425.8460} {1014.4590 1014.1520}}}
****************************************
Report : report_utilization
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Sep 12 00:49:59 2023
****************************************
Utilization Ratio:                      0.2638
Utilization options:
 - Area calculation based on:           site_row in region { {425.5430 425.8460} {425.5430 1014.1520} {1014.4590 1014.1520} {101
4.4590 425.8460} } of block ORCA
 - Categories of objects excluded:      hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                             346462.8163
Total Capacity Area:                    205744.1031
Total Area of cells:                    54284.7206
Area of excluded objects:
 - hard_macros         :                130644.7564
 - macro_keepouts      :                3008.8108
 - soft_macros         :                0.0000
 - io_cells            :                0.0000
 - hard_blockages      :                141527.9649

Utilization of site-rows with:
 - Site 'unit':                         0.2638

0.2638
```

  - **report_congestion after place_opt**

```
****************************************
Report : congestion
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Sep 12 00:46:20 2023
****************************************

Layer     |    overflow     |              # GRCs has
Name      |  total  |  max  | overflow (%)      | max overflow
------------------------------------------------------------
Both Dirs |     176 |     2 |     175  ( 0.01%) |       1
H routing |     176 |     2 |     175  ( 0.02%) |       1
V routing |       0 |     0 |       0  ( 0.00%) |       0

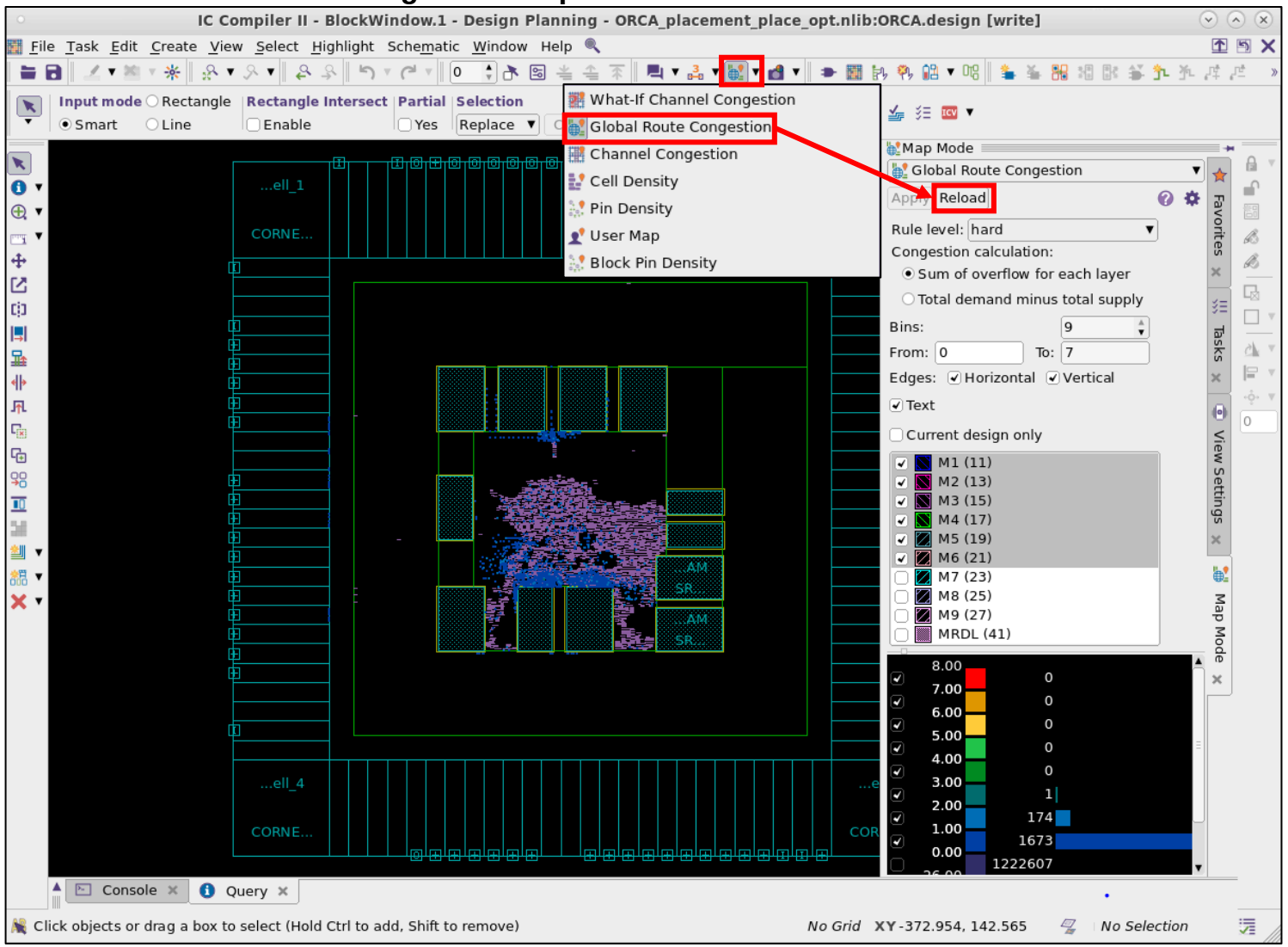
1
```


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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Check Global Route Congestion map**



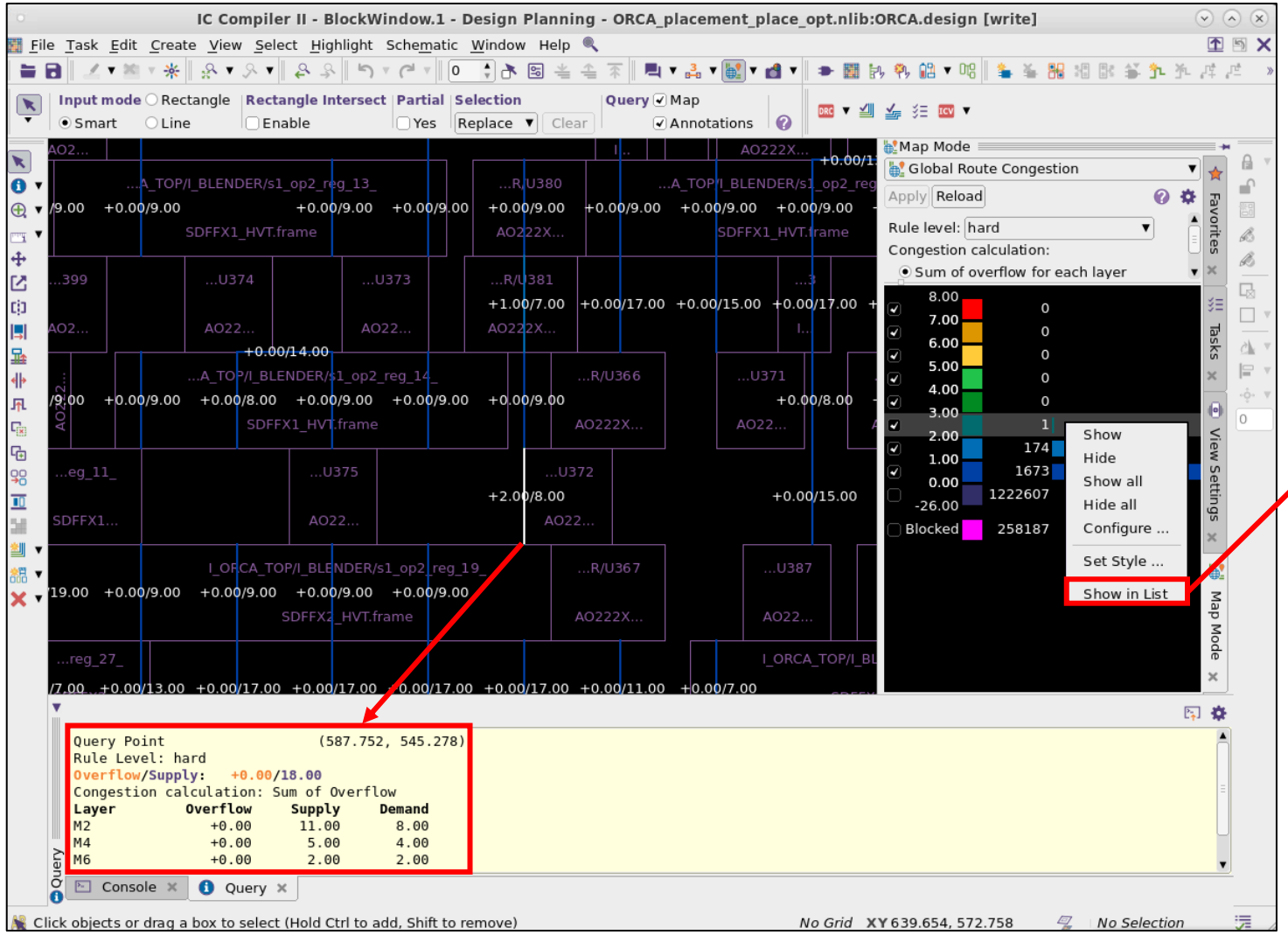
- The tool supports the Global Route Congestion map to highlight areas of the design which contain a high demand for global routes for the specified layers
- Congested areas might lead to problems during detail route step
- To generate the global route congestion debug map, click the Reload button (After completion, the tool displays the congestion map)

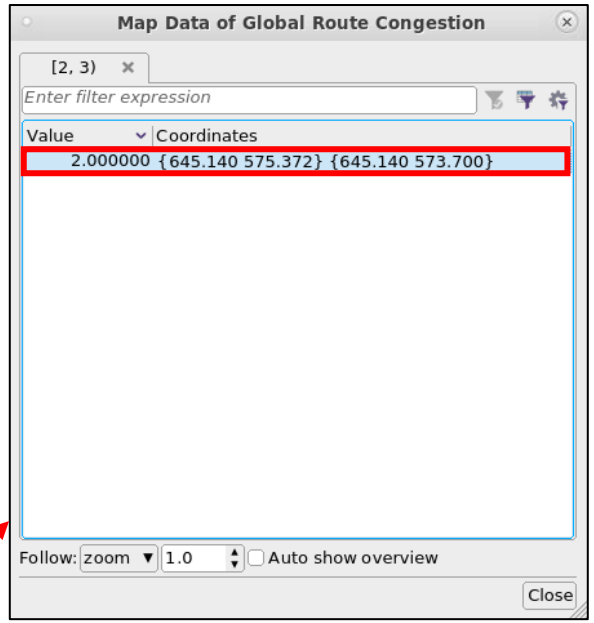
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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Check Global Route Congestion map**





- Click Right button on mouse at "sum of overflow for each layer"
- Click "show in List" & click value of Map Data of Global Route Congestion window
- You can show & analyze congestion about select value at location

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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Check placement after place_opt**
  - **report_timing after placement**

```
icc2_shell> report_timing -significant_digits 5
****************************************
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -report_by design
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Sep 12 23:10:06 2023
****************************************

  Startpoint: I_ORCA_TOP/I_RISC_CORE/Oprnd_A_reg_1_ (rising edge-triggered flip-flop clocked by sys_clk)
  Endpoint: I_ORCA_TOP/I_RISC_CORE/Neg_Flag_reg (rising edge-triggered flip-flop clocked by SYS_2x_CLK)
  Mode: func
  Corner: ss_125c
  Scenario: func.ss_125c
  Path Group: SYS_2x_CLK
  Path Type: max

  Point                                                 Incr         Path
  ---------------------------------------------------------------------------
  clock sys_clk (rise edge)                          0.00000      0.00000
  clock network delay (ideal)                       -0.03926     -0.03926

  I_ORCA_TOP/I_RISC_CORE/Oprnd_A_reg_1_/CLK (SDFFX1_RVT)
                                                     0.00000     -0.03926 r
  I_ORCA_TOP/I_RISC_CORE/Oprnd_A_reg_1_/Q (SDFFX1_RVT)
                                                     0.14971      0.11045 r
  I_ORCA_TOP/I_RISC_CORE/HFSBUF_1353_1843/Y (NBUFFX4_RVT)
                                                     0.07641      0.18686 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_1/C1 (HADDX1_RVT)      0.07551      0.26237 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_2/C1 (HADDX1_RVT)      0.06662      0.32899 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_3/C1 (HADDX1_RVT)      0.06773      0.39672 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_4/C1 (HADDX1_RVT)      0.06823      0.46495 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_5/C1 (HADDX1_RVT)      0.06721      0.53217 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_6/C1 (HADDX1_RVT)      0.07137      0.60354 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_7/C1 (HADDX1_RVT)      0.06857      0.67211 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_8/C1 (HADDX1_RVT)      0.06605      0.73816 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_9/C1 (HADDX1_RVT)      0.06456      0.80272 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_10/C1 (HADDX1_RVT)     0.06689      0.86961 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_11/C1 (HADDX1_RVT)     0.06670      0.93631 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_12/C1 (HADDX1_RVT)     0.06641      1.00273 r
  I_ORCA_TOP/I_RISC_CORE/U1_1_13/C1 (HADDX1_RVT)     0.06727      1.07000 r
```

```
I_ORCA_TOP/I_RISC_CORE/U1_1_14/C1 (HADDX1_RVT)     0.06735      1.13735 r
I_ORCA_TOP/I_RISC_CORE/U1103/Y (XOR2X1_RVT)        0.11547      1.25282 f
I_ORCA_TOP/I_RISC_CORE/U180/Y (AO222X1_RVT)        0.07364      1.32646 f
I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_1_2993/Y (OR4X1_RVT)
                                                   0.11463      1.44109 f
I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_4_4500/Y (INVX0_HVT)
                                                   0.02745      1.46854 r
I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_2_4503/Y (AND4X1_LVT)
                                                   0.05255      1.52109 r
I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_1_4502/Y (AND3X1_LVT)
                                                   0.04721      1.56829 r
I_ORCA_TOP/I_RISC_CORE/U9/Y (OR2X1_LVT)            0.04942      1.61771 r
I_ORCA_TOP/I_RISC_CORE/U31/Y (AO22X1_LVT)          0.04984      1.66755 r
I_ORCA_TOP/I_RISC_CORE/Neg_Flag_reg/D (SDFFARX1_RVT)
                                                   0.00002      1.66757 r
data arrival time                                               1.66757

clock SYS_2x_CLK (rise edge)                       2.00000      2.00000
clock network delay (ideal)                        0.19629      2.19629
I_ORCA_TOP/I_RISC_CORE/Neg_Flag_reg/CLK (SDFFARX1_RVT)
                                                   0.00000      2.19629 r
clock uncertainty                                 -0.40000      1.79629
library setup time                                -0.12873      1.66756
data required time                                              1.66755
---------------------------------------------------------------------------
data required time                                              1.66755
data arrival time                                              -1.66757
---------------------------------------------------------------------------
slack (VIOLATED)                                               -0.00002
```

How to Fix setup violation

- Add buffer or repeaters on long signal nets for reducing net delay
- Cell swapping for higher driving strength
- Cell resizing to a larger size for faster delay

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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Report placement after place_opt**

```
icc2_shell> update_timing -full
Information: Starting 'update_timing' (FLW-8000)
Information: Time: 2023-09-18 16:13:50 / Session: 0.04 hr / Command: 0.00 hr / Memory: 787 MB (FLW-8100)
Information: The net parasitics of block ORCA are cleared. (TIM-123)
Information: Corner ss_n40c: no PVT mismatches. (PVT-032)
Information: Corner ff_125c: no PVT mismatches. (PVT-032)
Information: Corner ss_125c: no PVT mismatches. (PVT-032)
Information: Corner ff_n40c: no PVT mismatches. (PVT-032)
```

- **Source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/user_script/placement_rpt.tcl**

```
icc2_shell> source ./user_script/placement_rpt.tcl
```

- **Go to the Placement reports directory**

```
  < user@linux placement >> pwd
/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/04_placement/reports/placement
  < user@linux placement >> ls
placement.congestion.rpt                  timing_report_max_placement_scan_capture.ff_125c  timing_report_min_placement_func.ss_n40c
placement.fanout.rpt                      timing_report_max_placement_scan_capture.ff_n40c  timing_report_min_placement_scan_capture.ff_125c
placement.legality.check.rpt              timing_report_max_placement_scan_capture.ss_125c  timing_report_min_placement_scan_capture.ff_n40c
placement.max_tran.rpt                    timing_report_max_placement_scan_capture.ss_n40c  timing_report_min_placement_scan_capture.ss_125c
placement.qor.snap.rpt                    timing_report_max_placement_scan_shift.ff_125c    timing_report_min_placement_scan_capture.ss_n40c
placement.runtime.rpt                     timing_report_max_placement_scan_shift.ff_n40c    timing_report_min_placement_scan_shift.ff_125c
placement.util.rpt                        timing_report_max_placement_scan_shift.ss_125c    timing_report_min_placement_scan_shift.ff_n40c
timing_report_max_placement_func.ff_125c  timing_report_max_placement_scan_shift.ss_n40c    timing_report_min_placement_scan_shift.ss_125c
timing_report_max_placement_func.ff_n40c  timing_report_min_placement_func.ff_125c          timing_report_min_placement_scan_shift.ss_n40c
timing_report_max_placement_func.ss_125c  timing_report_min_placement_func.ff_n40c
timing_report_max_placement_func.ss_n40c  timing_report_min_placement_func.ss_125c
```


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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Check reports**
  - **Check placement.max_tran.rpt**

```
****************************************
Report : constraint
         -all_violators
         -max_transition
Design : ORCA
Version: S-2021.06-SP5
Date   : Mon Sep 18 16:22:27 2023
****************************************


  Mode: func Corner: ff_n40c
  Scenario: func.ff_n40c
 --------------------------------------------------------------------------------
  Number of max_transition violation(s): 0

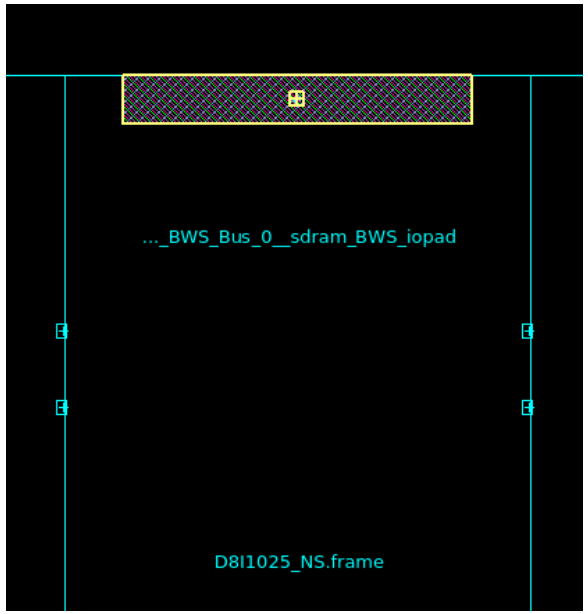
  Mode: scan_capture Corner: ff_n40c
  Scenario: scan_capture.ff_n40c
 --------------------------------------------------------------------------------
  Number of max_transition violation(s): 0

  Mode: scan_shift Corner: ff_n40c
  Scenario: scan_shift.ff_n40c
 --------------------------------------------------------------------------------
  Number of max_transition violation(s): 0

  Mode: func Corner: ss_125c
  Scenario: func.ss_125c
 --------------------------------------------------------------------------------
  Number of max_transition violation(s): 0

  Mode: scan_capture Corner: ss_125c
  Scenario: scan_capture.ss_125c
  max_transition
                            Required        Actual
  Net                      Transition     Transition        Slack  Violation
 -------------------------------------------------------------------------------
  sd_BWS[0]                    0.80           0.90          -0.10  (VIOLATED)
    PIN : SDRAM_BWS_Bus_0__sdram_BWS_iopad/PADIO
                               0.80           0.90          -0.10  (VIOLATED)
    PORT: sd_BWS[0]            0.80           0.90          -0.10  (VIOLATED)

  sd_A[9]                      0.80           0.90          -0.10  (VIOLATED)
    PIN : SDRAM_A_Bus_9__sdram_A_iopad/PADIO
                               0.80           0.90          -0.10  (VIOLATED)
    PORT: sd_A[9]              0.80           0.90          -0.10  (VIOLATED)
```



Violation that occurs between the virtual port and the pin of the PAD, so this violation can be ignored

SYNOPSYS
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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Check reports**
  - **Check placement.fanout.rpt**

```
****************************************
Report : net_fanout
         -high_fanout
Design : ORCA
Version: S-2021.06-SP5
Date   : Mon Sep 18 16:54:47 2023
****************************************

NetName                 Fanout Driver
-------                 ------ ------
buf_pclk                  1090 I_CLK_SOURCE_PCLK/Y
buf_sdram_clk             1197 DFTC_70/Y
1
```

buf_pclk, buf_sdram_clk are clk paths, so HFS (High Fanout Synthesis) proceeds in the CTS step

  - **Check placement.runtime.rpt (check runtime)**

```
copy_lib START              2023-09-06_22-44
copy_lib                    2023-09-06_22-44        00d:00h:00m:03s     00d:00h:00m:06s     527108
place_opt START             2023-09-06_22-44
place_opt                   2023-09-06_23-05        00d:00h:19m:18s     00d:00h:20m:14s     2192420
```


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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Check reports**
  - **Check placement.qor.snap.rpt**

```
****************************************
Report : qor
         -summary
Design : ORCA
Version: S-2021.06-SP5
Date   : Mon Sep 18 16:54:45 2023
****************************************

Timing
--------------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
--------------------------------------------------------------------------------
func.ff_125c       (Setup)          1.27560         0.00000               0
func.ff_n40c       (Setup)          1.46994         0.00000               0
func.ss_125c       (Setup)         -0.00006        -0.00006               1
func.ss_n40c       (Setup)         -0.00024        -0.00024               1
scan_capture.ff_125c (Setup)        9.65215         0.00000               0
scan_capture.ff_n40c (Setup)        9.68642         0.00000               0
scan_capture.ss_125c (Setup)        9.11218         0.00000               0
scan_capture.ss_n40c (Setup)        8.90735         0.00000               0
scan_shift.ff_125c (Setup)          9.76383         0.00000               0
scan_shift.ff_n40c (Setup)          9.79757         0.00000               0
scan_shift.ss_125c (Setup)          9.25449         0.00000               0
scan_shift.ss_n40c (Setup)          9.02169         0.00000               0
Design             (Setup)         -0.00024        -0.00030               2

func.ff_125c       (Hold)          -0.07335        -0.16560               3
func.ff_n40c       (Hold)          -0.09901        -0.18967               3
func.ss_125c       (Hold)          -0.02250        -0.04501               2
func.ss_n40c       (Hold)          -0.01581        -0.03163               2
scan_capture.ff_125c (Hold)         0.01718         0.00000               0
scan_capture.ff_n40c (Hold)         0.01777         0.00000               0
scan_capture.ss_125c (Hold)         0.19622         0.00000               0
scan_capture.ss_n40c (Hold)         0.22185         0.00000               0
scan_shift.ff_125c (Hold)          -0.05417        -0.30496              17
scan_shift.ff_n40c (Hold)          -0.04094        -0.16328              12
scan_shift.ss_125c (Hold)           0.04517         0.00000               0
scan_shift.ss_n40c (Hold)           0.05002         0.00000               0
Design             (Hold)          -0.09901        -0.49621              20
--------------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------------
Cell Area (netlist):                          750892.52290
Cell Area (netlist and physical only):        1174892.52300
Nets with DRC Violations:       22
1
```

Check setup summary at placement step

You can check also Cell Area & Nets with DRC violation (ex) max transition)

SYNOPSYS
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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Check reports**
  - **Check placement.legality.check.rpt**

```
****************************************
  Report : Legality
****************************************

VIOLATIONS BY CATEGORY:
   MOVABLE  APP-FIXED USER-FIXED  DESCRIPTION
         0          0          0  Two objects overlap.
         0          0          0  A cell violates a pnet.
         0          0          0  A cell is illegal at a site.
         0          0          0  A cell is not aligned with a site.
         0          0          0  A cell has an illegal orientation.
         0          0          0  A cell spacing rule is violated.
         0          0          0  A layer rule is violated.
         0          0          0  A cell is in the wrong region.
         0          0          0  Two cells violate cts margins.
         0          0          0  Two cells violate coloring.

         0          0          0  TOTAL

TOTAL 0 Violations.
```

The check_legality command checks the legality of the current placement and output a report of violation statistics (ex. Overlap, site, spacing, orientation)

```
VIOLATIONS BY SUBCATEGORY:
     MOVABLE  APP-FIXED USER-FIXED  DESCRIPTION

         0          0          0    Two objects overlap.
           0          0          0    Two cells overlap.
           0          0          0    Two cells have overlapping keepout margins.
           0          0          0    A cell overlaps a blockage.
           0          0          0    A cell keepout margin overlaps a blockage.

         0          0          0    A cell violates a pnet.

         0          0          0    A cell is illegal at a site.
           0          0          0    A cell violates pin-track alignment rules.
           0          0          0    A cell is illegal at a site.
           0          0          0    A cell violates legal index rule.
           0          0          0    A cell has the wrong variant for its location.

         0          0          0    A cell is not aligned with a site.
           0          0          0    A cell is not aligned with the base site.
           0          0          0    A cell is not aligned with an overlaid site.

         0          0          0    A cell has an illegal orientation.

         0          0          0    A cell spacing rule is violated.
           0          0          0    A spacing rule is violated in a row.
           0          0          0    A spacing rule is violated between adjacent rows.
           0          0          0    A cell violates vertical abutment rule.
           0          0          0    A cell violates metal spacing rule.

         0          0          0    A layer rule is violated.
           0          0          0    A layer VTH rule is violated.
           0          0          0    A layer OD rule is violated.
           0          0          0    A layer OD max-width rule is violated.
           0          0          0    A layer ALL_OD corner rule is violated.
           0          0          0    A layer max-vertical-length rule is violated.
           0          0          0    A layer TPO rule is violated.
           0          0          0    Filler cell insertion cannot satisfy layer rules.

         0          0          0    A cell is in the wrong region.
           0          0          0    A cell is outside its hard bound.
           0          0          0    A cell is in the wrong voltage area.
           0          0          0    A cell violates an exclusive movebound.

         0          0          0    Two cells violate cts margins.

         0          0          0    Two cells violate coloring.


check_legality for block design ORCA succeeded!


check_legality succeeded.
```


SYNOPSYS
Sync
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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Check reports**
  - **Check timing_report_max for all corners**

    Ex) open timing_report_max_placement_func.ss_n40c report and search (VIOLATED)

```
------------------------------
  slack (VIOLATED)
```

  - **Analyze VIOLATED Timing Path**

```
Startpoint: I_ORCA_TOP/I_BLENDER/s3_op2_reg_22_ (rising edge-triggered flip-flop clocked by sys_clk)
Endpoint: I_ORCA_TOP/I_BLENDER/s4_op2_reg_30  (rising edge-triggered flip-flop clocked by SYS_CLK)
Mode: func
Corner: ss_n40c
Scenario: func.ss_n40c
Path Group: SYS_CLK
Path Type: max

Attributes
    b - black-box (unknown)
    s - size_only
    d - dont_touch
    u - dont_use
    g - generic
    h - hierarchical
    i - ideal
    n - noncombinational
    E - extracted timing model
    Q - Quick timing model
```

Timing paths are grouped into path groups according to the clock associated with the Endpoint of the path

Max = setup

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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Analyze VIOLATED Timing Path**

Point	Fanout	Trans	Derate	Delta	Incr	Path	Location	Attributes
clock sys_clk (rise edge)					0.00000	0.00000		
clock network delay (ideal)					0.00000	0.00000		
I_ORCA_TOP/I_BLENDER/s3_op2_reg_22_/CLK (SDFFX1_HVT)		0.00000	1.00000		0.00000	0.00000 r	(599.27,605.64)	s, n
I_ORCA_TOP/I_BLENDER/s3_op2_reg_22_/Q (SDFFX1_HVT)		0.10345	1.00000		0.30659	0.30659 f	(602.75,606.29)	s, n
I_ORCA_TOP/I_BLENDER/s3_op2[22] (net)								
I_ORCA_TOP/I_BLENDER/HFSBUF_32_1920/A (NBUFFX4_LVT)		0.10345	1.00000		0.00031	0.30689 f	(562.26,599.72)	
I_ORCA_TOP/I_BLENDER/HFSBUF_32_1920/Y (NBUFFX4_LVT)		0.04179	1.00000		0.09090	0.39780 f	(561.60,599.61)	
I_ORCA_TOP/I_BLENDER/HFSNET_232 (net)	8							
I_ORCA_TOP/I_BLENDER/U6291/A1 (AND2X1_RVT)		0.04179	1.00000		0.00010	0.39789 f	(555.12,599.54)	
I_ORCA_TOP/I_BLENDER/U6291/Y (AND2X1_RVT)		0.02628	1.00000		0.06727	0.46516 f	(554.61,599.61)	
I_ORCA_TOP/I_BLENDER/n609 (net)	1							
I_ORCA_TOP/I_BLENDER/U2261/A0 (HADDX1_RVT)		0.02628	1.00000		0.00000	0.46516 f	(552.54,599.69)	
I_ORCA_TOP/I_BLENDER/U2261/SO (HADDX1_RVT)		0.03942	1.00000		0.12306	0.58823 r	(551.42,599.90)	
I_ORCA_TOP/I_BLENDER/n4201 (net)	1							
I_ORCA_TOP/I_BLENDER/U2231/B (FADDX1_HVT)		0.03942	1.00000		0.00002	0.58825 r	(551.24,597.77)	
I_ORCA_TOP/I_BLENDER/U2231/CO (FADDX1_HVT)		0.09262	1.00000		0.19201	0.78026 r	(549.84,597.83)	
I_ORCA_TOP/I_BLENDER/n4131 (net)	1							
I_ORCA_TOP/I_BLENDER/U2151/B (FADDX1_RVT)		0.09262	1.00000		0.00002	0.78028 r	(547.29,601.11)	
I_ORCA_TOP/I_BLENDER/U2151/S (FADDX1_RVT)		0.04587	1.00000		0.19655	0.97683 f	(546.11,601.25)	
I_ORCA_TOP/I_BLENDER/n3981 (net)	1							
I_ORCA_TOP/I_BLENDER/U2141/CI (FADDX1_HVT)		0.04587	1.00000		0.00002	0.97685 f	(546.35,597.72)	
I_ORCA_TOP/I_BLENDER/U2141/S (FADDX1_HVT)		0.07751	1.00000		0.26289	1.23974 r	(547.48,597.91)	
I_ORCA_TOP/I_BLENDER/n3961 (net)	1							
I_ORCA_TOP/I_BLENDER/U2415/B (FADDX1_RVT)		0.07751	1.00000		0.00002	1.23976 r	(547.67,594.42)	
I_ORCA_TOP/I_BLENDER/U2415/CO (FADDX1_RVT)		0.05232	1.00000		0.13052	1.37028 r	(549.17,594.49)	
I_ORCA_TOP/I_BLENDER/n2310 (net)	1							
I_ORCA_TOP/I_BLENDER/U2315/CI (FADDX1_RVT)		0.05232	1.00000		0.00002	1.37030 r	(547.54,596.50)	
I_ORCA_TOP/I_BLENDER/U2315/CO (FADDX1_RVT)		0.05114	1.00000		0.11351	1.48380 r	(546.10,596.38)	
I_ORCA_TOP/I_BLENDER/n2210 (net)	1							
I_ORCA_TOP/I_BLENDER/U2215/CI (FADDX1_LVT)		0.05114	1.00000		0.00000	1.48380 r	(544.05,596.50)	
I_ORCA_TOP/I_BLENDER/U2215/CO (FADDX1_LVT)		0.03796	1.00000		0.07526	1.55907 r	(542.60,596.38)	
I_ORCA_TOP/I_BLENDER/n2110 (net)	1							
I_ORCA_TOP/I_BLENDER/U2115/CI (FADDX1_RVT)		0.03796	1.00000		0.00002	1.55909 r	(544.35,594.37)	
I_ORCA_TOP/I_BLENDER/U2115/CO (FADDX1_RVT)		0.05430	1.00000		0.11118	1.67027 r	(542.90,594.49)	
I_ORCA_TOP/I_BLENDER/n2010 (net)	1							
I_ORCA_TOP/I_BLENDER/U2015/CI (FADDX1_RVT)		0.05430	1.00000		0.00002	1.67028 r	(544.35,589.81)	
I_ORCA_TOP/I_BLENDER/U2015/CO (FADDX1_RVT)		0.05251	1.00000		0.11581	1.78610 r	(542.90,589.70)	
I_ORCA_TOP/I_BLENDER/n1910 (net)	1							
I_ORCA_TOP/I_BLENDER/U1915/CI (FADDX1_RVT)		0.05251	1.00000		0.00002	1.78612 r	(544.83,587.68)	
I_ORCA_TOP/I_BLENDER/U1915/CO (FADDX1_RVT)		0.05339	1.00000		0.11608	1.90220 r	(546.28,587.80)	
I_ORCA_TOP/I_BLENDER/n1810 (net)	1							
I_ORCA_TOP/I_BLENDER/U1815/CI (FADDX1_RVT)		0.05339	1.00000		0.00002	1.90222 r	(544.83,584.34)	
I_ORCA_TOP/I_BLENDER/U1815/CO (FADDX1_RVT)		0.05106	1.00000		0.11385	2.01607 r	(546.28,584.45)	
I_ORCA_TOP/I_BLENDER/n1710 (net)	1							
I_ORCA_TOP/I_BLENDER/U1715/CI (FADDX1_RVT)		0.05106	1.00000		0.00000	2.01607 r	(545.57,583.12)	

Check ideal network

R to R path
CLK pin to D pin

Cell location

Cell Delay &
Path Delay

Timing derate numbers are ratios used to derate(increase/decrease) the delay numbers you get in your timing reports

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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Analyze VIOLATED Timing Path**

```
I_ORCA_TOP/I_BLENDER/U1715/CO (FADDX1_RVT)                      0.05445   1.00000   0.11658   2.13264 r   (544.12,583.01)
I_ORCA_TOP/I_BLENDER/n1610 (net)                          1
I_ORCA_TOP/I_BLENDER/U1615/CI (FADDX1_RVT)                      0.05445   1.00000   0.00002   2.13266 r   (547.39,579.78)
I_ORCA_TOP/I_BLENDER/U1615/CO (FADDX1_RVT)                      0.05424   1.00000   0.11772   2.25039 r   (545.94,579.66)
I_ORCA_TOP/I_BLENDER/n1510 (net)                          1
I_ORCA_TOP/I_BLENDER/U1515/CI (FADDX1_LVT)                      0.05424   1.00000   0.00002   2.25040 r   (545.74,574.31)
I_ORCA_TOP/I_BLENDER/U1515/CO (FADDX1_LVT)                      0.03916   1.00000   0.07746   2.32786 r   (547.19,574.42)
I_ORCA_TOP/I_BLENDER/n1410 (net)                          1
I_ORCA_TOP/I_BLENDER/U1415/CI (FADDX1_LVT)                      0.03916   1.00000   0.00002   2.32788 r   (548.02,569.75)
I_ORCA_TOP/I_BLENDER/U1415/CO (FADDX1_LVT)                      0.03704   1.00000   0.07114   2.39902 r   (549.47,569.63)
I_ORCA_TOP/I_BLENDER/n1310 (net)                          1
I_ORCA_TOP/I_BLENDER/U1315/CI (FADDX1_LVT)                      0.03704   1.00000   0.00000   2.39902 r   (549.39,567.62)
I_ORCA_TOP/I_BLENDER/U1315/CO (FADDX1_LVT)                      0.03857   1.00000   0.07210   2.47112 r   (550.84,567.73)
I_ORCA_TOP/I_BLENDER/n1210 (net)                          1
I_ORCA_TOP/I_BLENDER/U1215/CI (FADDX1_RVT)                      0.03857   1.00000   0.00002   2.47114 r   (554.56,566.40)
I_ORCA_TOP/I_BLENDER/U1215/CO (FADDX1_RVT)                      0.05255   1.00000   0.10956   2.58070 r   (556.01,566.29)
I_ORCA_TOP/I_BLENDER/n1110 (net)                          1
I_ORCA_TOP/I_BLENDER/U1115/CI (FADDX1_RVT)                      0.05255   1.00000   0.00002   2.58072 r   (555.47,563.06)
I_ORCA_TOP/I_BLENDER/U1115/CO (FADDX1_RVT)                      0.05447   1.00000   0.11719   2.69791 r   (556.92,562.94)
I_ORCA_TOP/I_BLENDER/n1010 (net)                          1
I_ORCA_TOP/I_BLENDER/U1015/CI (FADDX1_RVT)                      0.05447   1.00000   0.00002   2.69793 r   (557.75,557.59)
I_ORCA_TOP/I_BLENDER/U1015/CO (FADDX1_RVT)                      0.05360   1.00000   0.11711   2.81504 r   (559.20,557.70)
I_ORCA_TOP/I_BLENDER/n910 (net)                           1
I_ORCA_TOP/I_BLENDER/U915/CI (FADDX1_LVT)                       0.05360   1.00000   0.00002   2.81506 r   (560.77,554.24)
I_ORCA_TOP/I_BLENDER/U915/CO (FADDX1_LVT)                       0.04301   1.00000   0.08091   2.89597 r   (559.32,554.36)
I_ORCA_TOP/I_BLENDER/n810 (net)                           1
I_ORCA_TOP/I_BLENDER/U815/CI (FADDX1_RVT)                       0.04301   1.00000   0.00006   2.89602 r   (572.19,553.03)
I_ORCA_TOP/I_BLENDER/U815/CO (FADDX1_RVT)                       0.05045   1.00000   0.10902   3.00505 r   (573.64,552.91)
I_ORCA_TOP/I_BLENDER/n710 (net)                           1
I_ORCA_TOP/I_BLENDER/U715/CI (FADDX1_RVT)                       0.05045   1.00000   0.00000   3.00505 r   (575.08,553.03)
I_ORCA_TOP/I_BLENDER/U715/CO (FADDX1_RVT)                       0.05625   1.00000   0.11805   3.12309 r   (576.53,552.91)
I_ORCA_TOP/I_BLENDER/n610 (net)                           1
I_ORCA_TOP/I_BLENDER/U693/CI (FADDX1_RVT)                       0.05625   1.00000   0.00004   3.12313 r   (575.82,560.93)
I_ORCA_TOP/I_BLENDER/U693/CO (FADDX1_RVT)                       0.05320   1.00000   0.11721   3.24034 r   (574.37,561.05)
I_ORCA_TOP/I_BLENDER/n510 (net)                           1
I_ORCA_TOP/I_BLENDER/U5105/CI (FADDX1_RVT)                      0.05320   1.00000   0.00002   3.24036 r   (571.56,563.06)
I_ORCA_TOP/I_BLENDER/U5105/CO (FADDX1_RVT)                      0.05379   1.00000   0.11675   3.35711 r   (570.11,562.94)
I_ORCA_TOP/I_BLENDER/n428 (net)                           1
I_ORCA_TOP/I_BLENDER/U4105/CI (FADDX1_RVT)                      0.05379   1.00000   0.00002   3.35712 r   (570.65,567.62)
I_ORCA_TOP/I_BLENDER/U4105/CO (FADDX1_RVT)                      0.05438   1.00000   0.11757   3.47469 r   (569.20,567.73)
I_ORCA_TOP/I_BLENDER/n3100 (net)                          1
I_ORCA_TOP/I_BLENDER/U3105/CI (FADDX1_RVT)                      0.05438   1.00000   0.00002   3.47471 r   (569.74,573.09)
I_ORCA_TOP/I_BLENDER/U3105/CO (FADDX1_RVT)                      0.05434   1.00000   0.11774   3.59245 r   (568.29,572.98)
I_ORCA_TOP/I_BLENDER/n2101 (net)                          1
I_ORCA_TOP/I_BLENDER/U2109/CI (FADDX1_RVT)                      0.05434   1.00000   0.00002   3.59247 r   (569.45,577.65)
I_ORCA_TOP/I_BLENDER/U2109/S (FADDX1_RVT)                       0.03971   1.00000   0.16638   3.75885 f   (570.59,577.84)
I_ORCA_TOP/I_BLENDER/N508 (net)                           1
I_ORCA_TOP/I_BLENDER/U403/A1 (AO222X1_LVT)                      0.03971   1.00000   0.00000   3.75885 f   (573.27,577.94)
I_ORCA_TOP/I_BLENDER/U403/Y (AO222X1_LVT)                       0.03283   1.00000   0.08932   3.84817 f   (574.62,577.88)
I_ORCA_TOP/I_BLENDER/N572 (net)                           1
I_ORCA_TOP/I_BLENDER/s4_op2_reg_30 /D (SDFFX1_RVT)              0.03283   1.00000   0.00004   3.84821 f   (593.63,577.78)    s, n
data arrival time                                                                             3.84821
```

R to R path
CLK pin to D pin

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Placement : Use ICC2 (IC Compiler II tool)

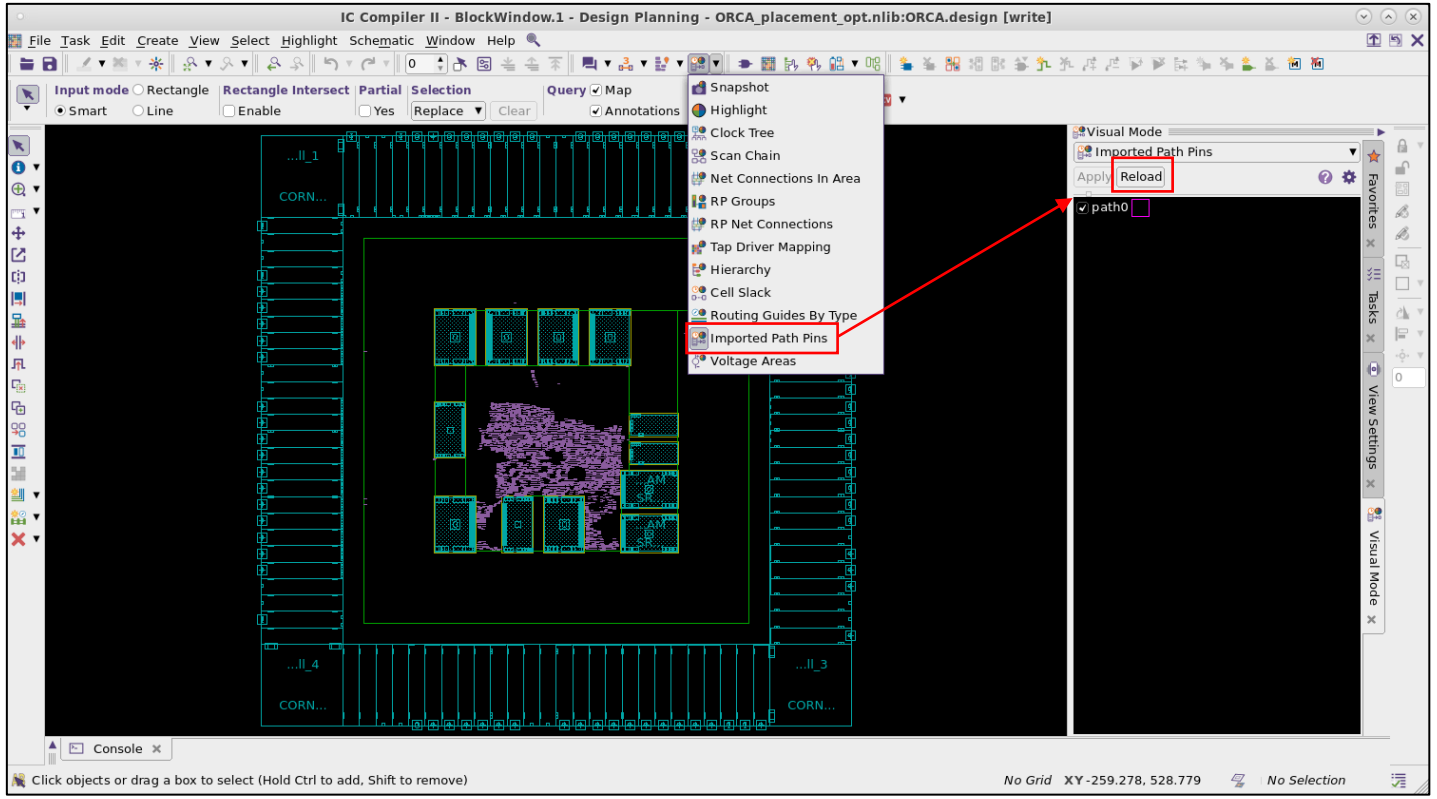
12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Analyze VIOLATED Timing Path**

```
clock sys_clk (rise edge)                                                    4.00000      4.00000
clock network delay (ideal)                                                  0.00000      4.00000
I_ORCA_TOP/I_BLENDER/s4_op2_reg_30_/CLK (SDFFX1_RVT)          0.00000        0.00000      4.00000 r   (594.55,578.54)    s, n
library setup time                                            1.00000      -0.15203     3.84797
data required time                                                                     3.84797
-------------------------------------------------------------------------------------------------------------------------------
data required time                                                                     3.84797
data arrival time                                                                     -3.84821
-------------------------------------------------------------------------------------------------------------------------------
slack (VIOLATED)                                                                      -0.00024
```

Setup Slack = DRT - DAT

- **Imported Path Pins**



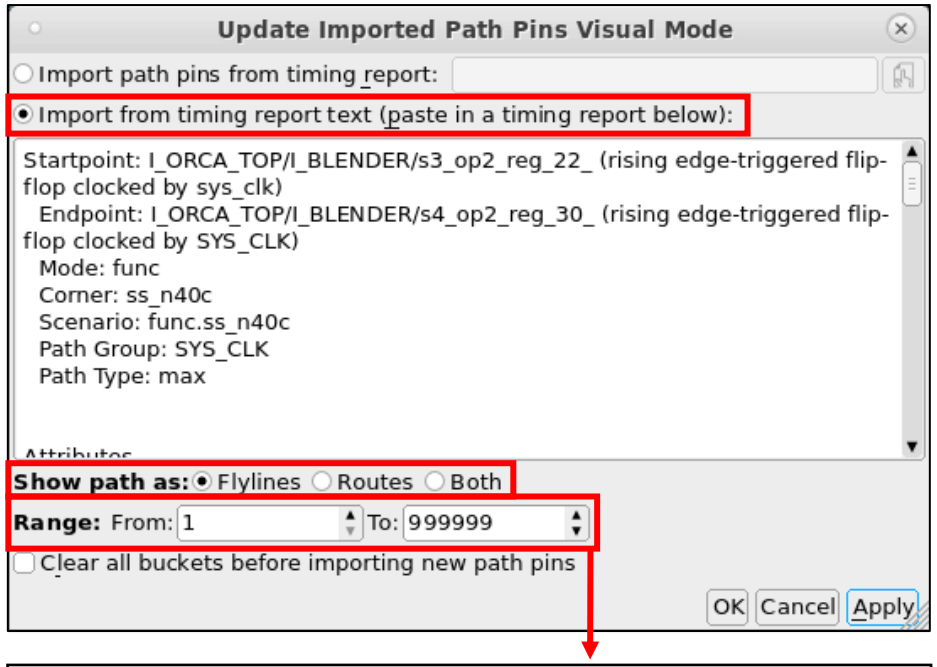
The "Imported Path Pins" visual mode takes timing path data from a timing report and highlights the timing path in the Layout (Paths can be viewed one-by-one)

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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Imported Path Pins**



When importing paths, use settings in Reload dialog box to specify the path number range to import

From : Generates imported path pins visual mode start from the specified path number

To : Import paths up to the specified path number

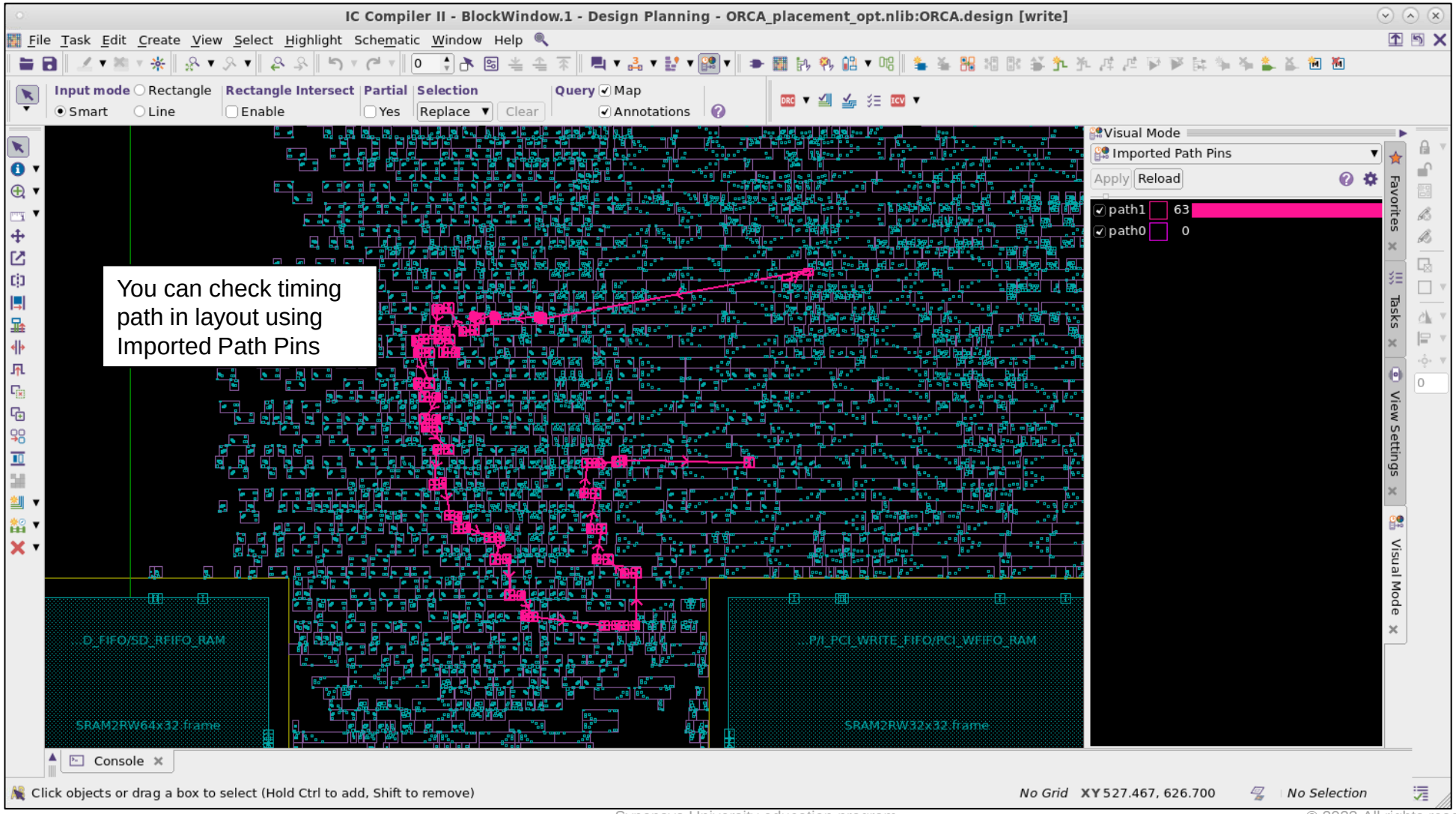
1. Click the Reload button
2. Select from timing report text
3. Select the timing path in timing reports and paste it
4. Choose the Show path as option to specify the path type
5. The Imported Path Pins Visual Mode panel displays a histogram bar for the path with the number of pins in the path (When choose Routes, timing report should include input_pins to see the shapes between pins)
6. Choose the Clear all buckets before importing new path pins option to delete the histogram before importing the new path

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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/09_place_opt.tcl

- **Imported Path Pins**



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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/10_refine_opt.tcl

- **set_app_option before refine_opt**

```
set_app_options -list {refine_opt.place.congestion_effort {high}}
refine_opt.place.congestion_effort high
icc2_shell> set_app_options -list {refine_opt.place.effort {high}}
refine_opt.place.effort high
icc2_shell> set_app_options -list {place_opt.final_place.effort {high}}
place_opt.final_place.effort high
```

- **Run refine_opt**

```
icc2_shell> sec_StartTimer refine_opt
1
icc2_shell>
icc2_shell> refine_opt
icc2_shell> sec_ReportResourceUtil refine_opt
```

refine_opt

- By running refine_opt command, you can optimize the place_opt completed db once again

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Placement : Use ICC2 (IC Compiler II tool)

11. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/11_insert_spare_cell.tcl

- **Insert spare cell after placement (BUF, INV, AND, FF, OR)**

```
icc2_shell> set core_bdry "{425.543 425.846} {1014.459 1014.152}"
{425.543 425.846} {1014.459 1014.152}
icc2_shell>
icc2_shell> #add NBUFFX2_RVT
icc2_shell> add_spare_cells -num_instances 100 \
? -boundary $core_bdry \
? -cell_name ORCA_spare_cell_NBUF \
? -lib_cell "saed32rvt/NBUFFX2_RVT"
Information: Starting 'add_spare_cells' (FLW-8000)
Information: Time: 2023-09-11 21:44:38 / Session: 2.26 hr / Command: 0.00 hr / Memory: 2129 MB (FLW-8100)
The total needed area to add new spare cells is 203.3152.
****************************************
Report : ECO add_spare_cells Summary
Design : ORCA
Version: S-2021.06-SP5
Date   : Mon Sep 11 21:44:38 2023
****************************************
Total 100 spare cells are added.

Library Cell saed32rvt/NBUFFX2_RVT:
    Add  100 cells in the voltage Area DEFAULT_VA
Information: Ending 'add_spare_cells' (FLW-8001)
Information: Time: 2023-09-11 21:44:38 / Session: 2.26 hr / Command: 0.00 hr / Memory: 2129 MB (FLW-8100)
1
icc2_shell>
icc2_shell> #add INV_RVT
icc2_shell> add_spare_cells -num_instances 100 \
? -boundary $core_bdry \
? -cell_name ORCA_spare_cell_INV \
? -lib_cell "saed32rvt/INVX2_RVT"
Information: Starting 'add_spare_cells' (FLW-8000)
Information: Time: 2023-09-11 21:44:38 / Session: 2.26 hr / Command: 0.00 hr / Memory: 2129 MB (FLW-8100)
The total needed area to add new spare cells is 152.4864.
****************************************
Report : ECO add_spare_cells Summary
Design : ORCA
Version: S-2021.06-SP5
Date   : Mon Sep 11 21:44:38 2023
****************************************
Total 100 spare cells are added.

Library Cell saed32rvt/INVX2_RVT:
    Add  100 cells in the voltage Area DEFAULT_VA
Information: Ending 'add_spare_cells' (FLW-8001)
Information: Time: 2023-09-11 21:44:38 / Session: 2.26 hr / Command: 0.00 hr / Memory: 2129 MB (FLW-8100)
1
```


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Placement : Use ICC2 (IC Compiler II tool)

11. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/11_insert_spare_cell.tcl

- **Insert spare cell after placement (BUF, INV, AND, FF, OR)**

```
icc2_shell> #add AND_RVT
icc2_shell> add_spare_cells -num_instances 50 \
? -boundary $core_bdry \
? -cell_name ORCA_spare_cell_AND \
? -lib_cell "saed32rvt/AND2X2_RVT"
Information: Starting 'add_spare_cells' (FLW-8000)
Information: Time: 2023-09-11 21:44:38 / Session: 2.26 hr / Command: 0.00 hr / Memory: 2129 MB (FLW-8100)
The total needed area to add new spare cells is 114.3648.
****************************************
Report : ECO add_spare_cells Summary
Design : ORCA
Version: S-2021.06-SP5
Date   : Mon Sep 11 21:44:38 2023
****************************************
Total 50 spare cells are added.

Library Cell saed32rvt/AND2X2_RVT:
    Add   50 cells in the voltage Area DEFAULT_VA
Information: Ending 'add_spare_cells' (FLW-8001)
Information: Time: 2023-09-11 21:44:38 / Session: 2.26 hr / Command: 0.00 hr / Memory: 2129 MB (FLW-8100)
1
icc2_shell>
icc2_shell> #add FF_RVT
icc2_shell> add_spare_cells -num_instances 50 \
? -boundary $core_bdry \
? -cell_name ORCA_spare_cell_FF \
? -lib_cell "saed32rvt/RSDFFX2_RVT"
Information: Starting 'add_spare_cells' (FLW-8000)
Information: Time: 2023-09-11 21:44:38 / Session: 2.26 hr / Command: 0.00 hr / Memory: 2129 MB (FLW-8100)
The total needed area to add new spare cells is 876.7968.
****************************************
Report : ECO add_spare_cells Summary
Design : ORCA
Version: S-2021.06-SP5
Date   : Mon Sep 11 21:44:38 2023
****************************************
Total 50 spare cells are added.

Library Cell saed32rvt/RSDFFX2_RVT:
    Add   50 cells in the voltage Area DEFAULT_VA
Information: Ending 'add_spare_cells' (FLW-8001)
Information: Time: 2023-09-11 21:44:38 / Session: 2.26 hr / Command: 0.00 hr / Memory: 2129 MB (FLW-8100)
1
```


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Placement : Use ICC2 (IC Compiler II tool)

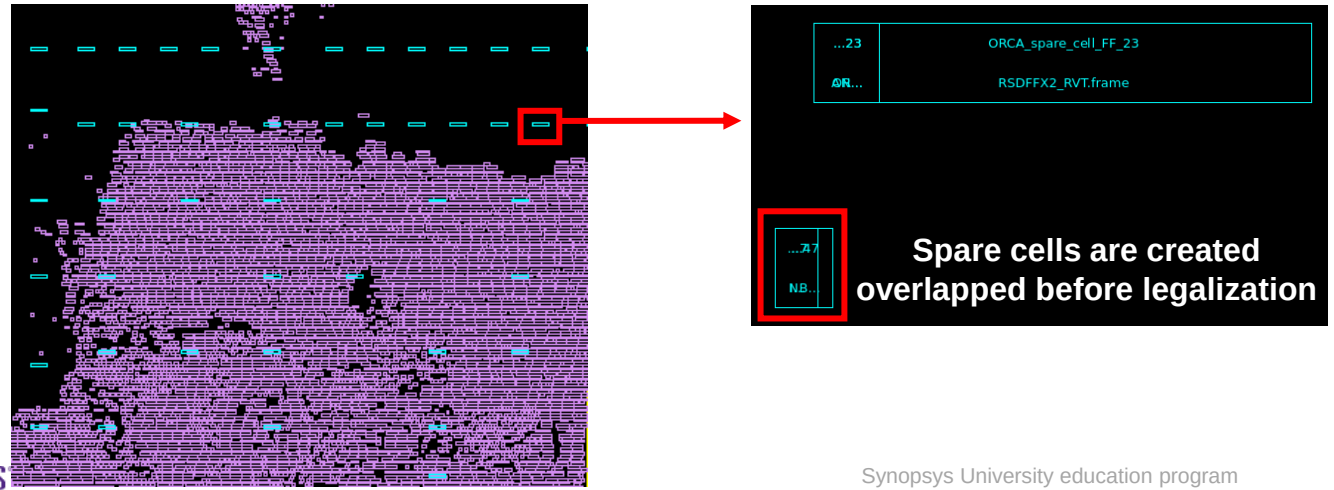
11. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/11_insert_spare_cell.tcl

- **Insert spare cell after placement (BUF, INV, AND, FF, OR)**

```
icc2_shell> #add OR2x2_RVT
icc2_shell> add_spare_cells -num_instances 50 \
? -boundary $core_bdry \
? -cell_name ORCA_spare_cell_XOR \
? -lib_cell "saed32rvt/OR2X2_RVT"
Information: Starting 'add_spare_cells' (FLW-8000)
Information: Time: 2023-09-11 21:44:38 / Session: 2.26 hr / Command: 0.00 hr / Memory: 2129 MB (FLW-8100)
The total needed area to add new spare cells is 114.3648.
****************************************
Report : ECO add_spare_cells Summary
Design : ORCA
Version: S-2021.06-SP5
Date   : Mon Sep 11 21:44:38 2023
****************************************
Total 50 spare cells are added.

Library Cell saed32rvt/OR2X2_RVT:
    Add   50 cells in the voltage Area DEFAULT_VA
Information: Ending 'add_spare_cells' (FLW-8001)
Information: Time: 2023-09-11 21:44:38 / Session: 2.26 hr / Command: 0.00 hr / Memory: 2129 MB (FLW-8100)
1
```

- **Check Spare cell after created**



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Placement : Use ICC2 (IC Compiler II tool)

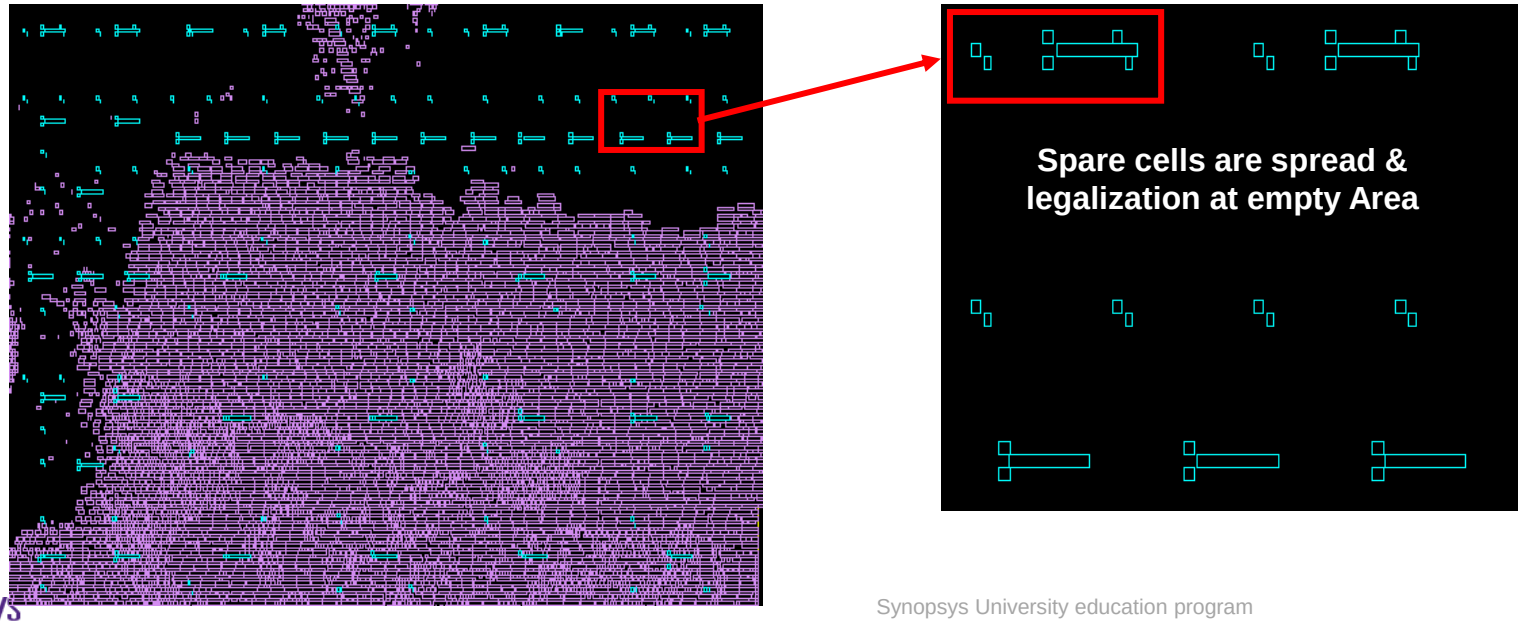
11. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/11_insert_spare_cell.tcl

- **Spread & legalize spare cells**

```
icc2_shell>
spread_spare_cells -cells [get_cells -phy -filter "full_name =~ ORCA_spare_cell*"] \
?    -boundary $core_bdry
The spare cells are spread successfully.
1
icc2_shell>
icc2_shell> legalize_placement -incremental


icc2_shell> set_placement_status legalize_only \
? [get_flat_cells -filter is_spare_cell]
1
```

- **Check Spare cell after spread & legalize**



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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/12_check_placement.tcl

- **Check placement after refine_opt**
  - **check_legality after placement (check the legality of the current placement)**

```
icc2_shell> check_legality
Information: The command 'check_legality' cleared the undo history. (UNDO-016)

************************

running check_legality

Warning: Routing direction of metal layer PO is neither "horizontal" nor "vertical".  PDC checks will not be performed on this layer.
PDC-003)
PDC app_options settings =========
        place.legalize.enable_prerouted_net_check: 1
        place.legalize.num_tracks_for_access_check: 1
        place.legalize.use_eol_spacing_for_access_check: 0
        place.legalize.allow_touch_track_for_access_check: 1
        place.legalize.reduce_conservatism_in_eol_check: 0
        place.legalize.preroute_shape_merge_distance: 0.0
        place.legalize.enable_non_preferred_direction_span_check: 0

Layer M1: cached 0 shapes out of 1369 total shapes.
Layer M2: cached 664 shapes out of 664 total shapes.
Cached 86128 vias out of 163438 total vias.

check_legality for block design ORCA ...
Information: Initializing classic cellmap without advanced rules enabled and without PDC enabled
Information: The following app options are used in cellmap
        place.legalize.enable_color_aware_placement : false
        place.legalize.use_nll_query_cm : false
        place.legalize.enable_advanced_legalizer : false
        place.legalize.enable_prerouted_net_check : true
        place.legalize.enable_advanced_prerouted_net_check : false
        place.legalize.always_continue : true
        place.legalize.limit_legality_checks : false
        place.common.pnet_aware_density : 1.0000
        place.common.pnet_aware_min_width : 0.0000
        place.common.pnet_aware_layers : {}
        place.common.use_placement_model : false
        place.common.enable_advanced_placement_model : true
        cts.placement.cell_spacing_rule_style : maximum
Total 0.1100 seconds to build cellmap data
Information: Creating classic rule checker.
Warning: Routing direction of metal layer PO is neither "horizontal" nor "vertical".  PDC checks will not be performed on this layer.
PDC-003)
=====> Processed 178 ref cells (19 fillers) from library
Design has no advanced rules
Checking legality
Checking cell legality:
0%....10%....20%....30%....40%....50%....60%....70%....80%....90%....100%
Sorting rows.
Checking spacing rule legality.
0%....10%....20%....30%....40%....50%....60%....70%....80%....90%....100%
Checking packing rule legality.
```


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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/12_check_placement.tcl

- **Check placement after refine_opt**
  - **check_legality after placement**

```
****************************************
  Report : Legality
****************************************

VIOLATIONS BY CATEGORY:
   MOVABLE  APP-FIXED USER-FIXED  DESCRIPTION
         0          0          0  Two objects overlap.
         0          0          0  A cell violates a pnet.
         0          0          0  A cell is illegal at a site.
         0          0          0  A cell is not aligned with a site.
         0          0          0  A cell has an illegal orientation.
         0          0          0  A cell spacing rule is violated.
         0          0          0  A layer rule is violated.
         0          0          0  A cell is in the wrong region.
         0          0          0  Two cells violate cts margins.
         0          0          0  Two cells violate coloring.

         0          0          0  TOTAL

TOTAL 0 Violations.
```

The check_legality command checks the legality of the current placement and output a report of violation statistics (ex. Overlap, site, spacing, orientation)

```
VIOLATIONS BY SUBCATEGORY:
     MOVABLE  APP-FIXED USER-FIXED  DESCRIPTION

         0          0          0    Two objects overlap.
           0          0          0    Two cells overlap.
           0          0          0    Two cells have overlapping keepout margins.
           0          0          0    A cell overlaps a blockage.
           0          0          0    A cell keepout margin overlaps a blockage.

         0          0          0    A cell violates a pnet.

         0          0          0    A cell is illegal at a site.
           0          0          0    A cell violates pin-track alignment rules.
           0          0          0    A cell is illegal at a site.
           0          0          0    A cell violates legal index rule.
           0          0          0    A cell has the wrong variant for its location.

         0          0          0    A cell is not aligned with a site.
           0          0          0    A cell is not aligned with the base site.
           0          0          0    A cell is not aligned with an overlaid site.

         0          0          0    A cell has an illegal orientation.

         0          0          0    A cell spacing rule is violated.
           0          0          0    A spacing rule is violated in a row.
           0          0          0    A spacing rule is violated between adjacent rows.
           0          0          0    A cell violates vertical abutment rule.
           0          0          0    A cell violates metal spacing rule.

         0          0          0    A layer rule is violated.
           0          0          0    A layer VTH rule is violated.
           0          0          0    A layer OD rule is violated.
           0          0          0    A layer OD max-width rule is violated.
           0          0          0    A layer ALL_OD corner rule is violated.
           0          0          0    A layer max-vertical-length rule is violated.
           0          0          0    A layer TPO rule is violated.
           0          0          0    Filler cell insertion cannot satisfy layer rules.

         0          0          0    A cell is in the wrong region.
           0          0          0    A cell is outside its hard bound.
           0          0          0    A cell is in the wrong voltage area.
           0          0          0    A cell violates an exclusive movebound.

         0          0          0    Two cells violate cts margins.

         0          0          0    Two cells violate coloring.


check_legality for block design ORCA succeeded!
```


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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/12_check_placement.tcl

- **Check placement after refine_opt**
  - **check_mv_deisgn after placement**

```
icc2_shell> check_mv_design
Information: The command 'check_mv_design' cleared the undo history. (UNDO-016)
****************************************
Report : check_mv_design
Design : ORCA
Version: S-2021.06-SP5
Date   : Mon Sep 11 22:36:40 2023
****************************************

---------- Power domain rule ----------
No errors or warnings.

---------- Supply set rule ----------
No errors or warnings.

---------- Supply net rule ----------
No errors or warnings.

---------- Supply port rule ----------
No errors or warnings.

---------- Isolation strategy rule ----------
No errors or warnings.

---------- Level shifter strategy rule ----------
No errors or warnings.

---------- Retention strategy rule ----------
No errors or warnings.

---------- Power switch strategy rule ----------
No errors or warnings.

---------- Repeater rule ----------
No errors or warnings.

---------- Terminal boundary rule ----------
No errors or warnings.

---------- Isolation cell rule ----------
No errors or warnings.

---------- Level shifter cell rule ----------
No errors or warnings.

---------- Retention cell rule ----------
No errors or warnings.

---------- Switch cell rule ----------
No errors or warnings.
```

```
---------- PGMUX rule ----------
No errors or warnings.

---------- Diode cell rule ----------
No errors or warnings.

---------- Model rule ----------
No errors or warnings.

---------- Isolation rule ----------
No errors or warnings.

---------- Voltage shifting rule ----------
Warning: Driver pin 'scan_en' (supply net 'VDD' [0.95V]) cannot drive load pin 'scan_en_iopad/PADIO' (supply net 'VD
DIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'test_mode' (supply net 'VDD' [0.95V]) cannot drive load pin 'test_mode_iopad/PADIO' (supply net
 'VDDIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'power_save' (supply net 'VDD' [0.95V]) cannot drive load pin 'powersave_iopad/PADIO' (supply ne
t 'VDDIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'sys_clk' (supply net 'VDD' [0.95V]) cannot drive load pin 'sys_clk_iopad/PADIO' (supply net 'VD
DIO' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'pclk' (supply net 'VDD' [0.95V]) cannot drive load pin 'pclk_iopad/PADIO' (supply net 'VDDIO' [
2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'prst_n' (supply net 'VDD' [0.95V]) cannot drive load pin 'prst_n_iopad/PADIO' (supply net 'VDDI
O' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'pidsel' (supply net 'VDD' [0.95V]) cannot drive load pin 'pidsel_iopad/PADIO' (supply net 'VDDI
O' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'pgnt_n' (supply net 'VDD' [0.95V]) cannot drive load pin 'pgnt_n_iopad/PADIO' (supply net 'VDDI
O' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'pm66en' (supply net 'VDD' [0.95V]) cannot drive load pin 'pm66en_iopad/PADIO' (supply net 'VDDI
O' [2.25V]) due to voltage difference. (MV-012)
Warning: Driver pin 'sdr_clk' (supply net 'VDD' [0.95V]) cannot drive load pin 'sdr_clk_iopad/PADIO' (supply net 'VD
DIO' [2.25V]) due to voltage difference. (MV-012)
Information: Total 10 MV-012 violations. (MV-080)

---------- Tie-off connection rule ----------
No errors or warnings.

---------- Analog net rule ----------
No errors or warnings.

---------- Physical block pin rule ----------
No errors or warnings.

---------- PG pin rule ----------
No errors or warnings.

---------- Signal pin rule ----------
No errors or warnings.

---------- Summary ----------
Information: Total 0 error(s) and 10 warning(s) from check_mv_design. (MV-082)
1
```

- The check_mv_design command checks multi-voltage related electrical rules for the design, and issues Error and Warning messages as appropriate
- By default, all multi-voltage checks except pg net and pg connection checks are included unless a specific type of checks or -all option is specified

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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/12_check_placement.tcl

- **Check placement after refine_opt**
  - **report_utilization about Core area after placement**

```
icc2_shell> report_utilization -region {{{425.5430 425.8460} {1014.4590 1014.1520}}}
****************************************
Report : report_utilization
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Sep 19 10:24:03 2023
****************************************
Utilization Ratio:                      0.2355
Utilization options:
 - Area calculation based on:           site_row in region { {425.5430 425.8460} {425.5430 1014.1520} {1014.4590 1014.1520} {1014.4590 425.846
0} } of block ORCA
 - Categories of objects excluded:      hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                             346462.8163
Total Capacity Area:                    205744.1031
Total Area of cells:                    48447.2871
Area of excluded objects:
 - hard_macros         :                130644.7564
 - macro_keepouts      :                3008.8108
 - soft_macros         :                0.0000
 - io_cells            :                0.0000
 - hard_blockages      :                141527.9649

Utilization of site-rows with:
 - Site 'unit':                         0.2355

0.2355
```

  - **report_congestion after placement**

```
****************************************
Report : congestion
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Sep 19 10:25:03 2023
****************************************

Layer     |    overflow     |              # GRCs has
Name      |  total  |  max  | overflow (%)      | max overflow
------------------------------------------------------------
Both Dirs |     216 |     2 |     215  ( 0.01%) |       1
H routing |     216 |     2 |     215  ( 0.03%) |       1
V routing |       0 |     0 |       0  ( 0.00%) |       0

1
```


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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/12_check_placement.tcl

- **Check placement after refine_opt**
  - **report_timing after placement**

```
icc2_shell> report_timing -significant_digits 5
****************************************
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -report_by design
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Sep 19 10:29:51 2023
****************************************

  Startpoint: I_ORCA_TOP/I_RISC_CORE/Oprnd_A_reg_2_ (rising edge-triggered flip-flop clocked by sys_clk)
  Endpoint: I_ORCA_TOP/I_RISC_CORE/Zro_Flag_reg (rising edge-triggered flip-flop clocked by SYS_2x_CLK)
  Mode: func
  Corner: ss_125c
  Scenario: func.ss_125c
  Path Group: SYS_2x_CLK
  Path Type: max

  Point                                                 Incr         Path
  ------------------------------------------------------------------------------
  clock sys_clk (rise edge)                          0.00000      0.00000
  clock network delay (ideal)                       -0.03926     -0.03926

  I_ORCA_TOP/I_RISC_CORE/Oprnd_A_reg_2_/CLK (SDFFX1_RVT)
                                                     0.00000     -0.03926 r
  I_ORCA_TOP/I_RISC_CORE/Oprnd_A_reg_2_/Q (SDFFX1_RVT)
                                                     0.15144      0.11219 r
  I_ORCA_TOP/I_RISC_CORE/HFSBUF_1141_1818/Y (NBUFFX4_LVT)
                                                     0.06933      0.18152 r
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_3_2918/Y (INVX1_LVT)
                                                     0.02670      0.20822 f
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_6_2921/Y (AND2X1_RVT)
                                                     0.06748      0.27570 f
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_5_2920/Y (OR2X1_RVT)
                                                     0.06506      0.34076 f
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_1_3066/Y (NAND2X0_RVT)
                                                     0.04753      0.38829 r
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_3_3068/Y (OA22X1_LVT)
                                                     0.06763      0.45593 r
  I_ORCA_TOP/I_RISC_CORE/U2_41/CO (FADDX1_LVT)       0.07332      0.52925 r
  I_ORCA_TOP/I_RISC_CORE/U2_51/CO (FADDX1_LVT)       0.07292      0.60216 r
  I_ORCA_TOP/I_RISC_CORE/U2_61/CO (FADDX1_LVT)       0.07120      0.67337 r
  I_ORCA_TOP/I_RISC_CORE/U2_71/CO (FADDX1_LVT)       0.08049      0.75386 r
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_1_3042/Y (AO21X1_LVT)
                                                     0.07133      0.82519 r
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_5_3046/Y (AO21X1_LVT)
                                                     0.06863      0.89382 r
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_1_3003/Y (AO21X1_LVT)
                                                     0.06855      0.96237 r
```

```
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_5_3046/Y (AO21X1_LVT)
                                                     0.06863      0.89382 r
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_1_3003/Y (AO21X1_LVT)
                                                     0.06855      0.96237 r
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_1_2985/Y (AO21X1_LVT)
                                                     0.06310      1.02546 r
  I_ORCA_TOP/I_RISC_CORE/U2_121/CO (FADDX1_LVT)      0.07360      1.09907 r
  I_ORCA_TOP/I_RISC_CORE/U2_131/CO (FADDX1_LVT)      0.07254      1.17160 r
  I_ORCA_TOP/I_RISC_CORE/U2_141/S (FADDX1_LVT)       0.09996      1.27157 f
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_2_3095/Y (AO221X1_LVT)
                                                     0.06418      1.33575 f
  I_ORCA_TOP/I_RISC_CORE/U157/Y (OR3X1_LVT)          0.06611      1.40186 f
  I_ORCA_TOP/I_RISC_CORE/SGI13_5242/Y (OR2X2_LVT)
                                                     0.04210      1.44395 f
  I_ORCA_TOP/I_RISC_CORE/ctmTdsLR_5_3133/Y (OR3X1_LVT)
                                                     0.04362      1.48758 f
  I_ORCA_TOP/I_RISC_CORE/SGI38_5234/Y (NOR3X0_LVT)
                                                     0.07404      1.56162 r
  I_ORCA_TOP/I_RISC_CORE/U9/Y (OR2X2_LVT)            0.04858      1.61020 r
  I_ORCA_TOP/I_RISC_CORE/U41/Y (AO22X1_LVT)          0.04753      1.65773 r
  I_ORCA_TOP/I_RISC_CORE/Zro_Flag_reg/D (SDFFARX1_RVT)
                                                     0.00000      1.65773 r
  data arrival time                                               1.65773

  clock SYS_2x_CLK (rise edge)                       2.00000      2.00000
  clock network delay (ideal)                        0.19629      2.19629
  I_ORCA_TOP/I_RISC_CORE/Zro_Flag_reg/CLK (SDFFARX1_RVT)
                                                     0.00000      2.19629 r
  clock uncertainty                                 -0.40000      1.79629
  library setup time                                -0.12775      1.66853
  data required time                                              1.66853
  ------------------------------------------------------------------------------
  data required time                                              1.66853
  data arrival time                                              -1.65773
  ------------------------------------------------------------------------------
  slack (MET)                                                     0.01080

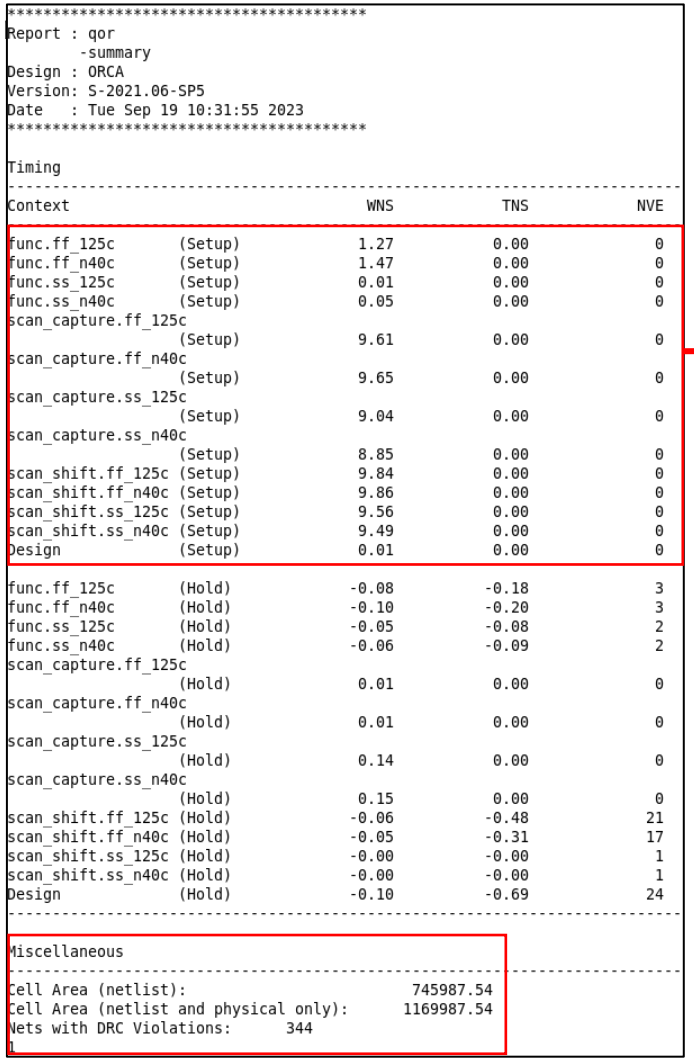

1
```


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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/12_check_placement.tcl

- **report_qor –summary after refine_opt**
  -



```
****************************************
Report : qor
         -summary
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Sep 19 10:31:55 2023
****************************************

Timing
------------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
------------------------------------------------------------------------------
func.ff_125c       (Setup)             1.27            0.00               0
func.ff_n40c       (Setup)             1.47            0.00               0
func.ss_125c       (Setup)             0.01            0.00               0
func.ss_n40c       (Setup)             0.05            0.00               0
scan_capture.ff_125c
                   (Setup)             9.61            0.00               0
scan_capture.ff_n40c
                   (Setup)             9.65            0.00               0
scan_capture.ss_125c
                   (Setup)             9.04            0.00               0
scan_capture.ss_n40c
                   (Setup)             8.85            0.00               0
scan_shift.ff_125c (Setup)             9.84            0.00               0
scan_shift.ff_n40c (Setup)             9.86            0.00               0
scan_shift.ss_125c (Setup)             9.56            0.00               0
scan_shift.ss_n40c (Setup)             9.49            0.00               0
Design             (Setup)             0.01            0.00               0
------------------------------------------------------------------------------
func.ff_125c       (Hold)             -0.08           -0.18               3
func.ff_n40c       (Hold)             -0.10           -0.20               3
func.ss_125c       (Hold)             -0.05           -0.08               2
func.ss_n40c       (Hold)             -0.06           -0.09               2
scan_capture.ff_125c
                   (Hold)              0.01            0.00               0
scan_capture.ff_n40c
                   (Hold)              0.01            0.00               0
scan_capture.ss_125c
                   (Hold)              0.14            0.00               0
scan_capture.ss_n40c
                   (Hold)              0.15            0.00               0
scan_shift.ff_125c (Hold)             -0.06           -0.48              21
scan_shift.ff_n40c (Hold)             -0.05           -0.31              17
scan_shift.ss_125c (Hold)             -0.00           -0.00               1
scan_shift.ss_n40c (Hold)             -0.00           -0.00               1
Design             (Hold)             -0.10           -0.69              24
------------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------------
Cell Area (netlist):                          745987.54
Cell Area (netlist and physical only):       1169987.54
Nets with DRC Violations:       344
1
```

You can check WNS (Worst Negative Slack), TNS(Total Negative Slack) and number of violation at qor report

No setup timing violation after refine_opt

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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/13_report_timing_and_constraints.tcl

- **Report placement after refine_opt**

```
icc2_shell> update_timing -full
Information: Starting 'update_timing' (FLW-8000)
Information: Time: 2023-09-18 16:13:50 / Session: 0.04 hr / Command: 0.00 hr / Memory: 787 MB (FLW-8100)
Information: The net parasitics of block ORCA are cleared. (TIM-123)
Information: Corner ss_n40c: no PVT mismatches. (PVT-032)
Information: Corner ff_125c: no PVT mismatches. (PVT-032)
Information: Corner ss_125c: no PVT mismatches. (PVT-032)
Information: Corner ff_n40c: no PVT mismatches. (PVT-032)
```

- **Source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/user_script/placement_rpt.tcl**

```
icc2_shell> source ./user_script/refine_opt_rpt.tcl
```

- **Go to the Placement reports directory**

```
  < user@linux refine_opt >> pwd
/slowfs/dgscratch2/asic/ASIC_PROJECT/2_BACKEND/1_pnr/rundir/04_placement/reports/refine_opt
  < user@linux refine_opt >> ls
refine_opt.congestion.rpt                 timing_report_max_refine_opt_scan_capture.ff_125c  timing_report_min_refine_opt_func.ss_n40c
refine_opt.fanout.rpt                     timing_report_max_refine_opt_scan_capture.ff_n40c  timing_report_min_refine_opt_scan_capture.ff_125c
refine_opt.legality.check.rpt             timing_report_max_refine_opt_scan_capture.ss_125c  timing_report_min_refine_opt_scan_capture.ff_n40c
refine_opt.max_tran.rpt                   timing_report_max_refine_opt_scan_capture.ss_n40c  timing_report_min_refine_opt_scan_capture.ss_125c
refine_opt.qor.snap.rpt                   timing_report_max_refine_opt_scan_shift.ff_125c    timing_report_min_refine_opt_scan_capture.ss_n40c
refine_opt.runtime.rpt                    timing_report_max_refine_opt_scan_shift.ff_n40c    timing_report_min_refine_opt_scan_shift.ff_125c
refine_opt.util.rpt                       timing_report_max_refine_opt_scan_shift.ss_125c    timing_report_min_refine_opt_scan_shift.ff_n40c
timing_report_max_refine_opt_func.ff_125c timing_report_max_refine_opt_scan_shift.ss_n40c    timing_report_min_refine_opt_scan_shift.ss_125c
timing_report_max_refine_opt_func.ff_n40c timing_report_min_refine_opt_func.ff_125c          timing_report_min_refine_opt_scan_shift.ss_n40c
timing_report_max_refine_opt_func.ss_125c timing_report_min_refine_opt_func.ff_n40c
timing_report_max_refine_opt_func.ss_n40c timing_report_min_refine_opt_func.ss_125c
```

Check the report in the directory just as you did in the place_opt reports

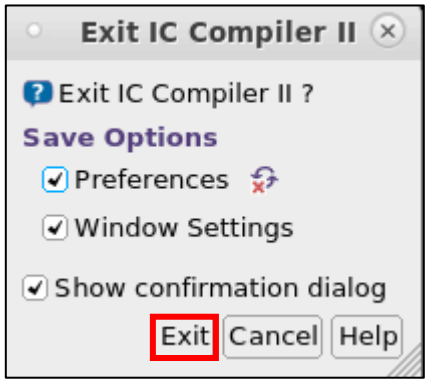
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Placement : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/04_placement/script/14_save_and_exit.tcl

- **save_lib & exit**

```
icc2_shell> save_lib -as ./nlib/ORCA_refine_opt
Saving library 'ORCA_refine_opt' as './nlib/ORCA_refine_opt'
1
icc2_shell> exit
```



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WEEK 7 : Exercise for Placement

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Placement : Use ICC2 (IC Compiler II tool) HomeWork

- **In the "Exercise for placement", Run the placement under several case, and check the results"**

  - **Using placement script & powerplan.nlib**

    placement script - ${WORK_DIR}/2_BACKEND/1_pnr/04_placement/script/*

    powerplan.nlib - ${WORK_DIR}/2_BACKEND/1_pnr/03_powerplan/nlib/ORCA_powerplan_homework.nlib

  - **Guidelines**

  - Exercise 1 - Create soft blockage in a specific area and run placement, then check congestion

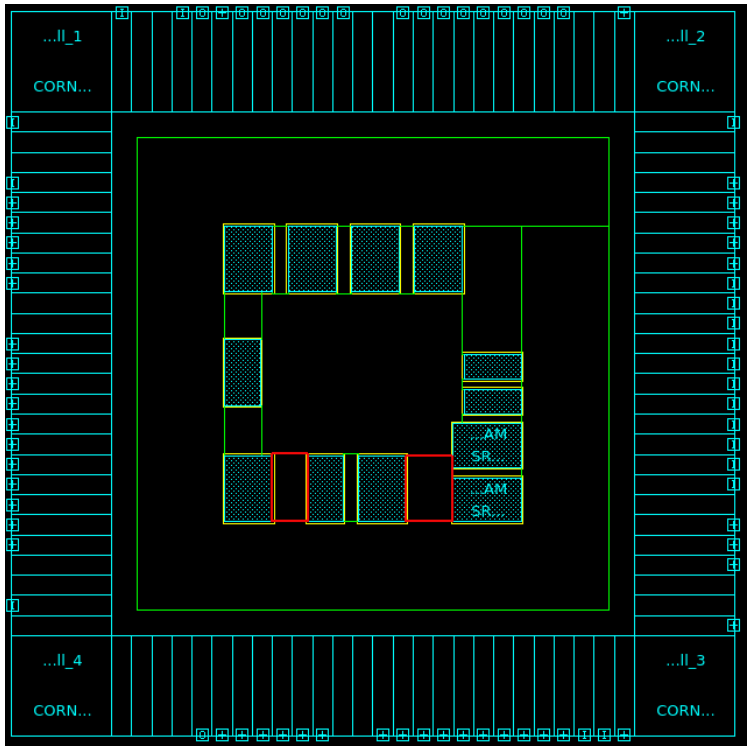
  - Exercise 2 – Compare and analyze result after run placement with changed floorplan

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Placement : Use ICC2 (IC Compiler II tool) HomeWork

- **Exercise 1 - Create soft blockage in a specific area and run placement, then check congestion**

  - **Create soft blockages in two areas between Macro**



- **bbox of soft_blockage**

  Soft_blockage1 : {520.4510 425.8460} {590.4510 560.2530}

  Soft_blockage2 : {786.1980 425.8460} {878.7860 557.2520}

```
create_placement_blockage -name "homework_sof_blk" -type soft -boundary $bbox
```

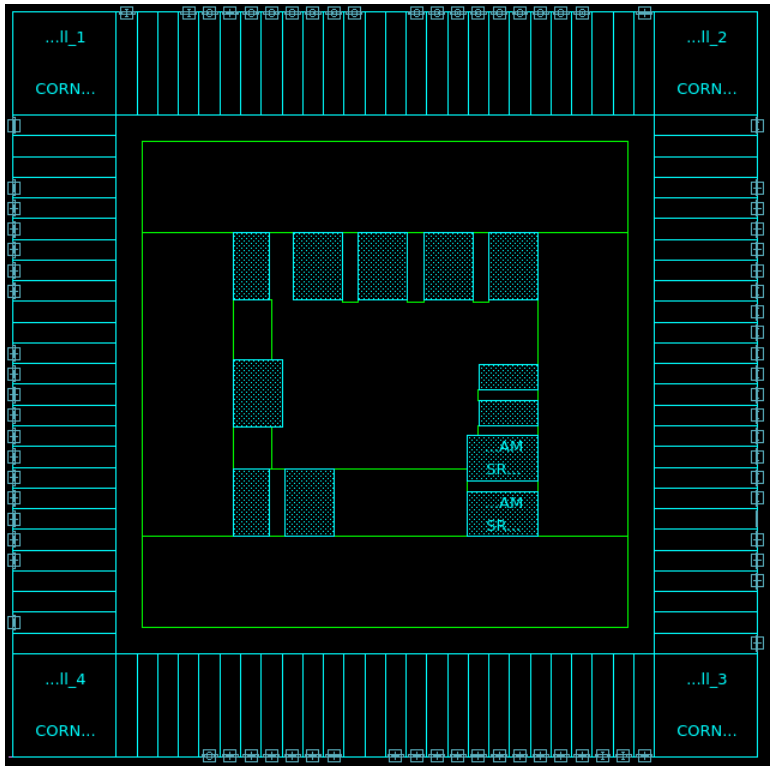
  - **After the placement, compare the congestion results with LAB**

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Placement : Use ICC2 (IC Compiler II tool) HomeWork

- **Exercise 2 – Compare and analyze result after run placement with changed floorplan**
  - **Running placement after complete the powerplan with the changed floorplan (changed at floorplan homework)**

    <Changed floorplan at floorplan homework>



  - **After the placement, compare & analyze reports with before changed floorplan (timing, transition, congestion, fanout, utilization, qor)**
  - **Save by name "ORCA_placement_homework.nlib" & "ORCA_placement_refine_opt_homework.nlib" after placement**

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Thank You

